Physics Lectures

Classical Field Theory I

Lecture Notes

Foundations, variational principles, and fundamental classical fields.

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Introduction

0.1 Why Classical Field Theory Matters

Classical field theory is the point at which many separate parts of physics begin to look like one subject. Waves on a string, electromagnetic radiation, fluids, elastic media, scalar fields, gauge fields, and even the metric of spacetime are all described by quantities that vary from point to point and evolve in time. At first these examples look unrelated. Field theory teaches us to see the common structure behind them.

The first idea is simple. A mechanical system has a finite list of generalized coordinates. A field theory replaces that finite list by a function, or by several functions, defined over space or spacetime.

$$q_i(t) \quad \longrightarrow \quad \phi(x,t).$$

This replacement changes the scale of the problem. Instead of asking how one coordinate changes with time, we ask how a whole spatially distributed object changes, propagates, stores energy, interacts with sources, and responds to boundary conditions.

This shift is not only technical. It changes how one thinks about physics. A field equation is local: it says how neighboring values of a field constrain one another. A conservation law is local: it says how density can flow through the boundary of a region. A symmetry is local or global: it says which changes of description or physical transformations leave the model intact. The language of field theory is therefore the language of locality, propagation, symmetry, and conservation.

These notes are written as a bridge. On one side is classical mechanics: action, energy, momentum, Hamiltonian evolution, and variational principles. On the other side are quantum field theory, gauge theory, and modern geometry. The aim here is not to quantize fields. The aim is to understand the classical layer so well that later quantum and geometric structures no longer appear as magic.

Remark

Classical field theory is not a preliminary obstacle before quantum field theory. It is the structural skeleton on which quantum field theory, gauge theory, and relativistic physics are built.

A good way to read these notes is to keep asking three questions. What is the field? What equation does it satisfy? What physical mechanism does that equation express? The formalism is introduced to answer these questions, not to hide them.

0.2 What Is a Field?

A field is a physical quantity spread over a continuous domain. The value of the field may be a number, a vector, a tensor, a spinor, or a more general geometric object. What matters is not the name of the object, but the fact that it assigns some data to every point of space, spacetime, or another continuous background.

A temperature distribution in a solid is a scalar field: each point has a number. The velocity of a fluid is a vector field: each point has a vector. The electromagnetic field is richer: electric and magnetic fields are observed in a chosen frame, while the relativistic theory packages them into the field strength of a gauge potential. In gravity, the metric itself is a field; it tells us how spacetime intervals are measured.

This is the conceptual move from particles to fields. A point particle has a position. A field has a value everywhere. A particle trajectory is a curve. A field history is an extended pattern in spacetime. A particle may move through a region; a field may oscillate, radiate, form waves, carry energy, satisfy constraints, or approach different boundary values at infinity.

The word "field" should therefore not be reduced to a formula. A formula such as $\phi(t, \mathbf{x})$ is only notation. The physical question is: what does ϕ measure? Is it a displacement, density, potential, phase, component of a connection, or something else? The same mathematical type can represent very different physics.

Definition

A *classical field* is a physical quantity represented by data assigned to each point of a continuous domain. The field is classical when it is treated as a definite object rather than as an operator or quantum state.

The simplest fields are scalar fields, but they are not the end of the story. Vector fields lead naturally to electromagnetism. Lie-algebra-valued fields lead to Yang–Mills theory. Spinor fields are needed for spin-one-half matter. The course begins with the simplest examples because they teach the grammar; later examples show why the grammar is powerful.

0.3 Models, Assumptions, Equations, and Interpretation

A field equation is not a theory by itself. It becomes physics only after we say what the field represents, what assumptions were made, what domain is being studied, what boundary or initial data are allowed, and which quantities are observable. Without that surrounding interpretation, an equation is only a formal object.

The same mathematical equation may occur in different physical settings. The wave equation may describe a vibrating string, sound in an idealized medium, a small perturbation of a field around equilibrium, or a massless relativistic scalar field. The equation alone does not tell us which one is meant. The model does.

This is why the notes repeatedly move between four layers:

$$\text{assumptions} \longrightarrow \text{fields} \longrightarrow \text{equations} \longrightarrow \text{interpretation}.$$

The order is important. Assumptions decide what is ignored. Fields decide what kind of data the model assigns to spacetime. Equations decide how that data is constrained and evolves. Interpretation tells us what the solutions mean.

A useful example is electromagnetism. One can write Maxwell's equations as four vector equations, or more compactly as equations for the tensor $F_{\mu\nu}$, or geometrically as equations for a two-form. These are not three different physical theories. They are three ways of organizing the same model. However, once the potential A_μ is introduced, gauge freedom appears. Then one must understand the difference between a change of physical field and a change of description.

Summary

The goal of these notes is not only to manipulate equations. The goal is to read a field theory as a model: what it assumes, what its variables mean, how its equations work, what it conserves, and what questions it can answer.

A technically correct calculation can still be physically empty if the symbols are not interpreted. Conversely, a clear physical idea needs formalism when it is to become precise. The course is about learning to move in both directions.

0.4 How to Read a Field Equation

A field equation should first be read as a local statement. It relates the value of a field and its derivatives at a point, or in an arbitrarily small neighborhood of that point. This locality is one of the defining ideas of field theory. What happens here is controlled by the field here and by how it changes nearby.

For example, an equation such as

$$(\Box + m^2)\phi = 0$$

is not merely a compact formula. It says that time variation, spatial variation, and the local value of the field balance one another. When $m = 0$, the equation has the character of wave propagation. When $m \neq 0$, the field also has a local restoring term. The parameter m therefore changes the physical behavior of solutions.

Whenever a new field equation appears, the following questions should be asked before doing calculations:

1. What is the unknown field, and what kind of object is it?
2. What is the domain: space, spacetime, or a curved manifold?
3. Which derivatives appear, and what local behavior do they measure?
4. What constants occur, and what are their dimensions?
5. What initial or boundary data are needed?
6. Does the equation come from an action principle?
7. Which symmetries and conservation laws does the equation express?
8. Which quantities are gauge-invariant or physically observable?

This habit prevents a common mistake: treating equations as isolated formulas. In physics, an equation is compressed information about a model. It must be unpacked.

Example

Maxwell's equations are not only equations for $\mathbf{E}$ and $\mathbf{B}$. They encode charge conservation, propagation of electromagnetic waves, gauge redundancy, local energy flow, and the coupling between fields and sources.

A good field theorist does not only solve equations. They ask what the equation is allowed to mean. What has been idealized? What is held fixed? What counts as a source? Which transformations change the physical state, and which merely change the description? These questions are as important as the algebra.

0.5 Notation as a Language

Field theory needs notation, but notation is not the subject. Symbols are useful only when they make structure visible. Index notation, differential forms, Fourier modes, covariant derivatives, and gamma matrices are introduced in these notes because they help us say something clearly about fields.

The reader should not try to memorize every convention at once. At the beginning, it is enough to know what kind of object is being discussed. Is it a scalar, a vector, a tensor, a spinor, or a gauge potential? Does an index label spacetime, space, an internal symmetry, or a component of a multiplet? Most mistakes in field theory come from mixing these roles.

Spatial vectors are written in boldface, for example $\mathbf{x}$, $\mathbf{E}$, and $\mathbf{B}$. Spacetime indices are usually Greek, such as μ, ν, ρ, σ, while spatial indices are usually Latin, such as i, j, k. Internal indices for Lie algebras are often written as a, b, c. The same letter should not be read mechanically; it should be read as part of the object's transformation meaning.

Repeated indices are summed when the Einstein convention is used. Thus an expression such as

$$\partial_\mu F^{\mu\nu} = J^\nu$$

contains one summed index, μ, and one free index, ν. The free index tells us that this is a family of equations, one for each value of ν.

Units are also part of the language. In more physical examples, constants such as c, $\hbar$, or electromagnetic constants may be displayed to clarify dimensions. In more theoretical sections, natural units may be used to reduce clutter. The point is not to hide dimensions, but to avoid letting constants obscure the structure being studied.

Remark

Notation should be read semantically. An index is not decoration; it tells us how an object transforms, what it can be contracted with, and what kind of equation is being written.

Detailed conventions are collected in Appendix A. The main text introduces only what is needed at the moment it becomes useful.

0.6 Recommended Mathematical Background

The notes assume some familiarity with classical mechanics, multivariable calculus, linear algebra, and ordinary differential equations. The most important skills are not exotic. The reader should be comfortable differentiating functions of several variables, integrating by parts, manipulating matrices, reading vector calculus identities, and following a variational argument.

Partial differential equations appear throughout the course. Full mathematical rigour about existence and uniqueness is not the main goal, but one should remember that a field equation is not solved by writing down the equation. A solution requires a domain, initial data, boundary data, and regularity or falloff conditions. These choices are part of the physical problem.

Special relativity enters because many important field theories are relativistic. Tensor notation is introduced as needed. Differential forms, Lie groups, and spinors appear later, but only to the extent needed for classical field theory. The appendices provide reference material for these tools.

The best preparation is conceptual discipline. When a new symbol appears, ask what object it denotes. When a derivative appears, ask with respect to which variable. When a conservation law appears, ask what can flow through the boundary. When a symmetry appears, ask whether it is physical or gauge.

Summary

The required mathematical attitude is more important than a long prerequisite list: follow definitions carefully, track assumptions, check dimensions, control boundary terms, and ask what each symbol represents physically.

0.7 Structure of the Course

The course is organized as a gradual widening of perspective. It begins with the question of what a field is, then builds the variational and Hamiltonian machinery, and only then studies the standard classical fields. This order is intentional. The examples become easier to understand when the reader already knows what an action is, what a symmetry does, and how a conserved current is found.

Part I develops the foundations. It explains how field theory grows out of mechanics, why fields have infinitely many degrees of freedom, how initial and boundary data enter, and why relativistic notation is useful. This part is meant to slow the reader down before the formalism accelerates.

Part II develops the general formalism. The central objects are the action functional, the Lagrangian density, Euler–Lagrange equations, boundary terms, Noether currents, conserved charges, Hamiltonian phase space, Poisson brackets, and constraints. This is the technical core of the notes, but it should always be read with the physical interpretation in mind.

Part III studies fundamental classical fields. The real scalar field is the simplest model. The complex scalar field introduces internal symmetry and charge. Electromagnetism introduces gauge freedom. Yang–Mills theory generalizes this to non-Abelian gauge fields. Spinor fields introduce the classical Dirac equation and prepare the reader for fermions in quantum field theory.

The appendices collect tools that should not interrupt the main line of the course: conventions, variational calculus, Lie groups, differential forms, Green's functions, and suggested problems. They are meant to be used repeatedly, not necessarily read straight through at the beginning.

Remark

The notes should be read actively. A field equation should be varied, differentiated, dimensionally checked, solved in simple cases, and interpreted. The point is not to admire the formalism, but to learn what it makes visible.

The intended outcome is not that the reader remembers many formulas. The intended outcome is that the reader can look at a field theory and ask the right questions: what is the field, what is the action, what are the symmetries, what are the conserved quantities, what is gauge, what is physical, and what boundary conditions make the problem meaningful?

Part I

Foundations of Classical Field Theory

Chapter 1

From Mechanics to Fields

Remark

This chapter explains how the finite-dimensional language of classical mechanics becomes the infinite-dimensional language of fields. The main idea is simple but profound: a field is not one coordinate depending on time, but a continuous family of coordinates, one for each point of space.

1.1 Generalized Coordinates in Classical Mechanics

Classical mechanics begins with the motion of bodies. In its most elementary form, one describes a particle by its position as a function of time,

$$\mathbf{x}(t) = (x^1(t), x^2(t), x^3(t)).$$

This description is already a function: the position is assigned to every time at which the motion is considered. However, the number of independent configuration variables at each time is finite. A single particle in three spatial dimensions has three coordinates. A system of N particles has $3N$ coordinates.

The language of generalized coordinates abstracts away from the particular choice of Cartesian coordinates. Instead of writing all particle positions explicitly, one introduces coordinates

$$q^1(t), q^2(t), \dots, q^n(t),$$

where n is the number of degrees of freedom of the mechanical system. These coordinates may be distances, angles, normal mode amplitudes, or any other variables that describe the configuration of the system.

The important point is that the configuration at a fixed time is represented by a point

$$q(t) = (q^1(t), \dots, q^n(t))$$

in an n-dimensional configuration space. The motion of the system is then a curve in this configuration space.

Definition

The *configuration space* of a mechanical system is the space of all possible instantaneous configurations of the system. If the system has n generalized coordinates, a configuration is represented locally by a point $(q^1, \dots, q^n)$.

In the Lagrangian formulation, the dynamics is encoded in a Lagrangian

$$L(q, \dot{q}, t),$$

where $\dot{q}^i = dq^i/dt$. The action is the functional

$$S[q] = \int_{t_1}^{t_2} L(q(t), \dot{q}(t), t)\, dt.$$

The physical trajectories are selected by the variational principle

$$\delta S = 0,$$

for variations of the path that keep the endpoints fixed. This gives the Euler–Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}^i}\right) - \frac{\partial L}{\partial q^i} = 0, \qquad i = 1, \ldots, n.$$

This structure will reappear almost unchanged in field theory. The difference is that the finite list of coordinates $q^i(t)$ will be replaced by fields such as $\phi(t, \mathbf{x})$. The ordinary derivative with respect to time will be supplemented by spatial derivatives, and the ordinary Lagrangian L will be replaced by a Lagrangian density $\mathcal{L}$.

1.2 Systems with Finitely Many Degrees of Freedom

A system has finitely many degrees of freedom when its instantaneous state can be described by finitely many independent numbers. A double pendulum, a finite chain of masses and springs, a rigid body, or a finite electrical circuit are examples of such systems.

For a system of N coupled oscillators arranged in a line, one may introduce coordinates

$$q_1(t), q_2(t), \ldots, q_N(t),$$

where $q_i(t)$ is the displacement of the i-th oscillator from equilibrium. A typical Lagrangian has the form

$$L = \sum_{i=1}^{N} \frac{m}{2}\dot{q}_i^2 - \sum_{i=1}^{N} \frac{k}{2}(q_{i+1} - q_i)^2,$$

with suitable boundary conventions. The first term describes kinetic energy; the second describes elastic energy stored in relative displacements.

This example already contains the seed of field theory. The label i tells us where the oscillator is located in the chain. If the oscillators are separated by a distance a, the position of the i-th oscillator is

$$x_i = ia.$$

The coordinate $q_i(t)$ may therefore be viewed as the value of a discrete field at the lattice point x_i:

$$q_i(t) \approx \phi(t, x_i).$$

Example

A finite chain of masses is not yet a field theory, because it has only finitely many degrees of freedom. Nevertheless, it is the most useful bridge to field theory: it shows how a spatially distributed system may be described by one dynamical variable at each position.

The equations of motion of a finite chain are ordinary differential equations. For example, away from the endpoints one obtains equations of the schematic form

$$m\ddot{q}_i = k(q_{i+1} - 2q_i + q_{i-1}).$$

The expression on the right-hand side is a discrete second derivative. This is why the continuum limit of such a system naturally produces partial differential equations.

1.3 The Continuum Limit

The continuum limit is the passage from a very large number of discrete degrees of freedom to a field with continuously many degrees of freedom. The basic idea is to let the spacing a between neighbouring points become small while the number of points becomes large.

For the chain of oscillators, we introduce a field $\phi(t, x)$ such that

$$q_i(t) \approx \phi(t, x_i), \qquad x_i = ia.$$

Then the finite difference

$$\frac{q_{i+1} - q_i}{a}$$

approximates the spatial derivative

$$\partial_x \phi(t, x).$$

Similarly,

$$\frac{q_{i+1} - 2q_i + q_{i-1}}{a^2}$$

approximates the second spatial derivative

$$\partial_x^2 \phi(t, x).$$

Thus an equation of motion involving nearest-neighbour couplings becomes, in an appropriate limit, a differential equation for $\phi(t, x)$.

The passage from sums to integrals is equally important. A discrete sum over oscillators becomes an integral over space:

$$\sum_i a\, f(x_i) \longrightarrow \int f(x)\, dx.$$

In mechanics, the action has the form

$$S[q] = \int L\, dt.$$

In field theory, the action has the form

$$S[\phi] = \int \mathcal{L}(\phi, \partial_t \phi, \partial_x \phi, t, x)\, dt\, dx.$$

In higher dimensions this becomes

$$S[\phi] = \int \mathcal{L}(\phi, \partial_\mu \phi, x)\, d^{d+1}x,$$

where d is the number of spatial dimensions and $d + 1$ is the dimension of spacetime.

Remark

The continuum limit is not merely the replacement of sums by integrals. It also changes the type of mathematics. Finite-dimensional configuration spaces are replaced by spaces of functions, and ordinary differential equations are replaced by partial differential equations.

A simple example is the wave equation. Starting from a chain of coupled oscillators and taking the continuum limit, one obtains an equation of the form

$$\partial_t^2 \phi - c^2 \partial_x^2 \phi = 0,$$

where c is the wave speed. This equation does not describe the motion of one particle. It describes the evolution of a whole spatial profile.

1.4 Fields as Infinite-Dimensional Dynamical Systems

A field assigns a value to every point of space, or more generally to every point of spacetime. For a real scalar field in space, a configuration at time t is a function

$$\phi_t : \mathbf{x} \mapsto \phi(t, \mathbf{x}).$$

Thus the instantaneous configuration is not a finite tuple of numbers, but an entire function.

This is the precise sense in which field theory is infinite-dimensional. In a mechanical system with coordinates $q^i(t)$, the index i runs over a finite set. In a field theory, the role of the index is partly played by the spatial point $\mathbf{x}$, which ranges over a continuum.

One may write the analogy informally as

$$q^i(t) \quad \longrightarrow \quad \phi(t, \mathbf{x}).$$

The discrete label i is replaced by a continuous label $\mathbf{x}$. The velocity $\dot{q}^i(t)$ becomes the time derivative $\partial_t \phi(t, \mathbf{x})$. The interaction between neighbouring degrees of freedom produces spatial derivatives such as $\nabla\phi$ or $\Delta\phi$.

Definition

A *classical field* is a classical dynamical variable whose value depends on spacetime position. A field configuration at a fixed time is a function on space, and the time evolution of the field is the evolution of that function.

The configuration space of a field theory is therefore a space of functions. For example, the configuration space of a real scalar field may be thought of as a suitable space of functions

$$\phi : \mathbb{R}^d \to \mathbb{R}.$$

The word "suitable" hides analytic details: one may need fields to be smooth, square-integrable, rapidly decreasing, periodic, or compatible with boundary conditions. These details matter in advanced treatments. At this stage, the main point is conceptual: a field theory is a dynamical system whose configuration space is infinite-dimensional.

1.5 Examples: String, Membrane, Fluid Density, Electromagnetic Field

Field theory is not limited to fundamental particle physics. It appears whenever one studies quantities distributed over space and changing in time.

Vibrating string

A vibrating string may be described by a displacement field

$$y(t, x),$$

where x labels the position along the string and y is the transverse displacement. The field equation is, in the simplest approximation,

$$\partial_t^2 y - c^2 \partial_x^2 y = 0.$$

The physical meaning is direct: the acceleration of a small piece of the string is determined by the curvature of the string profile.

Membrane

A membrane has two spatial labels. Its displacement may be written as

$$z(t, x, y),$$

and its dynamics may involve equations such as

$$\partial_t^2 z - c^2(\partial_x^2 z + \partial_y^2 z) = 0.$$

The field now depends on time and two spatial coordinates.

Fluid density

A fluid may be described by fields such as mass density $\rho(t, \mathbf{x})$, velocity $\mathbf{v}(t, \mathbf{x})$, and pressure $p(t, \mathbf{x})$. These fields are not optional decorations; they are the natural variables because a fluid has properties at each point in space.

The continuity equation

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

expresses local conservation of mass. It says that density changes in a region when mass flows into or out of that region.

Electromagnetic field

Electromagnetism is the central example of a fundamental classical field theory. The electric and magnetic fields

$$\mathbf{E}(t, \mathbf{x}), \qquad \mathbf{B}(t, \mathbf{x})$$

assign vectors to every point of space and time. Maxwell's equations determine how these fields are produced by charges and currents and how they propagate.

In relativistic notation, the electromagnetic field is described by the antisymmetric field strength tensor $F_{\mu\nu}$, or equivalently by a gauge potential A_μ. This course will return to electromagnetism as the first major example of a relativistic gauge field theory.

Example

The same mathematical pattern appears in very different physical systems: strings, membranes, fluids, elastic media, electromagnetic fields, and gauge fields. The details differ, but the structural idea is the same: the dynamical variables are functions on space or spacetime.

1.6 Initial Data and Boundary Data

A field equation usually has many solutions. To select a particular solution, one must specify appropriate data. For evolution equations, the most common data are initial data: the field and enough time derivatives at an initial time.

For the wave equation

$$\partial_t^2 \phi - c^2 \Delta \phi = 0,$$

one typically specifies

$$\phi(0, \mathbf{x}) = f(\mathbf{x}), \qquad \partial_t \phi(0, \mathbf{x}) = g(\mathbf{x}).$$

The first function gives the initial profile. The second gives the initial velocity profile. Together they determine the later evolution, under suitable conditions.

Boundary data are also essential when the spatial domain has a boundary. For a field on an interval $[0, L]$, one may impose fixed endpoint conditions

$$\phi(t, 0) = 0, \qquad \phi(t, L) = 0,$$

or periodic boundary conditions

$$\phi(t, 0) = \phi(t, L).$$

Different boundary conditions describe different physical systems and lead to different spectra of allowed modes.

Remark

Initial conditions select a history from many possible histories. Boundary conditions define the physical problem itself: they say what kind of system the field lives on and how it interacts with the edge of its domain.

Boundary terms will become especially important in the variational principle. When deriving field equations from an action, integrations by parts produce terms on the boundary. A variational principle is well-defined only when those terms vanish or are cancelled by appropriate boundary contributions.

1.7 Locality and Causality

Most classical field theories used in physics are local. Locality means that the equations of motion at a point involve the field and finitely many of its derivatives at that same point, not the values of the field everywhere else. For example, the wave equation

$$\partial_t^2 \phi - c^2 \Delta \phi = 0$$

is local because it relates derivatives of ϕ at the same spacetime point.

A nonlocal equation would involve expressions such as

$$\int K(\mathbf{x}, \mathbf{y}) \phi(t, \mathbf{y}) \, d^d y,$$

where the value of the equation at $\mathbf{x}$ depends directly on field values at many other points $\mathbf{y}$. Nonlocal models exist and can be useful, but locality is the organizing principle of standard classical field theory.

Definition

A field equation is *local* if the equation at a spacetime point depends only on the values of the fields and a finite number of their derivatives at that same point.

In relativistic theories, locality is closely connected with causality. Signals and disturbances should not propagate faster than light. The causal structure of Minkowski spacetime divides directions into timelike, lightlike, and spacelike. A relativistic field equation should respect this structure.

The wave equation illustrates the idea. If a disturbance is created at one point, it propagates with finite speed. In relativistic theories this speed is bounded by the speed of light. This requirement is one reason why relativistic field theories are naturally formulated on spacetime rather than on space alone.

1.8 What Makes a Theory a Field Theory?

A theory is a field theory when its fundamental dynamical variables are fields and its equations determine how those fields vary over space and time. The following features are typical.

1. The configuration at a fixed time is a function, or a collection of functions, on space.
2. The action is usually an integral of a Lagrangian density over spacetime.
3. The equations of motion are usually partial differential equations.
4. The theory has local degrees of freedom, meaning that different regions of space can carry different field values.
5. Boundary conditions and initial data are part of the mathematical and physical specification of the problem.

The transition from mechanics to field theory may therefore be summarized as

$$\text{finite coordinates } q^i(t) \quad \longrightarrow \quad \text{fields } \phi^A(t, \mathbf{x}),$$

where A labels the type or component of the field. For example, a scalar field has one component, a vector field has several components, and a gauge field carries spacetime and possibly internal indices.

Remark

The slogan "a field is infinitely many coupled oscillators" is useful but not complete. It captures the continuum of degrees of freedom, but a full field theory also includes locality, boundary conditions, symmetries, conservation laws, and often relativistic spacetime structure.

This chapter has introduced the conceptual bridge from mechanics to fields. The next chapter develops the mathematical language needed to work with fields: indices, tensors, derivatives, differential forms, variations, distributions, and spacetime notation.

Chapter 2

Mathematical Preliminaries

Remark

This chapter collects the mathematical tools needed before the action principle for fields can be used cleanly. The goal is not to give a complete course in geometry or analysis, but to fix the language that will be used throughout the notes: fields, indices, derivatives, integrals, distributions, variations, functional derivatives, boundary terms, and Lorentzian notation.

2.1 Scalar, Vector, Tensor, and Spinor Fields

A classical field is a quantity assigned to points of space or spacetime. The simplest example is a real scalar field

$$\phi : M \to \mathbb{R},$$

where M is the space or spacetime on which the theory is defined. A scalar field has one value at each point and does not carry a direction or an index. In ordinary language, a temperature distribution $T(t, \mathbf{x})$ is a scalar field. In relativistic field theory, the Klein–Gordon field $\phi(x)$ is the basic example.

A vector field assigns a vector to each point. In three-dimensional space one may write

$$\mathbf{v}(t, \mathbf{x}) = v^1(t, \mathbf{x})\mathbf{e}_1 + v^2(t, \mathbf{x})\mathbf{e}_2 + v^3(t, \mathbf{x})\mathbf{e}_3.$$

The velocity field of a fluid and the electric field $\mathbf{E}(t, \mathbf{x})$ are familiar examples. In relativistic notation, vector fields are written with spacetime indices, for example $A_\mu(x)$ or $V^\mu(x)$.

Tensor fields assign tensors to each point. A rank-two tensor field may be written as $T^{\mu\nu}(x)$, $T_{\mu\nu}(x)$, or $T^\mu{}_\nu(x)$, depending on the position of the indices. The electromagnetic field strength $F_{\mu\nu}$, the metric tensor $g_{\mu\nu}$, and the stress-energy tensor $T^{\mu\nu}$ are all tensor fields.

Spinor fields are more subtle. They are not ordinary vectors, although they also have components. Spinors are objects whose transformation law is tied to the spin representation of the Lorentz group. The Dirac field $\psi(x)$ is the main example. In this course, spinors will be introduced only after the scalar and gauge-field examples are understood.

Definition

A *field type* specifies what kind of object is assigned to each point: a scalar, vector, tensor, spinor, or a collection of such objects. The field equations must be compatible with the transformation properties of that object.

The notation $\phi^A(x)$ is often useful. The label A may represent one component, many tensor indices, an internal symmetry index, or a spinor index. With this convention, many formulas can be written once and then applied to a large class of fields.

2.2 Coordinates, Indices, and Summation Convention

Coordinates label points. In nonrelativistic field theory one often separates time and space,

$$(t, \mathbf{x}) = (t, x^1, x^2, x^3).$$

In relativistic field theory, it is convenient to combine them into spacetime coordinates

$$x^\mu = (x^0, x^1, x^2, x^3),$$

where x^0 denotes the time coordinate, often ct or simply t in units where $c = 1$. Greek indices such as μ, ν, ρ usually run over spacetime values $0, 1, 2, 3$. Latin spatial indices such as i, j, k usually run over $1, 2, 3$.

The Einstein summation convention will be used throughout the notes: whenever an index appears once upstairs and once downstairs in the same term, summation over that index is understood. Thus

$$A_\mu B^\mu$$

means

$$\sum_{\mu=0}^{3} A_\mu B^\mu.$$

Similarly,

$$T^\mu{}_\nu V^\nu$$

means that the index ν is summed over, while μ remains a free index.

Definition

An index is *free* if it appears in the final expression and is not summed over. An index is *dummy* if it is summed over. Dummy indices may be renamed; free indices may not.

For example,

$$A_\mu B^\mu = A_\nu B^\nu$$

because μ and ν are dummy indices. But

$$T^\mu{}_\nu V^\nu$$

is not the same kind of object as

$$T^\rho{}_\nu V^\nu,$$

unless one also changes the name of the free index consistently. The free index labels the component of the resulting vector.

Indices are not decoration. They encode the tensorial type of a formula. A valid tensor equation must have the same free indices on both sides. For example,

$$\partial_\mu F^{\mu\nu} = J^\nu$$

is consistent because both sides carry the free index ν. The expression $\partial_\mu F^{\mu\nu} = J_\mu$, by contrast, would not be a valid tensor equation as written, because the free indices do not match.

2.3 Partial Derivatives and Functional Dependence

Fields depend on spacetime coordinates. For a scalar field $\phi(x)$, the partial derivative is written as

$$\partial_\mu \phi = \frac{\partial \phi}{\partial x^\mu}.$$

The collection $\partial_\mu \phi$ is the gradient of the scalar field in spacetime notation. Higher derivatives are written as

$$\partial_\mu \partial_\nu \phi.$$

When the coordinates are ordinary flat coordinates and the fields are smooth, partial derivatives commute:

$$\partial_\mu \partial_\nu \phi = \partial_\nu \partial_\mu \phi.$$

In a Lagrangian field theory, the Lagrangian density usually depends on the fields and their first derivatives:

$$\mathcal{L} = \mathcal{L}(\phi^A, \partial_\mu \phi^A, x).$$

The symbol x at the end allows explicit coordinate dependence. Many important theories have no explicit coordinate dependence, in which case spacetime translation symmetry leads to energy-momentum conservation.

It is important to distinguish ordinary dependence from functional dependence. The expression $\mathcal{L}(\phi, \partial_\mu \phi, x)$ is a function of the local variables $\phi(x)$, $\partial_\mu \phi(x)$, and x. The action

$$S[\phi] = \int \mathcal{L}(\phi, \partial_\mu \phi, x)\, d^{d+1}x$$

is a functional: it takes an entire field configuration ϕ and returns a number.

Example

For a real scalar field in flat spacetime, a standard Lagrangian density is

$$\mathcal{L} = \frac{1}{2}\partial_\mu \phi\, \partial^\mu \phi - \frac{1}{2} m^2 \phi^2.$$

This expression is local: at each spacetime point it depends only on ϕ and its first derivatives at that same point.

2.4 Differential Forms: A Minimal Introduction

Differential forms provide a compact language for integration, Stokes' theorem, and gauge fields. A zero-form is just a function. A one-form is an object that can be written locally as

$$\alpha = \alpha_\mu dx^\mu.$$

A two-form has the form

$$\omega = \frac{1}{2}\omega_{\mu\nu} dx^\mu \wedge dx^\nu,$$

where $\wedge$ is the wedge product. The wedge product is antisymmetric:

$$dx^\mu \wedge dx^\nu = -dx^\nu \wedge dx^\mu.$$

This antisymmetry is why the components of a two-form satisfy $\omega_{\mu\nu} = -\omega_{\nu\mu}$.

The exterior derivative maps p-forms to $(p+1)$-forms. For a function f,

$$df = \partial_\mu f\, dx^\mu.$$

For a one-form $A = A_\mu dx^\mu$,

$$dA = \frac{1}{2}(\partial_\mu A_\nu - \partial_\nu A_\mu)dx^\mu \wedge dx^\nu.$$

This is the natural form-language version of an antisymmetric field strength. In electromagnetism one writes

$$F = dA,$$

where A is the electromagnetic potential one-form and F is the field strength two-form.

A crucial identity is

$$d^2 = 0.$$

Applied to electromagnetism, this gives

$$dF = d(dA) = 0,$$

which is the geometric form of the homogeneous Maxwell equations.

Remark

Differential forms are not introduced here for elegance alone. They make the boundary structure of field theory visible. The formula $d^2 = 0$, Stokes' theorem, and the compact expression $F = dA$ are the natural language of gauge fields.

2.5 Integration on Space and Spacetime

A field theory action is usually an integral over spacetime. In d spatial dimensions, the spacetime integration measure is written as

$$d^{d+1}x = dx^0\, dx^1 \cdots dx^d.$$

For a relativistic theory in four spacetime dimensions, one often writes

$$S[\phi] = \int \mathcal{L}\, d^4x.$$

If one separates time and space, the same integral is written as

$$S[\phi] = \int dt \int d^dx\, \mathcal{L}.$$

The Lagrangian density $\mathcal{L}$ should not be confused with the Lagrangian L. The relation is

$$L(t) = \int d^dx\, \mathcal{L}(t, \mathbf{x}).$$

Thus the action may be written either as

$$S = \int L(t)\, dt$$

or as

$$S = \int \mathcal{L}\, dt\, d^dx.$$

When the spacetime region has a boundary, one must keep track of boundary integrals. In ordinary vector calculus, the divergence theorem says that

$$\int_\Omega \nabla \cdot \mathbf{J}\, d^dx = \int_{\partial\Omega} \mathbf{J} \cdot \mathbf{n}\, d\Sigma.$$

In spacetime notation, an analogous statement is

$$\int_\Omega \partial_\mu J^\mu\, d^{d+1}x = \int_{\partial\Omega} J^\mu n_\mu\, d\Sigma.$$

This is the reason total derivative terms are often said to become boundary terms.

2.6 Delta Functions and Distributions

The Dirac delta is not an ordinary function. It is a distribution defined by its action under an integral:

$$\int_{-\infty}^{\infty} \delta(x-a) f(x)\, dx = f(a).$$

In d spatial dimensions,

$$\int \delta^{(d)}(\mathbf{x}-\mathbf{y}) f(\mathbf{x})\, d^d x = f(\mathbf{y}).$$

The delta distribution is used to represent point sources, Green's functions, and functional derivatives.

Point charge density, for example, can be written as

$$\rho(\mathbf{x}) = q\, \delta^{(3)}(\mathbf{x}-\mathbf{x}_0),$$

meaning that the total charge is localized at $\mathbf{x}_0$. Indeed,

$$\int \rho(\mathbf{x})\, d^3 x = q.$$

Derivatives of delta distributions are defined by integration by parts. In one dimension,

$$\int \delta'(x-a) f(x)\, dx = -f'(a),$$

assuming boundary terms vanish. This rule is essential when manipulating Green's functions and variational derivatives.

Definition

A *distribution* is an object defined by how it acts under an integral against smooth test functions. The Dirac delta is the distribution that extracts the value of a test function at a point.

2.7 Variations of Fields

A variation of a field is an infinitesimal change of the field configuration. For a scalar field one writes

$$\phi(x) \longrightarrow \phi(x) + \varepsilon\eta(x),$$

where ε is a small parameter and $\eta(x)$ is an arbitrary test variation. The variation of the field is

$$\delta\phi(x) = \eta(x).$$

The variation of a derivative is the derivative of the variation:

$$\delta(\partial_\mu \phi) = \partial_\mu(\delta\phi).$$

This simple identity is the mechanism behind the Euler–Lagrange equations for fields.

For an action

$$S[\phi] = \int \mathcal{L}(\phi, \partial_\mu\phi, x)\, d^{d+1}x,$$

the first variation is

$$\delta S = \int \left(\frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\delta(\partial_\mu\phi) \right) d^{d+1}x.$$

Using $\delta(\partial_\mu\phi) = \partial_\mu\delta\phi$, one obtains

$$\delta S = \int \left(\frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\mu\delta\phi \right) d^{d+1}x.$$

The second term is then integrated by parts. This produces a bulk term and a boundary term. The bulk term gives the field equation; the boundary term tells us what must be assumed or added for the variational principle to be well defined.

2.8 Functional Derivatives

A functional derivative measures how a functional changes when the value of a field changes at a point. For a functional $F[\phi]$, the functional derivative is defined by

$$\delta F = \int \frac{\delta F}{\delta\phi(x)}\,\delta\phi(x)\,d^{d+1}x,$$

up to boundary terms and under the chosen class of variations.

The simplest example is

$$F[\phi] = \int f(x)\phi(x)\,d^{d+1}x.$$

Then

$$\delta F = \int f(x)\delta\phi(x)\,d^{d+1}x,$$

so

$$\frac{\delta F}{\delta\phi(x)} = f(x).$$

A second important identity is

$$\frac{\delta\phi(y)}{\delta\phi(x)} = \delta^{(d+1)}(y-x).$$

This is the field-theoretic analogue of $\partial q^j/\partial q^i = \delta^j_i$.

For a local action of the form

$$S[\phi] = \int \mathcal{L}(\phi, \partial_\mu\phi, x)\,d^{d+1}x,$$

the functional derivative is

$$\frac{\delta S}{\delta\phi} = \frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right),$$

provided the boundary term is handled appropriately. The Euler–Lagrange equation is then simply

$$\frac{\delta S}{\delta\phi} = 0.$$

For several fields ϕ^A, one has one equation for each field component:

$$\frac{\delta S}{\delta\phi^A} = 0.$$

2.9 Boundary Terms and the Meaning of "Up to a Total Derivative"

Boundary terms are not a technical nuisance. They are one of the main structural features of field theory. Consider a total derivative in spacetime,

$$\partial_\mu K^\mu.$$

Its integral over a spacetime region Ω is

$$\int_\Omega \partial_\mu K^\mu \, d^{d+1}x = \int_{\partial\Omega} K^\mu n_\mu \, d\Sigma.$$

Thus adding a total derivative to the Lagrangian density changes the action by a boundary term:

$$\mathcal{L}' = \mathcal{L} + \partial_\mu K^\mu.$$

The bulk Euler–Lagrange equations are unchanged, but the variational principle, canonical momenta, conserved charges, and boundary conditions may change.

This is the precise meaning of the phrase "up to a total derivative." It does not mean that the term is zero. It means that the term does not affect the bulk field equations under the assumed boundary conditions. If the boundary is physically important, the term may become essential.

> **Remark**
>
> A total derivative is harmless only after the boundary problem has been stated. In theories with boundaries, asymptotic regions, gauge symmetry, gravity, or conserved charges, boundary terms often contain physical information.

In the variational derivation, the characteristic integration by parts is

$$\int_\Omega P^\mu \partial_\mu \delta\phi \, d^{d+1}x = -\int_\Omega (\partial_\mu P^\mu)\delta\phi \, d^{d+1}x + \int_{\partial\Omega} P^\mu n_\mu \delta\phi \, d\Sigma.$$

The first term contributes to the Euler–Lagrange equation. The second is the boundary term. One may remove it by fixing $\delta\phi = 0$ on the boundary, by choosing falloff conditions, or by adding compensating boundary terms to the action.

2.10 Lorentzian Geometry Needed for Field Theory

Relativistic field theory is formulated on spacetime. The simplest spacetime is Minkowski spacetime, equipped with the metric $\eta_{\mu\nu}$. In these notes we will mostly use the signature

$$\eta_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1),$$

unless stated otherwise. With this convention,

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2.$$

The metric is used to raise and lower indices:

$$V_\mu = \eta_{\mu\nu} V^\nu, \qquad V^\mu = \eta^{\mu\nu} V_\nu.$$

It is also used to form Lorentz-invariant contractions, such as

$$V_\mu V^\mu.$$

The derivative with a raised index is defined by

$$\partial^\mu = \eta^{\mu\nu}\partial_\nu.$$

The d'Alembert operator is

$$\Box = \partial_\mu \partial^\mu.$$

With the above signature this becomes

$$\Box = \partial_t^2 - \nabla^2$$

in units where $c = 1$. The Klein–Gordon equation may then be written compactly as

$$(\Box + m^2)\phi = 0.$$

In curved spacetime, the metric becomes a field $g_{\mu\nu}(x)$, partial derivatives are replaced by covariant derivatives, and the volume element is modified. For much of Classical Field Theory I, however, the main stage will be flat Minkowski spacetime. Curved spacetime will appear later as a natural extension and as a bridge toward gravity.

Summary

The mathematical transition from mechanics to field theory is the transition from coordinates $q^i(t)$ to fields $\phi^A(x)$, from ordinary derivatives to spacetime derivatives, from a Lagrangian L to a Lagrangian density $\mathcal{L}$, and from finite-dimensional derivatives to functional derivatives. Boundary terms are part of the structure, not an afterthought.

Chapter 3

Special Relativity and Spacetime Formulation

Remark

Classical field theory becomes structurally clear when it is written on spacetime rather than on space plus time separately. This chapter introduces the relativistic language needed for the rest of the course: Minkowski spacetime, Lorentz transformations, four-vectors, tensors, proper time, causality, covariant equations, and invariant actions.

3.1 Minkowski Spacetime

Special relativity is built on the idea that space and time are not independent backgrounds. They form a single four-dimensional spacetime. A point of spacetime is called an event and is labelled by coordinates

$$x^\mu = (x^0, x^1, x^2, x^3).$$

In units where $c = 1$, one usually takes $x^0 = t$. If the speed of light is kept explicit, one often writes $x^0 = ct$.

The geometric structure of flat relativistic spacetime is encoded in the Minkowski metric. In these notes we use the mostly-minus convention

$$\eta_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1).$$

The spacetime interval between neighbouring events is

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu = dt^2 - d\mathbf{x}^2$$

in units $c = 1$. With c explicit, this becomes

$$ds^2 = c^2 dt^2 - d\mathbf{x}^2.$$

The interval is not a Euclidean distance. Its sign has physical meaning. A separation is timelike if $ds^2 > 0$, lightlike if $ds^2 = 0$, and spacelike if $ds^2 < 0$. Timelike separated events can lie on the worldline of a massive particle. Lightlike separated events can be connected by a light ray. Spacelike separated events cannot influence each other by any signal travelling at or below the speed of light.

Definition

Minkowski spacetime is the flat spacetime of special relativity. It is equipped with the metric $\eta_{\mu\nu}$, which defines the invariant interval $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$.

The key difference between Euclidean geometry and Minkowski geometry is that the metric has mixed signs. This is not a minor convention. It is the algebraic origin of light cones, causal structure, relativistic wave equations, and the classification of particles by mass.

3.2 Lorentz Transformations

A Lorentz transformation is a linear change of inertial coordinates that preserves the Minkowski interval. If

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu},$$

then $\Lambda^{\mu}{}_{\nu}$ is Lorentz if

$$\eta_{\rho\sigma} \Lambda^{\rho}{}_{\mu} \Lambda^{\sigma}{}_{\nu} = \eta_{\mu\nu}.$$

Equivalently, in matrix notation,

$$\Lambda^T \eta \Lambda = \eta.$$

This is the relativistic analogue of the condition $R^T R = I$ for rotations in Euclidean space, but with the Euclidean metric replaced by the Minkowski metric.

A standard boost in the x^1-direction has the form

$$\begin{aligned} t' &= \gamma(t - vx^1), \\ x'^1 &= \gamma(x^1 - vt), \\ x'^2 &= x^2, \\ x'^3 &= x^3, \end{aligned}$$

where

$$\gamma = \frac{1}{\sqrt{1 - v^2}}$$

in units $c = 1$. This transformation mixes time and space. That mixing is why relativistic field theory should be formulated on spacetime from the beginning.

Example

For a boost in the x^1-direction, one can check directly that

$$t'^2 - (x'^1)^2 - (x'^2)^2 - (x'^3)^2 = t^2 - (x^1)^2 - (x^2)^2 - (x^3)^2.$$

The coordinates change, but the interval does not.

The set of Lorentz transformations forms the Lorentz group, usually denoted $O(1, 3)$. The transformations continuously connected to the identity and preserving the direction of time form the proper orthochronous Lorentz group $SO^+(1, 3)$. Most local relativistic field theories are built to transform naturally under this group.

3.3 Four-Vectors and Tensors

A four-vector is an object whose components transform under Lorentz transformations in the same way as spacetime coordinates:

$$V'^{\mu} = \Lambda^{\mu}{}_{\nu} V^{\nu}.$$

Examples include the displacement dx^{μ}, the four-momentum p^{μ}, and the four-current J^{μ}. A covector transforms with the inverse matrix, and is usually written with a lower index:

$$W'_{\mu} = (\Lambda^{-1})^{\nu}{}_{\mu} W_{\nu}.$$

The metric converts vectors into covectors:

$$V_{\mu} = \eta_{\mu\nu} V^{\nu}.$$

The contraction of a vector and a covector is Lorentz invariant:

$$W_\mu V^\mu = W'_\mu V'^\mu.$$

Similarly, the squared norm of a four-vector is

$$V_\mu V^\mu.$$

For four-momentum,

$$p^\mu = (E, \mathbf{p})$$

in units $c = 1$, the invariant relation is

$$p_\mu p^\mu = E^2 - \mathbf{p}^2 = m^2.$$

This is the relativistic energy-momentum relation.

Tensors generalize vectors by carrying more indices. A rank-two tensor $T^{\mu\nu}$ transforms as

$$T'^{\mu\nu} = \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma T^{\rho\sigma}.$$

The electromagnetic field strength $F_{\mu\nu}$ is an antisymmetric rank-two tensor. The stress-energy tensor $T^{\mu\nu}$ is a rank-two tensor encoding energy density, momentum density, energy flux, and stresses.

Definition

A relativistic equation is *covariant* if both sides transform in the same way under Lorentz transformations. Covariance ensures that the equation has the same geometric content in all inertial frames.

A useful rule is that a valid tensor equation must have the same free indices on both sides. For example,

$$\partial_\mu F^{\mu\nu} = J^\nu$$

is a valid vector equation, because the free index ν appears on both sides. The equation represents four component equations, but it is written in a form that makes Lorentz covariance manifest.

3.4 Proper Time, Worldlines, and Causal Structure

The history of a point particle is a curve in spacetime, called a worldline:

$$x^\mu = x^\mu(\lambda),$$

where λ is a parameter along the curve. For a massive particle, one can use proper time τ as the parameter. Proper time is defined by

$$d\tau^2 = ds^2$$

for timelike curves in units $c = 1$. Thus

$$d\tau^2 = dt^2 - d\mathbf{x}^2.$$

If $\mathbf{v} = d\mathbf{x}/dt$, then

$$d\tau = dt\sqrt{1 - \mathbf{v}^2} = \frac{dt}{\gamma}.$$

Proper time is the time measured by a clock moving along the worldline.

The four-velocity is defined as

$$u^\mu = \frac{dx^\mu}{d\tau}.$$

It satisfies

$$u_\mu u^\mu = 1$$

with our signature convention. The four-momentum is

$$p^\mu = m u^\mu,$$

and therefore

$$p_\mu p^\mu = m^2.$$

Causal structure is encoded by light cones. At each event, the light cone separates possible future-directed timelike and lightlike directions from spacelike directions. Relativistic field equations must respect this structure: physical disturbances should propagate inside or on the light cone, not outside it.

Remark

The light cone is not an optical illustration added to spacetime. It is the causal skeleton of relativistic physics. Hyperbolic field equations, retarded Green's functions, and relativistic signal propagation are all organized around this structure.

For field theory, worldlines are less fundamental than fields over spacetime, but they remain useful. Sources, detectors, observers, and particles moving in external fields are often represented by worldlines interacting with spacetime fields.

3.5 Relativistic Notation for Fields

A relativistic field is written as a function of spacetime coordinates:

$$\phi(x) = \phi(x^0, x^1, x^2, x^3).$$

For a scalar field, Lorentz transformations act by changing the argument:

$$\phi'(x') = \phi(x).$$

Equivalently,

$$\phi'(x) = \phi(\Lambda^{-1}x).$$

This means that the scalar value attached to a spacetime event is independent of the coordinate system used to describe that event.

A vector field $A_\mu(x)$ transforms both in its argument and in its index:

$$A'_\mu(x') = (\Lambda^{-1})^\nu{}_\mu A_\nu(x).$$

A tensor field transforms with one Lorentz matrix for each index. Spinor fields transform with spinor representations rather than ordinary vector representations; this will be discussed later in the chapter on the Dirac field.

The derivative operator is written as

$$\partial_\mu = \frac{\partial}{\partial x^\mu}.$$

Raising the index gives

$$\partial^\mu = \eta^{\mu\nu}\partial_\nu.$$

The d'Alembert operator is

$$\Box = \partial_\mu \partial^\mu.$$

With our signature and $c = 1$, this is

$$\Box = \partial_t^2 - \nabla^2.$$

A relativistic scalar field equation can therefore be written compactly as

$$(\Box + m^2)\phi = 0.$$

This is the Klein–Gordon equation. Its compact form displays its Lorentz covariance: $\Box$ is a Lorentz-invariant differential operator, and m^2 is a scalar parameter.

3.6 Covariant and Non-Covariant Forms of Equations

A covariant equation is written in a way that makes its transformation properties clear. A non-covariant equation may describe the same physics, but it hides the spacetime structure by separating time and space.

For example, the Klein–Gordon equation can be written covariantly as

$$(\Box + m^2)\phi = 0.$$

In components, using $\Box = \partial_t^2 - \nabla^2$, the same equation is

$$\partial_t^2\phi - \nabla^2\phi + m^2\phi = 0.$$

The second form is useful for solving initial-value problems, but the first form makes Lorentz covariance manifest.

Maxwell's equations provide an even stronger example. In vector notation they are

$$\begin{aligned}
\nabla \cdot \mathbf{E} &= \rho, \\
\nabla \cdot \mathbf{B} &= 0, \\
\nabla \times \mathbf{E} + \partial_t \mathbf{B} &= 0, \\
\nabla \times \mathbf{B} - \partial_t \mathbf{E} &= \mathbf{J},
\end{aligned}$$

in units adapted to the chosen conventions. In covariant notation these become

$$\partial_\mu F^{\mu\nu} = J^\nu, \qquad \partial_{[\lambda} F_{\mu\nu]} = 0.$$

The second equation can also be written as

$$dF = 0$$

in differential-form notation.

Example

A non-covariant form is often better for calculation in a chosen frame. A covariant form is better for understanding the structure of the theory. In good relativistic field theory, both descriptions are available and equivalent.

Covariance does not mean that all components look numerically the same in every frame. It means that the equation as a geometric statement is frame-independent. The components change, but the tensorial relation remains valid.

3.7 Invariant Actions

The action principle is the central organizing principle of this course. In a relativistic field theory, the action should be a Lorentz scalar. This means that its value should not depend on the inertial coordinate system used to compute it.

For a scalar field, a standard action is

$$S[\phi] = \int \mathcal{L}\, d^4x,$$

with

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2.$$

The term $\partial_\mu\phi\partial^\mu\phi$ is a Lorentz scalar because the index μ is contracted. The term ϕ^2 is also a scalar. Therefore the Lagrangian density is a scalar, and the action is invariant under Lorentz transformations.

More generally, relativistic Lagrangian densities are built by contracting all spacetime indices. For electromagnetism, the basic invariant is

$$F_{\mu\nu}F^{\mu\nu}.$$

The Maxwell action in vacuum is proportional to

$$S[A] = -\frac{1}{4}\int F_{\mu\nu}F^{\mu\nu}\, d^4x.$$

The factor $-1/4$ is conventional, but useful: it gives standard normalizations for the equations of motion and energy density.

Definition

A relativistic action is *Lorentz invariant* if it is unchanged under Lorentz transformations. In practice, this is achieved by building the Lagrangian density from Lorentz scalars and integrating over the invariant spacetime volume element.

In flat spacetime, d^4x is invariant under proper Lorentz transformations because $|\det\Lambda| = 1$. In curved spacetime, the invariant volume element is no longer simply d^4x, but $\sqrt{|g|}\, d^4x$, where g is the determinant of the metric. This generalization will become important when field theory is coupled to gravity.

3.8 Why Relativistic Field Theory Is Natural

Relativistic field theory is not just classical mechanics rewritten with more indices. It is the natural framework for local physics compatible with special relativity.

First, special relativity treats space and time as parts of one structure. Therefore a local dynamical variable should naturally be a function on spacetime:

$$\phi = \phi(x).$$

Second, causality restricts how disturbances may propagate. Relativistic field equations must respect the light-cone structure of Minkowski spacetime. Third, energy and momentum form a four-vector, and conservation laws are naturally written as divergence equations such as

$$\partial_\mu T^{\mu\nu} = 0.$$

Fourth, interactions that are local in spacetime are naturally described by local Lagrangian densities. A typical action has the form

$$S = \int \mathcal{L}(\phi^A, \partial_\mu \phi^A, x)\, d^4x.$$

This expression says that the total action is assembled from local spacetime contributions. That locality is one of the deepest structural assumptions of classical and quantum field theory.

Finally, the same language prepares the transition to quantum field theory. The objects that will later be quantized are not particle trajectories, but fields, actions, symmetries, currents, and propagators. Classical relativistic field theory is therefore the structural bridge between classical mechanics and quantum field theory.

Summary

Special relativity turns field theory into spacetime geometry. Fields are functions or tensorial objects on Minkowski spacetime; equations should be Lorentz covariant; actions should be Lorentz invariant; and causal propagation is organized by the light cone. This is the natural language for scalar fields, electromagnetism, Yang–Mills theory, spinor fields, and eventually gravity.

Part II

Lagrangian Field Theory

Chapter 4

The Action Principle for Fields

Remark

The action principle is the central organizing idea of classical field theory. Instead of postulating a differential equation directly, one postulates an action functional. The field equations are then obtained by demanding that the action be stationary under arbitrary allowed variations of the fields.

4.1 The Action Functional

In classical mechanics, the action is a functional of a path $q(t)$:

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, dt.$$

A physical trajectory is selected by the condition

$$\delta S = 0$$

for variations of the path that obey the chosen endpoint conditions.

In field theory, the dynamical variable is not a finite list of coordinates but a field, or a collection of fields,

$$\phi^A(x),$$

where A labels the components or species of the field. The action is a functional of the entire field configuration:

$$S[\phi] = \int_\Omega \mathcal{L}(\phi^A, \partial_\mu \phi^A, x)\, d^{d+1}x.$$

Here Ω is the spacetime region under consideration, and $\mathcal{L}$ is the Lagrangian density.

Definition

An *action functional* assigns a number to each allowed field configuration. In a local first-order field theory it usually has the form

$$S[\phi] = \int_\Omega \mathcal{L}(\phi^A, \partial_\mu \phi^A, x)\, d^{d+1}x.$$

The word functional is important. A function takes numbers as input and returns a number. A functional takes a function as input and returns a number. The action therefore depends on the entire history of the field over a spacetime region, not merely on the value of the field at one point.

The principle of stationary action states that the physical fields are those for which the first-order change of the action vanishes:

$$\delta S = 0.$$

This condition is not a minimum principle in general. The action may have a minimum, a maximum, or a saddle point. The correct statement is stationarity.

4.2 Lagrangian Density

The Lagrangian density is the local quantity from which the action is built. If space and time are separated, the ordinary Lagrangian is obtained by integrating the Lagrangian density over space:

$$L(t) = \int d^d x \, \mathcal{L}(t, \mathbf{x}).$$

Then

$$S = \int dt \, L(t) = \int dt \int d^d x \, \mathcal{L}.$$

For a real scalar field in flat spacetime, a standard Lagrangian density is

$$\mathcal{L} = \frac{1}{2}\partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2.$$

The first term is the kinetic-gradient term, and the second term is a mass term. Using the mostly-minus convention, the first term expands as

$$\frac{1}{2}\partial_\mu \phi \partial^\mu \phi = \frac{1}{2}(\partial_t \phi)^2 - \frac{1}{2}|\nabla \phi|^2.$$

Thus the Lagrangian density contains a time-derivative part and a spatial-gradient part with opposite signs.

A more general scalar-field Lagrangian often has the form

$$\mathcal{L} = \frac{1}{2}\partial_\mu \phi \partial^\mu \phi - V(\phi),$$

where $V(\phi)$ is the potential energy density. For example,

$$V(\phi) = \frac{1}{2} m^2 \phi^2 + \frac{\lambda}{4}\phi^4$$

leads to a nonlinear scalar field equation.

Remark

The Lagrangian density is not unique. Different Lagrangian densities can give the same bulk field equations if they differ by a total derivative. However, they may still differ in their boundary terms, canonical variables, and charges.

A Lagrangian density should be built from quantities compatible with the symmetries one wants the theory to have. In relativistic field theory, this usually means building Lorentz scalars by contracting all spacetime indices. In gauge theory, it also means building expressions compatible with gauge symmetry.

4.3 Variational Principle for Fields

To vary a field, one considers a one-parameter family of field configurations

$$\phi^A_\varepsilon(x) = \phi^A(x) + \varepsilon \eta^A(x),$$

where ε is a small parameter and $\eta^A(x)$ is an arbitrary allowed variation. The variation of the field is

$$\delta \phi^A(x) = \eta^A(x).$$

The variation of a derivative is

$$\delta(\partial_\mu \phi^A) = \partial_\mu(\delta \phi^A).$$

The first variation of the action is defined by

$$\delta S = \frac{d}{d\varepsilon} S[\phi_\varepsilon]\bigg|_{\varepsilon=0}.$$

For a first-order local Lagrangian density,

$$\mathcal{L} = \mathcal{L}(\phi^A, \partial_\mu \phi^A, x),$$

we obtain

$$\delta S = \int_\Omega \left(\frac{\partial \mathcal{L}}{\partial \phi^A} \delta\phi^A + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \delta(\partial_\mu \phi^A) \right) d^{d+1}x.$$

Using $\delta(\partial_\mu \phi^A) = \partial_\mu \delta\phi^A$, this becomes

$$\delta S = \int_\Omega \left(\frac{\partial \mathcal{L}}{\partial \phi^A} \delta\phi^A + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \partial_\mu \delta\phi^A \right) d^{d+1}x.$$

The repeated field label A and spacetime index μ are summed over.

The second term contains a derivative of the variation. To isolate $\delta\phi^A$, we integrate by parts:

$$\begin{aligned} \delta S = &\int_\Omega \left[\frac{\partial \mathcal{L}}{\partial \phi^A} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \right) \right] \delta\phi^A \, d^{d+1}x \\ &+ \int_{\partial\Omega} \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \delta\phi^A n_\mu \, d\Sigma. \end{aligned}$$

The first integral is the bulk variation. The second is the boundary variation.

4.4 Euler–Lagrange Equations for Fields

If the boundary term vanishes under the chosen class of variations, then the stationarity condition $\delta S = 0$ implies that the bulk coefficient of each independent $\delta\phi^A$ must vanish. This gives the Euler–Lagrange equations for fields:

$$\frac{\partial \mathcal{L}}{\partial \phi^A} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \right) = 0.$$

Equivalently,

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \right) - \frac{\partial \mathcal{L}}{\partial \phi^A} = 0.$$

Both conventions are common; they differ only by multiplying the equation by -1.

Definition

The *Euler–Lagrange equation for a field* is the differential equation obtained by setting the first variation of the action to zero for arbitrary allowed variations of the field.

In functional-derivative notation, the same statement is

$$\frac{\delta S}{\delta \phi^A} = 0.$$

For a first-order local Lagrangian density,

$$\frac{\delta S}{\delta \phi^A} = \frac{\partial \mathcal{L}}{\partial \phi^A} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \right).$$

This compact notation is useful, but it should not hide the boundary term that appears during the derivation.

4.5 Boundary Conditions in the Variational Principle

The variational principle is incomplete until its boundary conditions are specified. The integration by parts produces the boundary contribution

$$\int_{\partial\Omega} \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \delta\phi^A n_\mu \, d\Sigma.$$

A common choice is to fix the fields on the boundary:

$$\delta\phi^A|_{\partial\Omega} = 0.$$

Then the boundary term vanishes, and the bulk Euler–Lagrange equations follow.

This choice is not the only possible one. One may instead impose falloff conditions at infinity, periodic boundary conditions, Neumann-type boundary conditions, or add boundary terms to the action so that the total variation is well-defined. The correct choice depends on the physical problem.

Example

For a vibrating string on an interval $[0, L]$, fixed endpoints correspond to

$$\delta\phi(t, 0) = \delta\phi(t, L) = 0.$$

Free endpoints lead to different boundary conditions. The same bulk equation can therefore describe different physical systems depending on the boundary conditions.

Boundary terms become especially important in gauge theory and gravity. In such systems, surface terms may carry conserved charges, fluxes, or edge degrees of freedom. For now, the essential point is that the action principle always contains both a bulk statement and a boundary statement.

4.6 Total Derivatives and Equivalent Lagrangians

Suppose two Lagrangian densities differ by a total derivative:

$$\mathcal{L}' = \mathcal{L} + \partial_\mu K^\mu.$$

Then the corresponding actions differ by a boundary term:

$$S'[\phi] - S[\phi] = \int_\Omega \partial_\mu K^\mu \, d^{d+1}x = \int_{\partial\Omega} K^\mu n_\mu \, d\Sigma.$$

Therefore $\mathcal{L}$ and $\mathcal{L}'$ give the same bulk Euler–Lagrange equations when the boundary term does not contribute to the variation.

This is the precise sense in which two Lagrangians may be called equivalent. They are equivalent for the bulk equations under a specified class of boundary conditions. They are not necessarily equivalent as complete variational principles if the boundary is physically relevant.

Remark

The phrase "up to a total derivative" should always be read with care. It is safe for deriving local bulk equations under suitable boundary assumptions. It is not automatically safe for conserved charges, Hamiltonian generators, asymptotic symmetries, or gravitational actions.

A simple mechanical analogue is adding a total time derivative to a Lagrangian:

$$L'(q, \dot{q}, t) = L(q, \dot{q}, t) + \frac{dF(q, t)}{dt}.$$

The action changes by

$$F(q(t_2), t_2) - F(q(t_1), t_1),$$

which does not affect the Euler–Lagrange equations if the endpoints are fixed. Field theory is the spacetime version of the same idea.

4.7 On-Shell and Off-Shell Statements

A field configuration is said to be on shell if it satisfies the equations of motion. It is off shell if it is an arbitrary field configuration not assumed to satisfy those equations.

For example, for a scalar field with equation

$$(\Box + m^2)\phi = 0,$$

an on-shell field satisfies this differential equation. An off-shell field is any sufficiently regular function $\phi(x)$ in the configuration space of the theory.

Definition

An identity or formula is *on shell* if it holds after imposing the field equations. It is *off shell* if it holds without using the field equations.

This distinction is fundamental. The variational formula for δS is an off-shell statement. The Euler–Lagrange equations are obtained by demanding stationarity and then setting the bulk functional derivative to zero. A conservation law may hold only on shell, while a symmetry of the action is often an off-shell statement.

The notation

$$\frac{\delta S}{\delta \phi^A} = 0$$

therefore defines the shell: it specifies the subspace of field configurations that solve the equations of motion.

4.8 Worked Example: Real Scalar Field

Consider the real scalar field action in flat spacetime:

$$S[\phi] = \int \left(\frac{1}{2}\partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 \right) d^{d+1}x.$$

The Lagrangian density is

$$\mathcal{L} = \frac{1}{2}\partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2.$$

We compute

$$\frac{\partial \mathcal{L}}{\partial \phi} = -m^2 \phi.$$

The derivative with respect to $\partial_\mu \phi$ is

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi.$$

The Euler–Lagrange equation is therefore

$$-m^2\phi - \partial_\mu \partial^\mu \phi = 0.$$

Multiplying by -1, we obtain

$$(\Box + m^2)\phi = 0.$$

This is the Klein–Gordon equation.

Example

For $m = 0$, the equation becomes

$$\Box\phi = 0,$$

which is the relativistic wave equation. For $m \neq 0$, the field has a mass scale and the dispersion relation becomes $\omega^2 = |\mathbf{k}|^2 + m^2$.

If the Lagrangian density is instead

$$\mathcal{L} = \frac{1}{2}\partial_\mu \phi \partial^\mu \phi - V(\phi),$$

then the field equation is

$$\Box\phi + \frac{dV}{d\phi} = 0.$$

This form will be useful for interacting scalar field models.

4.9 Worked Example: Wave Equation from an Action

The ordinary wave equation in one spatial dimension can be obtained from the action

$$S[\phi] = \int dt\, dx\, \frac{1}{2}\left[(\partial_t \phi)^2 - c^2(\partial_x \phi)^2\right].$$

Here the Lagrangian density is

$$\mathcal{L} = \frac{1}{2}(\partial_t \phi)^2 - \frac{c^2}{2}(\partial_x \phi)^2.$$

There is no explicit dependence on ϕ, so

$$\frac{\partial \mathcal{L}}{\partial \phi} = 0.$$

The derivatives with respect to the field derivatives are

$$\frac{\partial \mathcal{L}}{\partial(\partial_t \phi)} = \partial_t \phi, \qquad \frac{\partial \mathcal{L}}{\partial(\partial_x \phi)} = -c^2 \partial_x \phi.$$

The Euler–Lagrange equation gives

$$\partial_t(\partial_t \phi) + \partial_x(-c^2 \partial_x \phi) = 0.$$

Thus

$$\partial_t^2 \phi - c^2 \partial_x^2 \phi = 0.$$

This is the wave equation.

This example shows the physical meaning of the two terms in the Lagrangian density. The term $(\partial_t \phi)^2/2$ is associated with kinetic energy, while $c^2(\partial_x \phi)^2/2$ is associated with the elastic cost of spatial variation. The sign difference between them is what gives a hyperbolic evolution equation rather than an elliptic equation.

4.10 Physical Interpretation of Field Equations

The Euler–Lagrange equations are not just formal consequences of calculus of variations. They express local balance laws for distributed degrees of freedom. In a mechanical system, the equation of motion balances inertia against forces. In a field theory, the equation balances time evolution, spatial gradients, sources, and self-interactions.

For the scalar field equation

$$\Box\phi + \frac{dV}{d\phi} = 0,$$

the operator $\Box$ describes propagation through spacetime, while $dV/d\phi$ describes the local restoring force or self-interaction coming from the potential. For the wave equation,

$$\partial_t^2\phi - c^2\partial_x^2\phi = 0,$$

the acceleration of the field is determined by the curvature of its spatial profile.

The action viewpoint also explains why field equations are so tightly connected to symmetries. Once the dynamics is encoded in an action, transformations of the fields and coordinates can be studied by asking whether they leave the action invariant. This is the starting point of Noether's theorem, which is the subject of the next chapter.

Summary

The action principle turns field dynamics into variational calculus. A local Lagrangian density defines an action; varying the action gives a bulk term and a boundary term; the vanishing of the bulk term gives the Euler–Lagrange field equations. Boundary conditions are part of the variational problem, and total derivatives must be interpreted relative to those boundary conditions.

Chapter 5

Symmetries and Noether's Theorems

Remark

Symmetry is not only a visual property of a solution. In field theory, symmetry is a structural property of the action. Noether's theorems explain why this matters: continuous global symmetries lead to conserved currents, while local symmetries reveal redundancies and identities among the field equations.

5.1 Continuous Symmetries in Field Theory

A symmetry is a transformation that preserves the physical content of a theory. In the action formulation, the most useful definition is that a transformation is a symmetry if it leaves the action invariant, possibly up to a boundary term. For fields $\phi^A(x)$, a continuous transformation can be written as

$$\phi^A(x) \longrightarrow \phi'^A(x) = \phi^A(x) + \varepsilon\Delta\phi^A(x) + O(\varepsilon^2),$$

where ε is a small parameter and $\Delta\phi^A$ is the infinitesimal change of the field.

The transformation is continuous because it is connected to the identity: when $\varepsilon = 0$, nothing changes. This is the situation covered by Noether's first theorem. Discrete symmetries, such as parity or charge conjugation, are important, but they do not directly produce conserved currents in the same way.

For a local Lagrangian density

$$\mathcal{L} = \mathcal{L}(\phi^A, \partial_\mu\phi^A, x),$$

the action is

$$S[\phi] = \int_\Omega \mathcal{L}\, d^{d+1}x.$$

A continuous symmetry is present if the transformation changes the Lagrangian density by at most a total derivative:

$$\delta\mathcal{L} = \partial_\mu K^\mu.$$

Then the action changes only by a boundary contribution,

$$\delta S = \int_{\partial\Omega} K^\mu n_\mu\, d\Sigma.$$

Under suitable boundary conditions, this is zero.

Definition

A *continuous variational symmetry* is an infinitesimal transformation of fields and possibly coordinates for which the Lagrangian density changes by a total derivative:

$$\delta\mathcal{L} = \partial_\mu K^\mu.$$

The phrase "up to a total derivative" is essential. Many physically important symmetries leave the action invariant without leaving the Lagrangian density strictly unchanged. The boundary term may be irrelevant for local equations of motion, but it can be important for charges and boundary physics.

5.2 Internal and Spacetime Symmetries

Symmetries in field theory come in two broad types. Internal symmetries act on the field values without moving spacetime points. Spacetime symmetries act on the coordinates and therefore also on the arguments of the fields.

An internal symmetry has the schematic form

$$\phi^A(x) \longrightarrow \phi'^A(x) = R^A{}_B \phi^B(x),$$

where $R^A{}_B$ acts on field components. For example, a complex scalar field may have a global $U(1)$ symmetry

$$\phi(x) \longrightarrow e^{i\alpha}\phi(x).$$

The spacetime point x is not changed. Only the internal phase of the field is changed.

A spacetime symmetry changes the coordinates:

$$x^\mu \longrightarrow x'^\mu = x^\mu + \varepsilon\xi^\mu(x).$$

Translations, rotations, Lorentz boosts, and scale transformations are examples. For a scalar field, the transformed field satisfies

$$\phi'(x') = \phi(x).$$

If one compares fields at the same coordinate value x, the infinitesimal change contains the derivative of the field:

$$\delta\phi(x) = -\varepsilon\xi^\mu\partial_\mu\phi(x).$$

Example

A spacetime translation by a constant vector a^μ sends

$$x^\mu \longrightarrow x'^\mu = x^\mu + a^\mu.$$

For a scalar field, this corresponds to the infinitesimal field variation

$$\delta\phi = -a^\mu\partial_\mu\phi.$$

The associated conserved quantities are energy and momentum.

Internal and spacetime symmetries may coexist. The Standard Model, for example, uses spacetime Poincare symmetry together with internal gauge symmetries. Even in purely classical field theory, the distinction is crucial: internal symmetries usually lead to charges such as electric charge, while spacetime symmetries lead to energy, momentum, and angular momentum.

5.3 Noether's First Theorem

Noether's first theorem states that every continuous global variational symmetry of the action gives a conserved current. We now derive the basic formula for fields ϕ^A with a first-order Lagrangian density

$$\mathcal{L} = \mathcal{L}(\phi^A, \partial_\mu\phi^A, x).$$

The variation of the Lagrangian density is

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi^A}\delta\phi^A + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^A)}\partial_\mu\delta\phi^A.$$

Integrating the second term by parts at the density level gives

$$\delta\mathcal{L} = \left[\frac{\partial\mathcal{L}}{\partial\phi^A} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^A)}\right)\right]\delta\phi^A \\ + \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^A)}\delta\phi^A\right).$$

The expression in square brackets is the Euler–Lagrange derivative. Denote it by

$$E_A(\mathcal{L}) = \frac{\partial\mathcal{L}}{\partial\phi^A} - \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^A)}\right).$$

Then

$$\delta\mathcal{L} = E_A(\mathcal{L})\delta\phi^A + \partial_\mu\Theta^\mu(\phi, \delta\phi),$$

where

$$\Theta^\mu(\phi, \delta\phi) = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^A)}\delta\phi^A$$

is the symplectic-potential current density for this simple first-order theory.

If the transformation is a symmetry, then

$$\delta\mathcal{L} = \partial_\mu K^\mu.$$

Therefore

$$E_A(\mathcal{L})\delta\phi^A + \partial_\mu\Theta^\mu = \partial_\mu K^\mu.$$

Rearranging gives

$$\partial_\mu J^\mu = -E_A(\mathcal{L})\delta\phi^A,$$

where

$$J^\mu = \Theta^\mu - K^\mu.$$

On shell, $E_A(\mathcal{L}) = 0$, so

$$\partial_\mu J^\mu = 0.$$

Theorem

Noether's first theorem. For every continuous global variational symmetry of the action, there exists a current J^μ whose divergence vanishes on shell:

$$\partial_\mu J^\mu = 0.$$

The associated charge is conserved under suitable boundary conditions.

This theorem is one of the central structural results of theoretical physics. It turns symmetry into conservation. Time translation symmetry gives energy conservation. Space translation symmetry gives momentum conservation. Rotational symmetry gives angular momentum conservation. Internal phase symmetry gives charge conservation.

5.4 Conserved Currents and Conserved Charges

A conserved current is a vector field J^μ satisfying

$$\partial_\mu J^\mu = 0.$$

Writing $J^\mu = (\rho, \mathbf{J})$, this equation becomes

$$\partial_t \rho + \nabla \cdot \mathbf{J} = 0.$$

This is a local conservation law. It says that the density ρ changes in a region only because the current $\mathbf{J}$ flows through the boundary of that region.

The associated charge on a constant-time slice Σ_t is

$$Q(t) = \int_{\Sigma_t} J^0(t, \mathbf{x})\, d^d x.$$

Its time derivative is

$$\frac{dQ}{dt} = \int_{\Sigma_t} \partial_t J^0\, d^d x = -\int_{\Sigma_t} \partial_i J^i\, d^d x.$$

Using the divergence theorem,

$$\frac{dQ}{dt} = -\int_{\partial \Sigma_t} J^i n_i\, d\Sigma.$$

Thus the charge is conserved if the flux through the boundary vanishes:

$$\frac{dQ}{dt} = 0.$$

Definition

A *conserved charge* is the spatial integral of the time component of a conserved current:

$$Q = \int_{\Sigma_t} J^0\, d^d x.$$

It is time-independent if the current has no net flux through the boundary of space.

The distinction between local and global conservation is important. The equation $\partial_\mu J^\mu = 0$ is local. It holds point by point. The conservation of Q is global and depends on boundary conditions. A locally conserved current may still produce a time-dependent charge in a finite region if current flows across the boundary.

5.5 Energy, Momentum, and Angular Momentum

Spacetime translations lead to energy and momentum. If the Lagrangian density has no explicit dependence on spacetime coordinates, then it is invariant under translations

$$x^\mu \longrightarrow x^\mu + a^\mu.$$

The corresponding conserved object is the stress-energy tensor $T^\mu{}_\nu$, satisfying

$$\partial_\mu T^\mu{}_\nu = 0$$

on shell.

The conserved four-momentum is

$$P_\nu = \int_{\Sigma_t} T^0{}_\nu\, d^d x.$$

The component associated with time translations is the energy. The components associated with spatial translations are the momenta.

Rotations and Lorentz boosts lead to angular momentum and boost charges. The corresponding current has the schematic form

$$M^{\mu\rho\sigma} = x^\rho T^{\mu\sigma} - x^\sigma T^{\mu\rho} + S^{\mu\rho\sigma},$$

where $S^{\mu\rho\sigma}$ is a possible spin contribution. For scalar fields, the spin term is absent. For vector and spinor fields, it is generally present.

The conservation law is

$$\partial_\mu M^{\mu\rho\sigma} = 0.$$

The associated charges are

$$J^{\rho\sigma} = \int_{\Sigma_t} M^{0\rho\sigma}\, d^d x.$$

Spatial components describe ordinary angular momentum; mixed time-space components describe Lorentz boost generators.

Remark

Energy, momentum, and angular momentum are not separate miracles. They are the Noether charges of spacetime symmetries: time translations, space translations, rotations, and Lorentz boosts.

This perspective is more powerful than memorizing conserved quantities one by one. Once the action and the symmetry are known, Noether's theorem tells us what kind of conservation law to expect.

5.6 Stress-Energy Tensor: Canonical Form

For a collection of fields ϕ^A with Lagrangian density $\mathcal{L}(\phi^A, \partial_\mu \phi^A)$ and no explicit coordinate dependence, the canonical stress-energy tensor is

$$T^\mu{}_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \partial_\nu \phi^A - \delta^\mu{}_\nu \mathcal{L}.$$

With the sign conventions used here, this tensor satisfies

$$\partial_\mu T^\mu{}_\nu = 0$$

on shell.

Let us check the structure. The first term is built from the momentum conjugate to $\partial_\mu \phi^A$ and the derivative $\partial_\nu \phi^A$. The second term subtracts the Lagrangian density when $\mu = \nu$. This is the direct field theory analogue of the mechanical relation between energy, momenta, velocities, and the Lagrangian.

For a real scalar field

$$\mathcal{L} = \frac{1}{2}\partial_\rho \phi \partial^\rho \phi - V(\phi),$$

one has

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi.$$

Therefore

$$T^\mu{}_\nu = \partial^\mu \phi\, \partial_\nu \phi - \delta^\mu{}_\nu \mathcal{L}.$$

With both indices raised, this is commonly written as

$$T^{\mu\nu} = \partial^\mu \phi\, \partial^\nu \phi - \eta^{\mu\nu} \mathcal{L}.$$

Example

For the real scalar field, the energy density is

$$T^{00} = \frac{1}{2}(\partial_t \phi)^2 + \frac{1}{2}|\nabla\phi|^2 + V(\phi).$$

This is positive for potentials bounded below, as expected for a stable scalar field theory.

The canonical stress-energy tensor is not always symmetric, and it is not always gauge invariant. This is not a contradiction. It means that the canonical tensor is not always the final or most physical stress-energy tensor.

5.7 Improvement Terms

A conserved current is not unique. If J^μ is conserved, then

$$J'^\mu = J^\mu + \partial_\nu B^{[\mu\nu]}$$

is also conserved, because

$$\partial_\mu \partial_\nu B^{[\mu\nu]} = 0$$

for antisymmetric $B^{[\mu\nu]} = -B^{[\nu\mu]}$. The added term is called an improvement term.

The same phenomenon occurs for the stress-energy tensor. One may add a total divergence that does not change the local conservation law but changes the local expression for the tensor:

$$T'^{\mu\nu} = T^{\mu\nu} + \partial_\rho B^{\rho\mu\nu},$$

with suitable antisymmetry in the indices of B. Under appropriate boundary conditions, the integrated charges remain the same.

Improvement terms are not merely cosmetic. They can make the stress-energy tensor symmetric, gauge invariant, or better adapted to coupling with gravity. For example, the Belinfante–Rosenfeld procedure combines the canonical stress-energy tensor with spin-current terms to produce a symmetric tensor.

Remark

Noether currents are defined up to improvement terms. Therefore one should not attach too much physical meaning to a particular local representative of a current without specifying boundary conditions and the class of allowed improvements.

This ambiguity is a preview of a deeper issue: conserved charges are often more robust than local current representatives. In gauge theory and gravity, the boundary part of a current can become the most important part of the charge.

5.8 Noether's Second Theorem and Gauge Symmetries

Noether's first theorem concerns global symmetries depending on finitely many constant parameters. Noether's second theorem concerns local symmetries depending on arbitrary functions. Gauge symmetries are the main example.

A simple gauge transformation in electromagnetism is

$$A_\mu \longrightarrow A_\mu + \partial_\mu \lambda,$$

where $\lambda(x)$ is an arbitrary function. The field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

is unchanged:

$$\delta F_{\mu\nu} = 0.$$

Therefore the Maxwell action is gauge invariant.

The key point is that $\lambda(x)$ is arbitrary. A gauge symmetry is not an ordinary physical transformation from one distinct state to another. It is a redundancy in the description. Two gauge-related configurations represent the same physical field configuration.

Noether's second theorem says that local gauge symmetries imply identities among the Euler–Lagrange equations. Schematically, if

$$\delta_\lambda \phi^A = R^A(\lambda, \partial\lambda, \dots)$$

is a gauge symmetry with arbitrary $\lambda(x)$, then the field equations cannot be independent. There must be a differential identity among them.

In electromagnetism, the field equation is

$$\partial_\mu F^{\mu\nu} = J^\nu.$$

Taking ∂_ν of both sides gives

$$\partial_\nu \partial_\mu F^{\mu\nu} = \partial_\nu J^\nu.$$

The left-hand side vanishes identically because $F^{\mu\nu}$ is antisymmetric and partial derivatives commute. Thus

$$\partial_\nu J^\nu = 0.$$

This expresses charge conservation as a consistency condition of the gauge field equation.

Theorem

Noether's second theorem. Local gauge symmetries imply differential identities among the Euler–Lagrange equations. These identities reflect the redundancy of the description and are not ordinary conservation laws of the same type as those produced by global symmetries.

This distinction is essential. A global symmetry usually produces a physical charge. A gauge symmetry usually signals that some variables are redundant and that constraints are present.

5.9 Global vs Local Symmetries

A global symmetry depends on constant parameters. For example,

$$\phi \longrightarrow e^{i\alpha}\phi$$

with constant α is a global $U(1)$ symmetry. Noether's first theorem then gives a conserved current and a conserved charge.

A local symmetry depends on arbitrary functions. For example,

$$\phi(x) \longrightarrow e^{i\alpha(x)}\phi(x)$$

with arbitrary $\alpha(x)$ is a local phase transformation. To make such a symmetry compatible with derivatives, one introduces a gauge field and replaces ordinary derivatives by covariant derivatives:

$$\partial_\mu \phi \longrightarrow D_\mu \phi.$$

The relation between global and local symmetries is subtle. A global symmetry is often associated with a conserved charge. A local symmetry is usually a redundancy. However, boundary conditions can convert part of a gauge symmetry into a genuine physical transformation at the boundary. This is why gauge charges and asymptotic symmetries play such an important role in modern field theory and gravity.

Definition

A *global symmetry* has parameters that are constant over spacetime. A *local symmetry* has parameters that are arbitrary functions of spacetime. Global symmetries usually generate conserved charges; local symmetries usually express redundancies and constraints.

The safest rule is this: global symmetries act on physical states, while gauge symmetries identify different descriptions of the same physical state. Boundary conditions and charges can complicate this rule, but it is the correct starting point.

5.10 Physical Meaning of Conservation Laws

A conservation law is not merely the statement that a number does not change. The local equation

$$\partial_\mu J^\mu = 0$$

expresses a balance between density and flux. In nonrelativistic notation,

$$\partial_t \rho + \nabla \cdot \mathbf{J} = 0.$$

This says that charge cannot disappear at a point; it can only flow from one place to another.

Energy conservation means that energy can move, transform between forms, or flow through boundaries, but it is not locally created or destroyed if the relevant symmetry assumptions hold. Momentum conservation expresses spatial homogeneity. Angular momentum conservation expresses rotational symmetry. Electric charge conservation expresses global phase symmetry.

The assumptions matter. If a Lagrangian has explicit time dependence, energy need not be conserved. If space translation symmetry is broken by an external background, momentum need not be conserved. If charge flows through the boundary of the region under study, the charge inside that region need not be constant.

Remark

Noether's theorem does not say that every quantity is conserved. It says that a specific conservation law follows from a specific continuous symmetry of the action, under specified boundary conditions.

This is why the action formulation is so economical. Instead of deriving every conservation law separately from the equations of motion, one studies the symmetries of the action and lets Noether's theorem organize the results.

5.11 Examples and Common Misinterpretations

Complex scalar field and phase symmetry

For a complex scalar field with Lagrangian density

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi,$$

there is a global $U(1)$ symmetry

$$\phi \longrightarrow e^{i\alpha}\phi, \qquad \phi^* \longrightarrow e^{-i\alpha}\phi^*.$$

The infinitesimal variations are

$$\delta\phi = i\alpha\phi, \qquad \delta\phi^* = -i\alpha\phi^*.$$

The associated Noether current is, up to the conventional normalization,

$$J^\mu = i(\phi\partial^\mu\phi^* - \phi^*\partial^\mu\phi).$$

On shell,

$$\partial_\mu J^\mu = 0.$$

This is the classical origin of charge conservation for a charged scalar field.

Misinterpretation: every symmetry gives a new particle

Noether's theorem says that a continuous global symmetry gives a conserved current. It does not say that every symmetry directly gives a new particle. The relationship between symmetries and particles belongs to quantum field theory and depends on representation theory, spontaneous symmetry breaking, and the spectrum of the theory.

Misinterpretation: gauge symmetry is an ordinary physical motion

Gauge symmetry is primarily a redundancy. Gauge-related configurations describe the same physical state. This is why gauge fixing is allowed and why gauge symmetries lead to constraints. Nevertheless, transformations that do not vanish at a boundary can generate physical charges. This is a boundary-sensitive refinement, not a contradiction.

Misinterpretation: total derivatives never matter

Total derivatives do not change the bulk Euler–Lagrange equations under suitable assumptions. But they can change boundary terms, canonical structures, and charges. Therefore total derivatives are harmless only after the boundary problem has been specified.

Summary

Noether's first theorem turns continuous global symmetries into conserved currents and charges. Spacetime translations give energy and momentum; rotations and Lorentz transformations give angular momentum and boost charges; internal symmetries give internal charges. Noether's second theorem explains why local gauge symmetries imply identities and constraints rather than ordinary physical motions. Conservation laws are local balance equations plus boundary conditions.

Chapter 6

Hamiltonian Formulation of Field Theory

Remark

The Hamiltonian formulation rewrites field theory as dynamics on phase space. It is less manifestly relativistic than the Lagrangian formulation, because it chooses a time coordinate, but it reveals the canonical variables, Poisson brackets, constraints, gauge freedom, and the role of energy as the generator of time evolution.

6.1 Canonical Momenta for Fields

The Lagrangian formulation treats fields and their spacetime derivatives in a nearly symmetric way. The Hamiltonian formulation begins by separating time from space. We write spacetime coordinates as

$$x^\mu = (t, \mathbf{x}),$$

and denote the time derivative of a field by

$$\dot{\phi}^A(t, \mathbf{x}) = \partial_t \phi^A(t, \mathbf{x}).$$

For a first-order Lagrangian density

$$\mathcal{L} = \mathcal{L}(\phi^A, \dot{\phi}^A, \partial_i \phi^A, t, \mathbf{x}),$$

the canonical momentum conjugate to ϕ^A is defined by

$$\pi_A(t, \mathbf{x}) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^A(t, \mathbf{x})}.$$

Thus each field component ϕ^A has a conjugate momentum density π_A.

Definition

The *canonical momentum* conjugate to a field ϕ^A is

$$\pi_A = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^A}.$$

It is a density on space and is evaluated at the same time and spatial point as ϕ^A.

For a real scalar field with

$$\mathcal{L} = \frac{1}{2}(\partial_t \phi)^2 - \frac{1}{2}|\nabla \phi|^2 - V(\phi),$$

the canonical momentum is simply

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}.$$

This example is regular: the velocity $\dot{\phi}$ can be recovered from the momentum π.

The situation is not always so simple. If the definition of the momenta cannot be inverted to express all velocities in terms of fields and momenta, the theory has constraints. Gauge theories and generally covariant theories almost always have this feature. The Hamiltonian formalism is therefore not merely an alternative notation; it is the natural language for understanding constraints and gauge redundancy.

6.2 Legendre Transform and Hamiltonian Density

In mechanics, the Hamiltonian is obtained from the Lagrangian by a Legendre transform:

$$H(q,p,t) = p_i \dot{q}^i - L(q,\dot{q},t),$$

where the velocities are expressed in terms of (q,p). In field theory, the same idea is applied pointwise in space. The Hamiltonian density is

$$\mathcal{H} = \pi_A \dot{\phi}^A - \mathcal{L}.$$

After the velocities have been expressed in terms of ϕ^A, π_A, and spatial derivatives of ϕ^A, the Hamiltonian is

$$H = \int d^d x\, \mathcal{H}.$$

The Legendre transform is valid in its simplest form only if the Hessian matrix with respect to the velocities is nondegenerate:

$$W_{AB} = \frac{\partial^2 \mathcal{L}}{\partial \dot{\phi}^A \partial \dot{\phi}^B}.$$

If W_{AB} is invertible, the momenta determine the velocities locally. If it is not invertible, the theory is singular and constraints appear.

Definition

A field theory is *regular* if the Legendre transform from velocities $\dot{\phi}^A$ to momenta π_A is locally invertible. It is *singular* if this transform is degenerate. Singular theories require constraints in the Hamiltonian formulation.

For the real scalar field,

$$\pi = \dot{\phi}.$$

Therefore

$$\begin{aligned}\mathcal{H} &= \pi\dot{\phi} - \mathcal{L} \\ &= \pi^2 - \left[\frac{1}{2}\pi^2 - \frac{1}{2}|\nabla\phi|^2 - V(\phi)\right].\end{aligned}$$

Thus

$$\mathcal{H} = \frac{1}{2}\pi^2 + \frac{1}{2}|\nabla\phi|^2 + V(\phi).$$

This is the energy density of the scalar field.

Example

For a free massive scalar field,

$$V(\phi) = \frac{1}{2}m^2\phi^2,$$

the Hamiltonian is

$$H = \int d^d x \left[\frac{1}{2}\pi^2 + \frac{1}{2}|\nabla\phi|^2 + \frac{1}{2}m^2\phi^2\right].$$

This Hamiltonian is positive for real fields and suitable boundary conditions.

6.3 Equal-Time Poisson Brackets

The Hamiltonian formulation requires a notion of Poisson bracket. For fields, the canonical Poisson brackets are imposed at equal time. The fundamental brackets are

$$\{\phi^A(t,\mathbf{x}),\phi^B(t,\mathbf{y})\}=0,$$

$$\{\pi_A(t,\mathbf{x}),\pi_B(t,\mathbf{y})\}=0,$$

and

$$\{\phi^A(t,\mathbf{x}),\pi_B(t,\mathbf{y})\}=\delta^A{}_B\delta^{(d)}(\mathbf{x}-\mathbf{y}).$$

The delta distribution appears because there is one canonical degree of freedom at each spatial point.

More generally, for two functionals $F[\phi,\pi]$ and $G[\phi,\pi]$, the field-theoretic Poisson bracket is

$$\{F,G\}=\int d^dx\left(\frac{\delta F}{\delta\phi^A(\mathbf{x})}\frac{\delta G}{\delta\pi_A(\mathbf{x})}-\frac{\delta F}{\delta\pi_A(\mathbf{x})}\frac{\delta G}{\delta\phi^A(\mathbf{x})}\right).$$

This is the infinite-dimensional analogue of the mechanical formula

$$\{F,G\}=\frac{\partial F}{\partial q^i}\frac{\partial G}{\partial p_i}-\frac{\partial F}{\partial p_i}\frac{\partial G}{\partial q^i}.$$

Definition

The *equal-time Poisson bracket* is the canonical Poisson bracket on the space of fields $\phi^A(t,\mathbf{x})$ and conjugate momenta $\pi_A(t,\mathbf{x})$ at a fixed time t.

Equal-time brackets are not manifestly Lorentz covariant, because one has chosen a time coordinate and a family of spatial hypersurfaces. This is the price paid for a canonical description. The reward is that time evolution, constraints, and canonical quantization become transparent.

6.4 Hamilton's Equations for Fields

Once the Hamiltonian is known, time evolution is generated by Poisson brackets. For any functional $F[\phi,\pi,t]$, one has

$$\frac{dF}{dt}=\{F,H\}+\frac{\partial F}{\partial t}.$$

Applying this to the basic fields gives Hamilton's equations:

$$\dot{\phi}^A(t,\mathbf{x})=\frac{\delta H}{\delta\pi_A(t,\mathbf{x})},$$

and

$$\dot{\pi}_A(t,\mathbf{x})=-\frac{\delta H}{\delta\phi^A(t,\mathbf{x})}.$$

These equations are equivalent to the Euler–Lagrange equations when the Legendre transform is regular.

For the real scalar field,

$$H=\int d^dx\left[\frac{1}{2}\pi^2+\frac{1}{2}|\nabla\phi|^2+V(\phi)\right].$$

The first Hamilton equation gives

$$\dot{\phi} = \frac{\delta H}{\delta \pi} = \pi.$$

The second gives

$$\dot{\pi} = -\frac{\delta H}{\delta \phi}.$$

To compute the functional derivative, vary the Hamiltonian:

$$\delta H = \int d^d x \left[\pi \delta \pi + \nabla \phi \cdot \nabla \delta \phi + V'(\phi) \delta \phi\right].$$

After integrating the gradient term by parts and assuming the boundary term vanishes,

$$\delta H = \int d^d x \left[\pi \delta \pi + (-\nabla^2 \phi + V'(\phi)) \delta \phi\right].$$

Therefore

$$\frac{\delta H}{\delta \phi} = -\nabla^2 \phi + V'(\phi),$$

and

$$\dot{\pi} = \nabla^2 \phi - V'(\phi).$$

Using $\pi = \dot{\phi}$, we obtain

$$\ddot{\phi} - \nabla^2 \phi + V'(\phi) = 0,$$

which is the Euler–Lagrange equation for the scalar field.

Example

For $V(\phi) = m^2 \phi^2 / 2$, Hamilton's equations imply

$$\ddot{\phi} - \nabla^2 \phi + m^2 \phi = 0,$$

that is,

$$(\Box + m^2) \phi = 0.$$

6.5 Phase Space of a Field Theory

The configuration space of a field theory is a space of field configurations $\phi^A(\mathbf{x})$. The phase space is the space of pairs

$$(\phi^A(\mathbf{x}), \pi_A(\mathbf{x})).$$

Thus a point of phase space specifies both the field profile and its canonical momentum profile at one time.

This phase space is infinite-dimensional. In mechanics, a system with n generalized coordinates has a $2n$-dimensional phase space with coordinates (q^i, p_i). In field theory, the spatial point $\mathbf{x}$ plays the role of a continuous label, so phase space consists of functions rather than finite tuples of numbers.

The canonical symplectic structure is encoded in the equal-time Poisson brackets. Equivalently, one may write the canonical symplectic form formally as

$$\Omega = \int d^d x \, \delta \pi_A(\mathbf{x}) \wedge \delta \phi^A(\mathbf{x}).$$

Here δ denotes the exterior derivative on the infinite-dimensional space of fields, not a spacetime variation. This notation is more advanced but useful: it emphasizes that Hamiltonian field theory is symplectic geometry on an infinite-dimensional space.

Remark

A field-theoretic phase space is not the same thing as spacetime. Spacetime is the domain on which fields are defined. Phase space is the space of all possible instantaneous field and momentum configurations.

In a well-posed Hamiltonian evolution problem, initial data define a point in phase space, and Hamilton's equations define a trajectory through phase space. This is the canonical version of the initial-value problem for fields.

6.6 Constraints: First Encounter

A constraint is a relation among canonical variables that must hold on the allowed phase space. Constraints arise when the canonical momenta are not all independent. This happens when the Legendre transform is degenerate.

A simple diagnostic is the Hessian matrix

$$W_{AB} = \frac{\partial^2 \mathcal{L}}{\partial \dot{\phi}^A \partial \dot{\phi}^B}.$$

If W_{AB} has zero modes, then not all velocities can be solved in terms of momenta. The momentum definitions imply relations

$$\chi_m(\phi, \pi) \approx 0.$$

The symbol $\approx$ is Dirac's weak equality. It means that the equation holds on the constraint surface, but should not necessarily be imposed before Poisson brackets are computed.

Definition

A *constraint* is a relation $\chi(\phi, \pi) \approx 0$ that restricts the allowed canonical data. The set of points satisfying all constraints is called the constraint surface.

Electromagnetism gives the standard example. The Maxwell Lagrangian density is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}.$$

The field component A_0 has no time derivative in $F_{\mu\nu}$ in the same way as the spatial components. Its canonical momentum is

$$\pi^0 = \frac{\partial \mathcal{L}}{\partial \dot{A}_0} = 0.$$

This is a constraint:

$$\pi^0 \approx 0.$$

It reflects the fact that A_0 is not an ordinary propagating degree of freedom. It is tied to Gauss's law and gauge freedom.

6.7 Primary and Secondary Constraints

Constraints that arise directly from the definition of the canonical momenta are called primary constraints. If

$$\chi_m(\phi, \pi) \approx 0$$

is a primary constraint, it must remain true during time evolution. Therefore one imposes the consistency condition

$$\dot{\chi}_m \approx 0.$$

Using the Hamiltonian, this becomes

$$\{\chi_m, H\} \approx 0,$$

or, more generally, a condition involving the total Hamiltonian.

The total Hamiltonian includes the canonical Hamiltonian plus primary constraints multiplied by arbitrary functions called Lagrange multipliers:

$$H_T = H_c + \int d^d x \, u^m(\mathbf{x})\chi_m(\mathbf{x}).$$

The functions u^m are not determined by the Legendre transform. Their presence signals that the evolution may not be unique until gauge freedom is fixed.

The consistency condition may lead to three possibilities:

1. it may be automatically satisfied;
2. it may determine some Lagrange multipliers;
3. it may produce new constraints, called secondary constraints.

In electromagnetism, the primary constraint $\pi^0 \approx 0$ leads, through consistency, to Gauss's law as a secondary constraint:

$$\partial_i \pi^i - J^0 \approx 0,$$

up to sign conventions. This is the Hamiltonian form of the electric charge constraint.

Remark

The distinction between primary and secondary constraints is procedural, not physical. Primary constraints arise from the momentum definitions; secondary constraints arise from demanding preservation of constraints under time evolution.

The systematic treatment of constraints is known as Dirac's constraint algorithm. This course only needs the first conceptual layer: constraints are not optional extra equations. They are part of the canonical structure of singular Lagrangian systems.

6.8 Gauge Freedom and Degenerate Legendre Transform

Gauge freedom and degeneracy of the Legendre transform are closely related. If a Lagrangian has a local gauge symmetry, then some apparent field components do not represent independent physical degrees of freedom. This redundancy appears in the Hamiltonian formulation as constraints.

The intuitive reason is simple. Gauge-related configurations describe the same physical state. Therefore the Lagrangian cannot assign independent physical velocities to all field components. Some directions in velocity space correspond not to physical motion but to motion along gauge orbits. Along such directions, the Legendre transform is degenerate.

Constraints are often classified as first class or second class. A constraint is first class if its Poisson bracket with all constraints vanishes on the constraint surface. Schematically,

$$\{\chi_m, \chi_n\} \approx 0.$$

First-class constraints generate gauge transformations. Second-class constraints do not generate gauge freedom; instead, they remove variables in a different way and require replacing Poisson brackets by Dirac brackets.

Definition

A *first-class constraint* has weakly vanishing Poisson brackets with all constraints. First-class constraints are associated with gauge transformations. A *second-class constraint* is a constraint that fails this condition.

For gauge theories, the true physical phase space is not the full constraint surface. One must also identify points related by gauge transformations. In words,

$$\text{physical phase space} = \frac{\text{constraint surface}}{\text{gauge transformations}}.$$

This quotient viewpoint is essential in advanced field theory, gauge theory, and gravity.

6.9 Energy as a Generator of Time Evolution

In Hamiltonian mechanics, the Hamiltonian generates time evolution. The same is true in field theory. If $F[\phi, \pi, t]$ is a phase-space functional, then

$$\frac{dF}{dt} = \{F, H\} + \frac{\partial F}{\partial t}$$

for regular theories. Thus the Hamiltonian is not merely the numerical energy; it is the generator of time translations on phase space.

For the basic fields,

$$\dot{\phi}^A = \{\phi^A, H\}, \qquad \dot{\pi}_A = \{\pi_A, H\}.$$

These are Hamilton's equations in bracket form. They show that the Hamiltonian vector field associated with H gives the dynamical flow on phase space.

If the Hamiltonian has no explicit time dependence and the boundary conditions allow the relevant integrations by parts, then

$$\frac{dH}{dt} = \{H, H\} = 0.$$

Thus energy is conserved. This is the Hamiltonian expression of time-translation symmetry.

Example

For the free scalar field, the Hamiltonian

$$H = \int d^d x \left[\frac{1}{2}\pi^2 + \frac{1}{2}|\nabla\phi|^2 + \frac{1}{2}m^2\phi^2\right]$$

generates the Klein–Gordon evolution through the equal-time Poisson brackets.

In constrained systems, the generator of time evolution is the total Hamiltonian H_T, not merely the canonical Hamiltonian. The arbitrary multipliers in H_T encode gauge freedom. Therefore Hamiltonian evolution in a gauge theory is unique only up to gauge transformations.

6.10 Comparison with the Lagrangian Formulation

The Lagrangian and Hamiltonian formulations describe the same classical dynamics when the Legendre transform is regular and boundary conditions are handled properly. However, they emphasize different structures.

The Lagrangian formulation is manifestly spacetime-oriented. It is usually the best language for relativistic covariance, action principles, Noether's theorems, path integrals, and coupling to geometry. A typical Lagrangian action has the compact form

$$S[\phi] = \int \mathcal{L}(\phi^A, \partial_\mu \phi^A, x)\, d^{d+1}x.$$

The Euler–Lagrange equations follow from varying S.

The Hamiltonian formulation is phase-space-oriented. It requires a choice of time coordinate and describes evolution of canonical data

$$(\phi^A(\mathbf{x}), \pi_A(\mathbf{x})).$$

Its central objects are the Hamiltonian, Poisson brackets, constraints, and canonical transformations. It is the natural starting point for canonical quantization, constraint analysis, and the study of generators.

Feature	Lagrangian formulation	Hamiltonian formulation
Basic variables	$\phi^A, \partial_\mu \phi^A$	ϕ^A, π_A
Central object	action S	Hamiltonian H
Dynamics	Euler–Lagrange equations	Hamilton's equations
Geometry	spacetime/covariant	phase space/canonical
Symmetries	Noether currents	generators via Poisson brackets
Gauge theories	variational identities	constraints and gauge orbits

Neither formulation is intrinsically superior. The Lagrangian formulation makes covariance and the action principle transparent. The Hamiltonian formulation makes the phase-space structure and constraints transparent. A mature use of classical field theory requires both.

Summary

The Hamiltonian formulation begins by choosing a time coordinate, defining canonical momenta, and performing a Legendre transform. For regular theories it leads to a Hamiltonian density, equal-time Poisson brackets, and Hamilton's equations. For singular theories it reveals constraints, gauge freedom, and the need for Dirac's constraint analysis. The Hamiltonian is both energy and the generator of time evolution, provided the boundary conditions make the generator well defined.

Part III

Fundamental Classical Fields

Chapter 7

The Real Scalar Field

Remark

The real scalar field is the simplest relativistic field theory with local degrees of freedom. It is simple enough to solve explicitly in the free case, but rich enough to introduce the essential structures of classical field theory: a Lorentz-invariant action, a wave equation, a dispersion relation, conserved energy-momentum, Green's functions, sources, interactions, stability, and small fluctuations around backgrounds.

7.1 Klein–Gordon Field

A real scalar field is a real-valued function on spacetime,

$$\phi : M \to \mathbb{R}.$$

In flat Minkowski spacetime, with coordinates $x^\mu = (t, \mathbf{x})$, the standard free massive scalar-field action is

$$S[\phi] = \int d^{d+1}x\, \mathcal{L},$$

where

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2.$$

With the mostly-minus convention,

$$\partial_\mu\phi\partial^\mu\phi = (\partial_t\phi)^2 - |\nabla\phi|^2.$$

Thus the Lagrangian density can also be written as

$$\mathcal{L} = \frac{1}{2}(\partial_t\phi)^2 - \frac{1}{2}|\nabla\phi|^2 - \frac{1}{2}m^2\phi^2.$$

The first term is the kinetic term, the second is the spatial-gradient term, and the third is the mass term.

Varying the action gives

$$\delta S = \int d^{d+1}x \left(\partial_\mu\phi\partial^\mu\delta\phi - m^2\phi\delta\phi\right).$$

Integrating the first term by parts and assuming that the boundary term vanishes, we obtain

$$\delta S = -\int d^{d+1}x \left(\partial_\mu\partial^\mu\phi + m^2\phi\right)\delta\phi.$$

The stationarity condition $\delta S = 0$ for arbitrary $\delta\phi$ gives

$$(\Box + m^2)\phi = 0,$$

where

$$\Box = \partial_\mu\partial^\mu = \partial_t^2 - \nabla^2.$$

This is the Klein–Gordon equation.

Definition

The *Klein–Gordon field* is a real scalar field satisfying

$$(\Box + m^2)\phi = 0.$$

It is the relativistic field equation associated with the invariant mass parameter m.

The equation is linear. Therefore, if ϕ_1 and ϕ_2 are solutions, then any linear combination

$$a\phi_1 + b\phi_2$$

is also a solution. This superposition property will fail once nonlinear interactions are added.

7.2 Massless and Massive Scalar Fields

The parameter m controls the relation between frequency and wavelength, the range of static response, and the interpretation of excitations. When $m = 0$, the equation becomes

$$\Box\phi = 0,$$

the relativistic wave equation. Disturbances propagate at the speed of light in flat spacetime.

When $m \neq 0$, the equation is

$$\partial_t^2\phi - \nabla^2\phi + m^2\phi = 0.$$

The mass term acts as a local restoring term: even a spatially homogeneous field oscillates,

$$\ddot{\phi} + m^2\phi = 0.$$

Thus a massive scalar field combines wave propagation with local oscillation.

The mass parameter also defines a length scale,

$$\ell_m = \frac{1}{m}$$

in units $\hbar = c = 1$. In classical static problems, this length scale controls the decay of the response to localized sources. The larger the mass, the shorter the range of the field.

Remark

The word "mass" has a relativistic meaning: the plane-wave excitations of the field satisfy the invariant relation $p_\mu p^\mu = m^2$. Classically, the same parameter also controls oscillation frequency and static screening length.

The massless case has additional infrared subtleties, especially in low spatial dimensions or in the presence of long-range sources. The massive case is often technically cleaner because the mass term suppresses long-distance response.

7.3 Plane-Wave Solutions

Because the free Klein–Gordon equation has constant coefficients, it is natural to look for plane-wave solutions. Let

$$\phi(x) = Ae^{-ik_\mu x^\mu},$$

where

$$k_\mu x^\mu = \omega t - \mathbf{k} \cdot \mathbf{x}$$

with our conventions. Acting with the d'Alembert operator gives

$$\Box e^{-ik_\mu x^\mu} = -k_\mu k^\mu e^{-ik_\mu x^\mu}.$$

The Klein–Gordon equation therefore requires

$$-k_\mu k^\mu + m^2 = 0,$$

or

$$k_\mu k^\mu = m^2.$$

Since

$$k_\mu k^\mu = \omega^2 - |\mathbf{k}|^2,$$

we get

$$\omega^2 = |\mathbf{k}|^2 + m^2.$$

A real scalar field cannot be a single complex exponential unless one takes the real part. A general real solution can be written as a superposition of modes, for example

$$\phi(t, \mathbf{x}) = \int \frac{d^d k}{(2\pi)^d} \left[a(\mathbf{k}) e^{-i\omega_{\mathbf{k}} t + i\mathbf{k}\cdot\mathbf{x}} + a^*(\mathbf{k}) e^{i\omega_{\mathbf{k}} t - i\mathbf{k}\cdot\mathbf{x}} \right],$$

where

$$\omega_{\mathbf{k}} = \sqrt{|\mathbf{k}|^2 + m^2}.$$

The complex conjugate term ensures that ϕ is real.

Example

In one spatial dimension, a real standing-wave solution of the massive equation is

$$\phi(t, x) = A\cos(kx)\cos(\omega t), \qquad \omega^2 = k^2 + m^2.$$

For $m = 0$, this reduces to the familiar wave relation $\omega = |k|$.

Plane waves are not localized. They are idealized basis solutions. Localized wave packets are constructed by superposing many plane waves with different wave vectors.

7.4 Dispersion Relation

The dispersion relation of the free massive scalar field is

$$\omega^2 = |\mathbf{k}|^2 + m^2.$$

This relation determines how different Fourier modes oscillate. It is the field theory version of the relativistic energy-momentum relation

$$E^2 = |\mathbf{p}|^2 + m^2.$$

The analogy becomes exact in quantum field theory, where field modes are quantized and interpreted in terms of particles. In the present classical context, it is already useful: it tells us how frequency depends on spatial wave number.

The phase velocity is

$$v_{\text{ph}} = \frac{\omega}{|\mathbf{k}|} = \sqrt{1 + \frac{m^2}{|\mathbf{k}|^2}}.$$

For $m \neq 0$, this is larger than one. This does not violate relativity, because phase velocity is not the signal velocity. The group velocity is

$$v_{\text{g}} = \frac{d\omega}{d|\mathbf{k}|} = \frac{|\mathbf{k}|}{\sqrt{|\mathbf{k}|^2 + m^2}}.$$

For massive modes,

$$v_{\text{g}} < 1.$$

In the high-momentum limit $|\mathbf{k}| \gg m$, the group velocity approaches one. In the low-momentum limit $|\mathbf{k}| \ll m$, the group velocity is small.

Remark

The massive Klein–Gordon field is dispersive: different wavelengths propagate with different group velocities. The massless field in flat spacetime has $\omega = |\mathbf{k}|$, so wave packets propagate without this mass-induced dispersion.

Dispersion is a crucial qualitative feature. A localized massive wave packet spreads because its Fourier components travel with different velocities. This is already visible at the classical level, before quantization.

7.5 Energy and Momentum of the Scalar Field

The canonical stress-energy tensor for the scalar field is

$$T^{\mu\nu} = \partial^\mu\phi\partial^\nu\phi - \eta^{\mu\nu}\mathcal{L}.$$

For the Lagrangian density

$$\mathcal{L} = \frac{1}{2}\partial_\rho\phi\partial^\rho\phi - V(\phi),$$

the energy density is

$$T^{00} = \frac{1}{2}(\partial_t\phi)^2 + \frac{1}{2}|\nabla\phi|^2 + V(\phi).$$

For the free massive field,

$$V(\phi) = \frac{1}{2}m^2\phi^2,$$

so

$$T^{00} = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}|\nabla\phi|^2 + \frac{1}{2}m^2\phi^2.$$

This is positive definite for real ϕ and $m^2 > 0$.

The total energy is

$$E = \int d^dx\, T^{00}.$$

The momentum density is

$$T^{0i} = \partial^0\phi\partial^i\phi.$$

Since $\partial^0 = \partial_t$ and $\partial^i = -\partial_i$, signs depend on index placement. A commonly used spatial momentum is

$$P^i = \int d^dx\, \dot{\phi}\, \partial_i\phi$$

up to the convention for raising or lowering the spatial index. The important structural statement is that spacetime translation invariance gives a conserved stress-energy tensor:

$$\partial_\mu T^{\mu\nu} = 0.$$

Definition

For a scalar field with potential $V(\phi)$, the energy density is

$$\mathcal{E} = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}|\nabla\phi|^2 + V(\phi).$$

Finite-energy configurations must usually approach minima of V sufficiently rapidly at spatial infinity.

The condition of finite energy is physically important. If space is noncompact, one cannot allow arbitrary field behavior at infinity. For finite total energy, $\nabla\phi$ and $V(\phi)$ must be integrable over space. This often forces the field to approach a vacuum value at infinity.

7.6 Green's Functions for the Klein–Gordon Operator

The Klein–Gordon operator is

$$\Box + m^2.$$

A Green's function for this operator is a distribution $G(x-y)$ satisfying

$$(\Box_x + m^2)G(x-y) = \delta^{(d+1)}(x-y).$$

Green's functions allow one to solve inhomogeneous equations of the form

$$(\Box + m^2)\phi(x) = J(x),$$

where $J(x)$ is a source.

Formally, a solution is

$$\phi(x) = \int d^{d+1}y\, G(x-y)J(y),$$

provided the Green's function and boundary conditions are chosen appropriately. There is not just one Green's function. The operator admits different inverses corresponding to different causal or boundary prescriptions.

The retarded Green's function G_{ret} satisfies

$$G_{\text{ret}}(x-y) = 0 \quad \text{unless } x \text{ lies in or on the future light cone of } y.$$

It describes causal response: the field at x depends only on sources in its past. The advanced Green's function has support in the opposite direction and is usually not used for causal initial-value problems.

Definition

A *retarded Green's function* is a Green's function whose support enforces causal propagation from source to response. It gives the solution determined by past sources, not future ones.

In Fourier representation, one formally writes

$$G(x) = \int \frac{d^{d+1}k}{(2\pi)^{d+1}} \frac{e^{-ik_\mu x^\mu}}{-k_\mu k^\mu + m^2}$$

with a prescription for how to treat the poles. Different pole prescriptions produce different Green's functions. This issue becomes central in quantum field theory, where the Feynman propagator appears.

7.7 Sources and Response

A scalar field can be coupled linearly to an external source $J(x)$ by adding a source term to the Lagrangian density:

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 + J\phi.$$

Varying the action gives

$$(\Box + m^2)\phi = J.$$

The sign of the source term is a convention; what matters is consistency between the Lagrangian and the resulting field equation.

For a static source, the equation becomes

$$(-\nabla^2 + m^2)\phi(\mathbf{x}) = J(\mathbf{x}).$$

For a point source in three spatial dimensions,

$$J(\mathbf{x}) = q\delta^{(3)}(\mathbf{x}),$$

the solution is proportional to the Yukawa potential,

$$\phi(r) \propto \frac{e^{-mr}}{r}.$$

This shows explicitly that the massive scalar field mediates a short-range response with characteristic length $1/m$. In the massless limit, the response becomes Coulomb-like:

$$\phi(r) \propto \frac{1}{r}.$$

Example

The massive scalar Green's function in the static three-dimensional problem shows the physical meaning of mass: mass suppresses long-distance response. The field sourced at one point decays exponentially rather than purely as $1/r$.

The source formalism is useful even when the source is not physical. It lets us probe the response of a field theory and define Green's functions operationally: the Green's function is the response to a point source.

7.8 Interacting Scalar Fields

The free scalar field is linear. Interactions are introduced by replacing the quadratic potential with a more general potential:

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi).$$

The field equation is

$$\Box\phi + \frac{dV}{d\phi} = 0.$$

If V is not purely quadratic, the equation is nonlinear.

Nonlinearity changes the structure of the theory dramatically. Superposition no longer holds: if ϕ_1 and ϕ_2 are solutions, their sum is generally not a solution. Interactions allow energy transfer between modes, nonlinear wave phenomena, solitons in some models, spontaneous symmetry breaking in suitable potentials, and nontrivial perturbation theory.

A common polynomial potential is

$$V(\phi) = \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4}\phi^4.$$

The equation of motion is then

$$\Box\phi + m^2\phi + \lambda\phi^3 = 0.$$

This is the classical ϕ^4 equation with a single-well potential when $m^2 > 0$ and $\lambda > 0$.

Remark

The free theory is exactly solvable because Fourier modes evolve independently. Interactions couple modes to each other. This mode coupling is the classical ancestor of interaction vertices in quantum field theory.

Interactions must be chosen with care. A formally Lorentz-invariant equation may still define an unstable theory if its potential is unbounded below.

7.9 ϕ^4 Theory as a Classical Model

The name ϕ^4 theory refers to scalar field models with a quartic self-interaction. A standard Lagrangian density is

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \left(\frac{1}{2}m^2\phi^2 + \frac{\lambda}{4}\phi^4\right).$$

The field equation is

$$\Box\phi + m^2\phi + \lambda\phi^3 = 0.$$

For $m^2 > 0$ and $\lambda > 0$, the potential has a single minimum at $\phi = 0$. Small oscillations around $\phi = 0$ behave approximately like a free massive scalar field.

Another important form is the double-well potential

$$V(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2, \qquad \lambda > 0.$$

This potential has two minima,

$$\phi = \pm v.$$

The equation of motion is

$$\Box\phi + \lambda\phi(\phi^2 - v^2) = 0.$$

This model has degenerate vacua and supports nontrivial classical structures in one spatial dimension, such as kink solutions.

Example

For the double-well potential, finite-energy configurations in one spatial dimension may approach different vacua at opposite infinities:

$$\phi(x \to -\infty) = -v, \qquad \phi(x \to +\infty) = v.$$

Such configurations cannot be continuously deformed to the trivial vacuum while preserving finite energy and boundary behavior.

Even when this course does not solve kink equations in detail, the lesson is important: nonlinear scalar fields can have qualitatively different sectors of solutions distinguished by boundary conditions.

7.10 Stability of the Potential

The energy density of a scalar field with potential V is

$$\mathcal{E} = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}|\nabla\phi|^2 + V(\phi).$$

For the theory to have a stable ground state, the potential should be bounded below. If $V(\phi) \to -\infty$ as $|\phi| \to \infty$, the energy can decrease without bound, and the classical theory has no stable vacuum.

For example,

$$V(\phi) = \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4}\phi^4$$

is bounded below when $\lambda > 0$. If $\lambda < 0$, it is unbounded below at large $|\phi|$, even if $m^2 > 0$. Thus the sign of the highest-power term is crucial.

A vacuum is a constant field configuration ϕ_0 minimizing the potential:

$$V'(\phi_0) = 0, \qquad V''(\phi_0) > 0.$$

The value of $V(\phi_0)$ is the vacuum energy density in this classical model. Often one shifts the potential by a constant so that the minimum has zero energy:

$$V(\phi_0) = 0.$$

This shift does not affect the scalar equation of motion in flat spacetime, because the equation depends on V', not on the absolute value of V.

Definition

A *classical vacuum* of a scalar field model is a constant field value ϕ_0 at which the potential is minimized. Small disturbances around a stable vacuum have positive squared mass $V''(\phi_0) > 0$.

In gravity, absolute vacuum energy matters because it contributes to the stress-energy tensor. In nongravitational flat-spacetime field theory, only differences in potential energy usually affect the classical scalar equation.

7.11 Small Fluctuations Around a Background Solution

Let $\bar{\phi}(x)$ be a background solution of the interacting scalar equation

$$\Box\bar{\phi} + V'(\bar{\phi}) = 0.$$

Now consider a small fluctuation $\eta(x)$ around this background:

$$\phi(x) = \bar{\phi}(x) + \eta(x),$$

where η is assumed small. Substituting into the field equation gives

$$\Box(\bar{\phi} + \eta) + V'(\bar{\phi} + \eta) = 0.$$

Expanding the potential derivative to first order,

$$V'(\bar{\phi} + \eta) = V'(\bar{\phi}) + V''(\bar{\phi})\eta + O(\eta^2).$$

Using the background equation, the linearized equation becomes

$$\left[\Box + V''(\bar{\phi}(x))\right]\eta = 0.$$

Thus small fluctuations see an effective spacetime-dependent mass squared

$$m_{\text{eff}}^2(x) = V''(\bar{\phi}(x)).$$

For a constant vacuum $\bar{\phi} = \phi_0$, this reduces to

$$(\Box + m_{\text{eff}}^2)\eta = 0, \qquad m_{\text{eff}}^2 = V''(\phi_0).$$

This formula is central: the mass of small excitations around a vacuum is the curvature of the potential at that vacuum.

Example

For

$$V(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2,$$

one has

$$V''(\phi) = \lambda(3\phi^2 - v^2).$$

At the vacua $\phi_0 = \pm v$,

$$m_{\text{eff}}^2 = V''(\pm v) = 2\lambda v^2.$$

Thus fluctuations around either vacuum are massive.

Linearization is not just an approximation technique. It is how one reads the local particle-like content of a classical field model, identifies stability, and studies propagation on nontrivial backgrounds.

Summary

The real scalar field is the simplest relativistic field model. The free theory is governed by the Klein–Gordon equation, has plane-wave solutions with $\omega^2 = |\mathbf{k}|^2 + m^2$, and carries positive energy when the potential is bounded below. Sources are handled using Green's functions, while interactions come from nonlinear potentials. Around a vacuum ϕ_0, the mass of small fluctuations is determined by the curvature of the potential: $m_{\text{eff}}^2 = V''(\phi_0)$.

Chapter 8

Complex Scalar Field and Internal Symmetry

Remark

A complex scalar field is the simplest field theory with a nontrivial internal symmetry. It can be understood as two real scalar fields, but its complex form makes a global $U(1)$ phase symmetry manifest. Noether's theorem then gives a conserved current and charge. By promoting the phase symmetry from global to local, one is naturally led to the gauge covariant derivative and to coupling with electromagnetism.

8.1 Complex Field as Two Real Fields

A complex scalar field is a function

$$\phi : M \to \mathbb{C}.$$

It may be decomposed into two real fields,

$$\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2), \qquad \phi^* = \frac{1}{\sqrt{2}}(\phi_1 - i\phi_2),$$

where ϕ_1 and ϕ_2 are real scalar fields. The factor $1/\sqrt{2}$ is conventional and gives convenient normalizations.

The free complex scalar field is described by the Lagrangian density

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi.$$

In terms of ϕ_1 and ϕ_2, this becomes

$$\mathcal{L} = \frac{1}{2}\partial_\mu \phi_1 \partial^\mu \phi_1 + \frac{1}{2}\partial_\mu \phi_2 \partial^\mu \phi_2 - \frac{1}{2}m^2\phi_1^2 - \frac{1}{2}m^2\phi_2^2.$$

Thus a free complex scalar field is equivalent to two real scalar fields with the same mass.

Definition

A *complex scalar field* is a scalar field with complex values. As a real field theory it contains two real scalar degrees of freedom, which may be taken as the real and imaginary parts of the field.

The complex notation is not merely aesthetic. It makes the internal rotation between ϕ_1 and ϕ_2 transparent. This internal rotation is the origin of the conserved $U(1)$ charge.

The Euler–Lagrange equations may be obtained by treating ϕ and ϕ^* as independent variables during the variation. Varying with respect to ϕ^* gives

$$(\Box + m^2)\phi = 0.$$

Varying with respect to ϕ gives the complex conjugate equation

$$(\Box + m^2)\phi^* = 0.$$

Equivalently, ϕ_1 and ϕ_2 each satisfy the real Klein–Gordon equation.

8.2 Global $U(1)$ Symmetry

The free complex scalar Lagrangian is invariant under constant phase rotations

$$\phi \longrightarrow e^{i\alpha}\phi, \qquad \phi^* \longrightarrow e^{-i\alpha}\phi^*,$$

where α is a real constant. This is a global $U(1)$ symmetry.

For infinitesimal α,

$$e^{i\alpha} = 1 + i\alpha + O(\alpha^2),$$

so the field variations are

$$\delta\phi = i\alpha\phi, \qquad \delta\phi^* = -i\alpha\phi^*.$$

The derivatives transform in the same way:

$$\delta(\partial_\mu\phi) = i\alpha\partial_\mu\phi, \qquad \delta(\partial_\mu\phi^*) = -i\alpha\partial_\mu\phi^*.$$

Substituting these variations into

$$\mathcal{L} = \partial_\mu\phi^*\partial^\mu\phi - m^2\phi^*\phi$$

shows that

$$\delta\mathcal{L} = 0.$$

Thus the symmetry is exact and leaves the Lagrangian density itself invariant, not merely up to a total derivative.

Definition

The group $U(1)$ is the group of complex phases $e^{i\alpha}$. A global $U(1)$ symmetry acts by multiplying the field by a constant phase at every spacetime point.

In terms of the two real fields, this symmetry is an internal rotation:

$$\begin{pmatrix}\phi_1\\ \phi_2\end{pmatrix} \longrightarrow \begin{pmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{pmatrix}\begin{pmatrix}\phi_1\\ \phi_2\end{pmatrix}.$$

It is called internal because it rotates field components into one another without rotating physical space.

8.3 Conserved Current and Charge

Noether's theorem associates a conserved current with the global $U(1)$ symmetry. Since

$$\mathcal{L} = \partial_\mu\phi^*\partial^\mu\phi - m^2\phi^*\phi,$$

we have

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = \partial^\mu\phi^*, \qquad \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)} = \partial^\mu\phi.$$

Using the infinitesimal variations without the common parameter α,

$$\Delta\phi = i\phi, \qquad \Delta\phi^* = -i\phi^*,$$

the Noether current is

$$J^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\Delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)}\Delta\phi^*.$$

Therefore

$$J^\mu = i\phi\partial^\mu\phi^* - i\phi^*\partial^\mu\phi.$$

Equivalently,

$$J^\mu = i(\phi\partial^\mu\phi^* - \phi^*\partial^\mu\phi).$$

Some authors use the opposite sign convention. The convention is not important as long as the definition of charge is used consistently.

On shell,

$$\partial_\mu J^\mu = 0.$$

The conserved charge is

$$Q = \int d^dx\, J^0.$$

Using $\partial^0 = \partial_t$, one has

$$J^0 = i(\phi\dot{\phi}^* - \phi^*\dot{\phi}).$$

Thus

$$Q = i\int d^dx\,(\phi\dot{\phi}^* - \phi^*\dot{\phi}).$$

Theorem

The global phase symmetry of the complex scalar field gives the conserved Noether current

$$J^\mu = i(\phi\partial^\mu\phi^* - \phi^*\partial^\mu\phi), \qquad \partial_\mu J^\mu = 0$$

on solutions of the field equations.

This current is the classical precursor of charge current for a charged scalar field. In a quantum theory, the corresponding charge counts particles and antiparticles with opposite signs. In the classical theory, it measures the oriented internal rotation carried by the field configuration.

8.4 Coupling to an External Electromagnetic Potential

A charged scalar field can be coupled to a prescribed electromagnetic potential $A_\mu(x)$. At first, we treat A_μ as an external background, not as a dynamical field. The guiding principle is that ordinary derivatives should be replaced by covariant derivatives:

$$\partial_\mu\phi \longrightarrow D_\mu\phi.$$

For a field of charge q, a common convention is

$$D_\mu\phi = (\partial_\mu + iqA_\mu)\phi.$$

The complex conjugate field transforms with the opposite charge, so

$$D_\mu\phi^* = (\partial_\mu - iqA_\mu)\phi^*.$$

The Lagrangian density becomes

$$\mathcal{L} = (D_\mu\phi)^* D^\mu\phi - m^2\phi^*\phi.$$

Here

$$(D_\mu\phi)^* = D_\mu\phi^*$$

with the above convention. Expanding the covariant derivative shows that the background field couples to the scalar current and also introduces a term quadratic in A_μ.

Remark

The electromagnetic potential does not couple to a real scalar field in the same minimal way, because a real scalar field has no continuous phase that can carry $U(1)$ charge. Charged scalar matter is naturally complex.

If A_μ is fixed externally, the scalar field evolves in a prescribed electromagnetic background. If A_μ is dynamical, one must add the Maxwell term to the action and vary with respect to both ϕ and A_μ.

8.5 Local Phase Symmetry

The global phase symmetry uses a constant parameter α. A local phase transformation allows the parameter to depend on spacetime:

$$\phi(x) \longrightarrow \phi'(x) = e^{i\alpha(x)}\phi(x).$$

If one uses ordinary derivatives, the derivative transforms as

$$\partial_\mu \phi'(x) = e^{i\alpha(x)}\partial_\mu \phi(x) + i(\partial_\mu \alpha)e^{i\alpha(x)}\phi(x).$$

The second term prevents $\partial_\mu \phi$ from transforming in the same way as ϕ. Therefore the free Lagrangian with ordinary derivatives is not invariant under local phase transformations.

The remedy is to introduce a gauge field A_μ whose transformation cancels the extra derivative of α. With the convention

$$D_\mu = \partial_\mu + iqA_\mu,$$

one takes

$$A_\mu \longrightarrow A'_\mu = A_\mu - \frac{1}{q}\partial_\mu \alpha.$$

Then

$$D'_\mu \phi' = e^{i\alpha} D_\mu \phi.$$

This is the defining covariance property of the gauge covariant derivative.

Definition

A *local phase symmetry* is a transformation with spacetime-dependent phase $\alpha(x)$. It requires a gauge field because ordinary derivatives of the field do not transform covariantly under local phase rotations.

This is the conceptual origin of gauge theory: demanding local internal symmetry forces the introduction of a compensating connection-like field.

8.6 Minimal Coupling

Minimal coupling is the prescription

$$\partial_\mu \longrightarrow D_\mu = \partial_\mu + iqA_\mu.$$

Applied to the free complex scalar field, it gives

$$\mathcal{L}_{\text{matter}} = (D_\mu \phi)^* D^\mu \phi - m^2 \phi^* \phi.$$

If the electromagnetic field is dynamical, the full scalar electrodynamics Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\phi)^* D^\mu\phi - m^2\phi^*\phi,$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

This theory is invariant under

$$\phi \longrightarrow e^{i\alpha(x)}\phi, \qquad A_\mu \longrightarrow A_\mu - \frac{1}{q}\partial_\mu\alpha.$$

Example

The replacement $\partial_\mu \to D_\mu$ is not an arbitrary trick. It is forced by the requirement that derivatives of charged fields transform covariantly under local phase rotations.

The word minimal means that one introduces the gauge field through the derivative and does not add extra higher-order or nonminimal terms such as $F_{\mu\nu}F^{\mu\nu}\phi^*\phi$. Such terms can be considered in effective field theories, but they are not part of the minimal coupling prescription.

8.7 Gauge Covariant Derivative

The gauge covariant derivative is designed to transform like the field itself:

$$D'_\mu\phi' = e^{i\alpha}D_\mu\phi.$$

This property ensures that

$$(D_\mu\phi)^* D^\mu\phi$$

is invariant under local $U(1)$ transformations. The mass term

$$m^2\phi^*\phi$$

is also invariant, because the phase cancels between ϕ and ϕ^*.

The commutator of two covariant derivatives gives the field strength. With

$$D_\mu = \partial_\mu + iqA_\mu,$$

one finds

$$[D_\mu, D_\nu]\phi = iqF_{\mu\nu}\phi.$$

Thus the electromagnetic field strength measures the noncommutativity of gauge covariant derivatives.

Definition

For a charged scalar field, the gauge covariant derivative is

$$D_\mu\phi = (\partial_\mu + iqA_\mu)\phi.$$

Its commutator satisfies

$$[D_\mu, D_\nu]\phi = iqF_{\mu\nu}\phi.$$

This formula is the Abelian prototype of Yang–Mills theory. In non-Abelian gauge theory, the gauge potential and the field strength take values in a Lie algebra, and the commutator term becomes structurally essential.

8.8 Physical Meaning of Charge

In the complex scalar theory, charge is the Noether charge associated with internal phase rotations. The field has a two-dimensional internal plane spanned by ϕ_1 and ϕ_2. A charged configuration carries a sense of rotation in that internal plane.

Using

$$\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2),$$

the charge density can be written in terms of real fields as

$$J^0 = \phi_2\dot{\phi}_1 - \phi_1\dot{\phi}_2,$$

up to the overall sign convention chosen for J^μ. This is analogous to angular momentum in the internal (ϕ_1, ϕ_2)-plane.

Remark

A complex scalar field is not "more mysterious" than two real scalar fields. What is new is the internal rotational symmetry between them. The conserved charge is the Noether charge of that internal rotation.

When the field is coupled to electromagnetism, this internal charge becomes the source of the electromagnetic field. In scalar electrodynamics, varying the action with respect to A_μ gives Maxwell equations with a matter current constructed from ϕ and $D_\mu\phi$.

8.9 Charged Scalar Field as a Classical Field Theory

The charged scalar field provides the simplest classical model of matter coupled to a gauge field. Its dynamical variables are

$$\phi, \qquad \phi^*, \qquad A_\mu.$$

The action is

$$S = \int d^{d+1}x \left[-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\phi)^* D^\mu\phi - m^2\phi^*\phi\right].$$

The scalar field equation is

$$D_\mu D^\mu\phi + m^2\phi = 0,$$

with the complex conjugate equation for ϕ^*. The gauge field equation has the form

$$\partial_\mu F^{\mu\nu} = j^\nu,$$

where the matter current is obtained by varying the matter action with respect to A_ν.

This theory illustrates several themes that will recur in the next chapters:

1. internal symmetry produces charge;
2. local symmetry requires a gauge field;
3. the gauge potential is not itself gauge invariant;
4. the field strength is gauge invariant in the Abelian theory;
5. the covariant derivative is the correct derivative for charged fields.

Summary

The complex scalar field is equivalent to two real scalar fields but displays a natural global $U(1)$ symmetry. Noether's theorem gives a conserved current and charge. Promoting the phase symmetry to a local symmetry forces the introduction of a gauge field and the covariant derivative $D_\mu = \partial_\mu + iqA_\mu$. This is the simplest route from scalar field theory to electromagnetism and, ultimately, to Yang–Mills gauge theory.

Chapter 9

Electromagnetic Field

Remark

Electromagnetism is the prototype of gauge field theory. It contains all the central ideas that later reappear in Yang–Mills theory: potentials, gauge freedom, field strength, an action principle, coupling to conserved matter currents, constraints, propagating transverse degrees of freedom, local energy and momentum, waves, and radiation.

9.1 Maxwell Equations in Vector Form

In units where the electric and magnetic constants are absorbed into the definitions of fields and sources, Maxwell's equations are

$$\nabla \cdot \mathbf{E} = \rho,$$

$$\nabla \cdot \mathbf{B} = 0,$$

$$\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0,$$

and

$$\nabla \times \mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}.$$

Here ρ is the charge density and $\mathbf{J}$ is the current density. The first and fourth equations contain sources. The second and third are homogeneous equations and express the absence of magnetic monopoles and the Faraday induction law.

The equations imply charge conservation. Taking the divergence of

$$\nabla \times \mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}$$

gives

$$\nabla \cdot \mathbf{J} = \nabla \cdot (\nabla \times \mathbf{B}) - \partial_t (\nabla \cdot \mathbf{E}).$$

Since $\nabla \cdot (\nabla \times \mathbf{B}) = 0$ and $\nabla \cdot \mathbf{E} = \rho$, one obtains

$$\partial_t \rho + \nabla \cdot \mathbf{J} = 0.$$

Thus Maxwell's equations are consistent only with locally conserved charge.

Definition

The electromagnetic field is described by an electric field $\mathbf{E}$ and a magnetic field $\mathbf{B}$ satisfying Maxwell's equations. In relativistic field theory these fields are not independent objects but components of a single antisymmetric spacetime tensor.

The vector form is physically transparent and useful for calculations in a chosen inertial frame. Its limitation is that Lorentz covariance is hidden. The covariant formulation reveals that electric and magnetic fields are frame-dependent parts of a single electromagnetic field strength.

9.2 Electromagnetic Potentials

The homogeneous Maxwell equations can be solved by introducing potentials. The equation

$$\nabla \cdot \mathbf{B} = 0$$

implies, at least locally in a topologically simple region, that

$$\mathbf{B} = \nabla \times \mathbf{A},$$

where $\mathbf{A}$ is the vector potential. Substituting this into Faraday's law

$$\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$$

gives

$$\nabla \times (\mathbf{E} + \partial_t \mathbf{A}) = 0.$$

Therefore there exists a scalar potential Φ such that

$$\mathbf{E} + \partial_t \mathbf{A} = -\nabla \Phi.$$

Thus

$$\mathbf{E} = -\nabla \Phi - \partial_t \mathbf{A}, \qquad \mathbf{B} = \nabla \times \mathbf{A}.$$

The potentials are naturally combined into the four-potential

$$A^\mu = (\Phi, \mathbf{A}).$$

With the mostly-minus metric convention,

$$A_\mu = (\Phi, -\mathbf{A}).$$

The electromagnetic field strength tensor is then defined by

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

This definition automatically incorporates the homogeneous Maxwell equations.

Remark

The potentials are not merely calculational conveniences. In the action formulation, the potential A_μ, not $\mathbf{E}$ and $\mathbf{B}$ separately, is the fundamental field variable.

The price of using potentials is redundancy: many different potentials describe the same electric and magnetic fields. This redundancy is gauge freedom.

9.3 Gauge Freedom

The electric and magnetic fields are unchanged under the transformation

$$\mathbf{A} \longrightarrow \mathbf{A}' = \mathbf{A} + \nabla \chi,$$

$$\Phi \longrightarrow \Phi' = \Phi - \partial_t \chi,$$

where $\chi(t, \mathbf{x})$ is an arbitrary smooth function. Indeed,

$$\mathbf{B}' = \nabla \times \mathbf{A}' = \nabla \times \mathbf{A} + \nabla \times \nabla \chi = \mathbf{B},$$

and

$$
\begin{aligned}
\mathbf{E}' &= -\nabla\Phi' - \partial_t \mathbf{A}' \\
&= -\nabla(\Phi - \partial_t\chi) - \partial_t(\mathbf{A} + \nabla\chi) \\
&= -\nabla\Phi - \partial_t\mathbf{A} = \mathbf{E}.
\end{aligned}
$$

In covariant notation, this is

$$A_\mu \longrightarrow A'_\mu = A_\mu + \partial_\mu\chi.$$

Then

$$F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu = F_{\mu\nu}.$$

Definition

A *gauge transformation* of the electromagnetic potential is

$$A_\mu \longrightarrow A_\mu + \partial_\mu\chi.$$

It changes the potential but leaves the field strength $F_{\mu\nu}$ unchanged.

Gauge freedom means that the potential contains more components than physical degrees of freedom. This is not a defect. It is the organizing principle of the theory. The redundant description makes locality and Lorentz covariance compatible with the existence of massless spin-one radiation.

9.4 Maxwell Theory from an Action

The free electromagnetic action is

$$S[A] = \int d^{d+1}x\, \mathcal{L}_{\text{EM}},$$

with

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}.$$

In four spacetime dimensions this is the standard Lorentz-invariant Maxwell Lagrangian density. The factor $-1/4$ gives the conventional normalization of the equations of motion.

Let us vary the action. Since

$$\delta F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu,$$

one finds, using antisymmetry of $F^{\mu\nu}$,

$$\delta\mathcal{L}_{\text{EM}} = -\frac{1}{2}F^{\mu\nu}\delta F_{\mu\nu} = -F^{\mu\nu}\partial_\mu\delta A_\nu.$$

After integration by parts,

$$\delta S = \int d^{d+1}x\, \partial_\mu F^{\mu\nu}\delta A_\nu + \text{boundary term}.$$

For variations that make the boundary term vanish, the field equation is

$$\partial_\mu F^{\mu\nu} = 0.$$

This is Maxwell's equation in vacuum.

If an external conserved current J^μ is present, one adds the interaction term

$$\mathcal{L}_{\text{int}} = -J_\mu A^\mu.$$

The field equation becomes

$$\partial_\mu F^{\mu\nu} = J^\nu.$$

Theorem

The Maxwell action
$$S[A] = \int d^{d+1}x \left(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - J_\mu A^\mu\right)$$
gives the sourced Maxwell equation
$$\partial_\mu F^{\mu\nu} = J^\nu$$
provided the external current is conserved and boundary terms are controlled.

Gauge invariance of the source term requires current conservation. Under $A_\mu \to A_\mu + \partial_\mu \chi$,
$$\delta S_{\text{int}} = -\int d^{d+1}x\, J^\mu \partial_\mu \chi = \int d^{d+1}x\, \chi \partial_\mu J^\mu + \text{boundary term}.$$
Thus gauge invariance requires
$$\partial_\mu J^\mu = 0.$$

9.5 Field Strength Tensor $F_{\mu\nu}$

The electromagnetic field strength is the antisymmetric tensor
$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$
It has six independent components in four spacetime dimensions. These components are the three electric and three magnetic components.

With the conventions
$$A^\mu = (\Phi, \mathbf{A}), \qquad A_\mu = (\Phi, -\mathbf{A}),$$
one has
$$F_{0i} = E_i.$$
The purely spatial components encode the magnetic field. A common convention is
$$F_{ij} = -\epsilon_{ijk} B_k,$$
with signs depending on the precise placement of indices and the convention for ϵ_{ijk}. What is invariant is the tensorial object $F_{\mu\nu}$, not a particular matrix convention.

The two Lorentz scalars built from the electromagnetic field are
$$F_{\mu\nu}F^{\mu\nu} = 2(|\mathbf{B}|^2 - |\mathbf{E}|^2),$$
and
$$F_{\mu\nu}\widetilde{F}^{\mu\nu} = -4\mathbf{E}\cdot\mathbf{B},$$
where the dual field strength is
$$\widetilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}.$$
These invariants show that $\mathbf{E}$ and $\mathbf{B}$ are frame-dependent parts of one field, while certain combinations are Lorentz invariant.

Definition

The electromagnetic field strength tensor $F_{\mu\nu}$ is gauge invariant and antisymmetric. It is the curvature associated with the gauge potential A_μ.

In differential-form notation, the potential is a one-form

$$A = A_\mu dx^\mu,$$

and the field strength is the two-form

$$F = dA.$$

Gauge transformations are

$$A \longrightarrow A + d\chi,$$

and the identity $d^2 = 0$ gives

$$dF = 0.$$

9.6 Maxwell Equations in Covariant Form

The inhomogeneous Maxwell equations are

$$\partial_\mu F^{\mu\nu} = J^\nu.$$

The homogeneous Maxwell equations are encoded by the Bianchi identity

$$\partial_{[\lambda} F_{\mu\nu]} = 0.$$

Equivalently,

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0.$$

In differential-form notation this is simply

$$dF = 0.$$

The covariant equations are therefore

$$dF = 0, \qquad \partial_\mu F^{\mu\nu} = J^\nu.$$

In four-dimensional differential-form notation, the second equation can be written schematically as

$$d \star F = \star J,$$

where $\star$ is the Hodge dual.

Charge conservation follows immediately from the inhomogeneous equation:

$$\partial_\nu J^\nu = \partial_\nu \partial_\mu F^{\mu\nu} = 0,$$

because $F^{\mu\nu}$ is antisymmetric while $\partial_\nu \partial_\mu$ is symmetric.

Summary

Maxwell's four vector equations are compactly expressed as one Bianchi identity and one sourced field equation:

$$\partial_{[\lambda} F_{\mu\nu]} = 0, \qquad \partial_\mu F^{\mu\nu} = J^\nu.$$

The first follows from $F = dA$; the second follows from the Maxwell action.

9.7 Coupling to Charged Matter

Electromagnetism couples to charged matter through a conserved current. At the level of the action, the coupling is

$$S_{\text{int}} = -\int d^{d+1}x\, J_\mu A^\mu.$$

Gauge invariance requires

$$\partial_\mu J^\mu = 0.$$

This statement is not optional: a nonconserved external current would make the gauge coupling inconsistent.

For a charged complex scalar field, the current is not imposed externally but is constructed from the matter field. Using the covariant derivative

$$D_\mu \phi = (\partial_\mu + iqA_\mu)\phi,$$

the scalar electrodynamics Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\phi)^* D^\mu\phi - m^2\phi^*\phi.$$

Varying with respect to A_μ gives a Maxwell equation with a matter current. The current is covariantly built from ϕ, ϕ^*, and their gauge covariant derivatives.

Remark

The electromagnetic field does not couple to arbitrary matter variables. It couples to a conserved $U(1)$ current. This is why the previous chapter on the complex scalar field is the natural bridge into Maxwell theory.

When matter is dynamical, conservation of the current is a consequence of the matter field equations and gauge symmetry. When matter is external, conservation must be imposed as a consistency condition.

9.8 Lorenz Gauge and Coulomb Gauge

Gauge freedom allows one to impose convenient conditions on A_μ. The Lorenz gauge condition is

$$\partial_\mu A^\mu = 0.$$

It is Lorentz covariant and simplifies the Maxwell equation. Since

$$\partial_\mu F^{\mu\nu} = \partial_\mu(\partial^\mu A^\nu - \partial^\nu A^\mu) = \Box A^\nu - \partial^\nu(\partial_\mu A^\mu),$$

the Lorenz gauge reduces the sourced equation to

$$\Box A^\nu = J^\nu.$$

In vacuum,

$$\Box A^\nu = 0.$$

The Coulomb gauge condition is

$$\nabla \cdot \mathbf{A} = 0.$$

It is not manifestly Lorentz covariant, but it is useful for separating constraint-like and propagating parts of the electromagnetic field. In Coulomb gauge, the scalar potential is determined by the instantaneous Poisson equation

$$-\nabla^2\Phi = \rho,$$

while the transverse part of $\mathbf{A}$ contains the radiative degrees of freedom.

Definition

The *Lorenz gauge* is $\partial_\mu A^\mu = 0$. The *Coulomb gauge* is $\nabla \cdot \mathbf{A} = 0$. They are gauge choices, not physical restrictions on $\mathbf{E}$ and $\mathbf{B}$.

A gauge condition does not remove the physical electromagnetic field. It removes redundancy in the potential description. Residual gauge transformations may remain even after a gauge condition is imposed.

9.9 Electromagnetic Stress-Energy Tensor

The electromagnetic stress-energy tensor is

$$T^{\mu\nu} = -F^{\mu\rho}F^{\nu}{}_{\rho} + \frac{1}{4}\eta^{\mu\nu}F_{\rho\sigma}F^{\rho\sigma}.$$

With the mostly-minus convention, this tensor gives the standard positive energy density

$$T^{00} = \frac{1}{2}(|\mathbf{E}|^2 + |\mathbf{B}|^2).$$

It is symmetric, gauge invariant, and conserved in vacuum:

$$\partial_\mu T^{\mu\nu} = 0.$$

In the presence of matter, the electromagnetic stress-energy tensor alone is not conserved; energy and momentum are exchanged between field and matter.

The trace of the electromagnetic stress-energy tensor in four spacetime dimensions is zero:

$$T^{\mu}{}_{\mu} = 0.$$

This reflects the classical scale invariance of source-free Maxwell theory in four dimensions.

Definition

The electromagnetic stress-energy tensor is the local density and flux tensor of energy and momentum carried by the electromagnetic field. It is constructed from $F_{\mu\nu}$, not directly from the gauge potential A_μ, and is therefore gauge invariant.

The stress-energy tensor is the object that couples to gravity. In general relativity, electromagnetic fields curve spacetime through their stress-energy, not merely through their energy density.

9.10 Energy, Momentum, and Poynting Vector

The electromagnetic energy density is

$$u = \frac{1}{2}(|\mathbf{E}|^2 + |\mathbf{B}|^2).$$

The energy flux is the Poynting vector

$$\mathbf{S} = \mathbf{E} \times \mathbf{B}.$$

The local energy conservation law is Poynting's theorem:

$$\partial_t u + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}.$$

The right-hand side is the rate at which electromagnetic energy is transferred to charged matter.

In vacuum, $\mathbf{J} = 0$, so

$$\partial_t u + \nabla \cdot \mathbf{S} = 0.$$

Thus electromagnetic energy is locally conserved and flows with flux $\mathbf{S}$.

The electromagnetic momentum density is

$$\mathbf{g} = \mathbf{E} \times \mathbf{B}$$

in the same units. Thus the Poynting vector is also the momentum density in relativistic units. This is consistent with the fact that electromagnetic waves carry both energy and momentum.

Example

For a plane electromagnetic wave in vacuum, $\mathbf{E}$, $\mathbf{B}$, and the propagation direction are mutually orthogonal. The energy flux points in the direction of propagation:

$$\mathbf{S} = \mathbf{E} \times \mathbf{B}.$$

The conservation laws of electromagnetism are not separate from Maxwell's equations. They follow from the field equations and the stress-energy tensor, just as Noether's theorem predicts from spacetime translation invariance.

9.11 Electromagnetic Waves

In vacuum, Maxwell's equations imply wave equations for both $\mathbf{E}$ and $\mathbf{B}$. Taking the curl of Faraday's law gives

$$\nabla \times (\nabla \times \mathbf{E}) + \partial_t (\nabla \times \mathbf{B}) = 0.$$

Using $\nabla \cdot \mathbf{E} = 0$, the vector identity

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$$

and $\nabla \times \mathbf{B} = \partial_t \mathbf{E}$, one obtains

$$\partial_t^2 \mathbf{E} - \nabla^2 \mathbf{E} = 0.$$

Similarly,

$$\partial_t^2 \mathbf{B} - \nabla^2 \mathbf{B} = 0.$$

Thus electromagnetic disturbances propagate at the speed of light in these units.

A plane wave solution has the form

$$\mathbf{E}(t, \mathbf{x}) = \mathbf{E}_0 e^{-i\omega t + i\mathbf{k}\cdot\mathbf{x}},$$

$$\mathbf{B}(t, \mathbf{x}) = \mathbf{B}_0 e^{-i\omega t + i\mathbf{k}\cdot\mathbf{x}}.$$

Maxwell's equations imply

$$\omega^2 = |\mathbf{k}|^2,$$

$$\mathbf{k} \cdot \mathbf{E}_0 = 0, \qquad \mathbf{k} \cdot \mathbf{B}_0 = 0,$$

and

$$\mathbf{B}_0 = \frac{1}{\omega} \mathbf{k} \times \mathbf{E}_0.$$

Therefore electromagnetic waves are transverse.

Definition

A free electromagnetic wave has two transverse physical polarizations. The longitudinal and time-like components of the potential are removed by gauge freedom and constraints.

The transversality of electromagnetic radiation is one of the clearest physical signatures of gauge theory. The potential has four components, but the free massless electromagnetic field propagates only two local physical degrees of freedom.

9.12 Retarded Potentials and Radiation

In Lorenz gauge, the sourced Maxwell equation becomes

$$\Box A^{\mu} = J^{\mu}.$$

This equation can be solved using a retarded Green's function:

$$A^{\mu}(x) = \int d^{d+1}y \, G_{\mathrm{ret}}(x-y) J^{\mu}(y).$$

The retarded prescription enforces causality: the potential at x depends only on sources in the past light cone of x.

For localized time-dependent sources, the far field contains radiation. The radiative part falls off with distance in such a way that the energy flux through a large sphere remains finite. Physically, radiation is the part of the electromagnetic field that carries energy and momentum away to infinity.

Remark

Static fields store energy around sources, but radiation transports energy to large distances. The distinction is tied to boundary behavior: radiation is identified by its flux through surfaces at infinity.

A complete treatment of radiation requires more detailed analysis of retarded solutions, multipole expansion, and asymptotic fields. For the present course, the key point is structural: hyperbolic field equations plus retarded boundary conditions give causal propagation from sources to fields.

9.13 Physical Degrees of Freedom of the Electromagnetic Field

The four-potential A_{μ} has four components. However, not all four describe physical degrees of freedom. Gauge transformations

$$A_{\mu} \longrightarrow A_{\mu} + \partial_{\mu}\chi$$

identify potentials that describe the same electromagnetic field. In addition, not all components of Maxwell's equations are evolution equations; Gauss's law acts as a constraint on initial data.

In vacuum, after accounting for gauge freedom and constraints, the electromagnetic field has two propagating degrees of freedom at each spatial momentum. These are the two transverse polarizations of light.

This counting is easiest to see for a plane wave. The condition

$$\mathbf{k} \cdot \mathbf{E}_0 = 0$$

forces the electric field to be perpendicular to the propagation direction. A vector perpendicular to $\mathbf{k}$ in three-dimensional space has two independent components. The magnetic field is then determined by

$$\mathbf{B}_0 = \frac{1}{\omega}\mathbf{k} \times \mathbf{E}_0.$$

Thus the independent radiative data consist of two polarizations.

Summary

Electromagnetism is the fundamental Abelian gauge theory. The gauge potential A_μ is redundant, while the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is gauge invariant. Maxwell's equations follow from the action $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$, coupling to matter requires a conserved current, and the free electromagnetic field propagates two transverse physical polarizations.

Chapter 10

Gauge Fields and Yang–Mills Theory

Remark

Yang–Mills theory is the non-Abelian generalization of electromagnetism. The main new feature is that the gauge field itself carries charge and therefore interacts with itself. Mathematically, the electromagnetic potential is replaced by a Lie-algebra-valued connection, and the field strength becomes its curvature.

10.1 From Electromagnetism to Non-Abelian Gauge Fields

Electromagnetism is a gauge theory with gauge group $U(1)$. Its gauge potential is a one-form

$$A = A_\mu dx^\mu,$$

and its field strength is

$$F = dA.$$

In components,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Because $U(1)$ is Abelian, there is no commutator term.

Yang–Mills theory starts from the same idea but replaces $U(1)$ by a general Lie group G, such as $SU(2)$ or $SU(3)$. The gauge potential is no longer an ordinary real-valued one-form. It is a Lie-algebra-valued one-form,

$$A = A_\mu dx^\mu, \qquad A_\mu = A_\mu^a T_a,$$

where T_a are generators of the Lie algebra $\mathfrak{g}$ of G.

In this chapter we use an anti-Hermitian generator convention,

$$[T_a, T_b] = f_{ab}{}^c T_c,$$

with real structure constants $f_{ab}{}^c$. With this convention, the gauge coupling appears in the covariant derivative as

$$D_\mu = \partial_\mu + gA_\mu.$$

Other texts often use Hermitian generators and write factors of i. The physics is the same, but signs and factors of i move between definitions.

Definition

A *Yang–Mills field* is a gauge field whose potential takes values in a non-Abelian Lie algebra. Its field strength contains both derivative terms and a commutator term.

The essential jump from electromagnetism to Yang–Mills theory is therefore

$$F = dA \qquad \longrightarrow \qquad F = dA + gA \wedge A.$$

The second term is absent in Abelian electromagnetism and is responsible for the self-interaction of non-Abelian gauge fields.

10.2 Lie Groups and Lie Algebras: Minimal Toolkit

A Lie group is a group that is also a smooth manifold. Its elements can be multiplied and inverted smoothly. A Lie algebra is the tangent space to the Lie group at the identity, equipped with a bracket. For a matrix group, the bracket is the commutator.

Let G be a Lie group and $\mathfrak{g}$ its Lie algebra. A basis of $\mathfrak{g}$ will be denoted by T_a. The Lie bracket is

$$[T_a, T_b] = f_{ab}{}^c T_c.$$

The constants $f_{ab}{}^c$ are the structure constants. They are antisymmetric in the lower indices:

$$f_{ab}{}^c = -f_{ba}{}^c.$$

They also satisfy the Jacobi identity,

$$[T_a, [T_b, T_c]] + [T_b, [T_c, T_a]] + [T_c, [T_a, T_b]] = 0.$$

In components this gives

$$f_{ab}{}^d f_{dc}{}^e + f_{bc}{}^d f_{da}{}^e + f_{ca}{}^d f_{db}{}^e = 0.$$

Example

For $SU(2)$, the Lie algebra is three-dimensional and the structure constants are proportional to the Levi-Civita symbol:

$$f_{ab}{}^c = \epsilon_{ab}{}^c$$

in a standard normalization. This is why $SU(2)$ gauge theory has three gauge potentials $A^1_\mu, A^2_\mu, A^3_\mu$.

Yang–Mills theory also requires an invariant inner product on the Lie algebra. For matrix groups this is often built from a trace. We write schematically

$$\langle X, Y \rangle = -\operatorname{tr}(XY)$$

for anti-Hermitian matrices, up to a conventional normalization. The invariance condition is

$$\langle [Z, X], Y \rangle + \langle X, [Z, Y] \rangle = 0.$$

This property allows us to build gauge-invariant actions from expressions such as $\langle F_{\mu\nu}, F^{\mu\nu} \rangle$.

10.3 Gauge Connection

The gauge potential of Yang–Mills theory is best understood as a connection. In coordinates it is written

$$A_\mu = A^a_\mu T_a.$$

The role of a connection is to define a covariant derivative for fields that transform under the gauge group.

Suppose a matter field ψ takes values in a representation space of G. The gauge covariant derivative is

$$D_\mu \psi = \partial_\mu \psi + g A_\mu \psi,$$

where A_μ acts on ψ through the chosen representation. The ordinary derivative $\partial_\mu \psi$ does not transform covariantly under local gauge transformations. The connection term corrects this failure.

Under a local gauge transformation

$$U(x) \in G,$$

the matter field transforms as

$$\psi \longrightarrow \psi' = U\psi.$$

We demand that the covariant derivative transform in the same way:

$$D'_\mu \psi' = U D_\mu \psi.$$

This requirement determines the transformation law of the connection:

$$A'_\mu = U A_\mu U^{-1} - \frac{1}{g}(\partial_\mu U)U^{-1}.$$

Definition

A *gauge connection* is a Lie-algebra-valued one-form $A = A_\mu dx^\mu$ whose transformation law makes the covariant derivative transform covariantly under local gauge transformations.

For an infinitesimal gauge transformation

$$U = 1 + \alpha + O(\alpha^2), \qquad \alpha = \alpha^a T_a,$$

the connection transforms as

$$\delta_\alpha A_\mu = -\frac{1}{g}\partial_\mu \alpha + [\alpha, A_\mu].$$

Equivalently,

$$\delta_\alpha A_\mu = -\frac{1}{g} D_\mu \alpha,$$

where the covariant derivative in the adjoint representation is

$$D_\mu \alpha = \partial_\mu \alpha + g[A_\mu, \alpha].$$

10.4 Curvature / Field Strength

The Yang–Mills field strength is the curvature of the connection. In differential-form notation,

$$F = dA + gA \wedge A.$$

In components,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + g[A_\mu, A_\nu].$$

Writing

$$F_{\mu\nu} = F^a_{\mu\nu} T_a,$$

we get

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f_{bc}{}^a A^b_\mu A^c_\nu.$$

The final term is the non-Abelian term. It is zero for an Abelian Lie algebra.

The field strength also appears as the commutator of covariant derivatives. If

$$D_\mu = \partial_\mu + gA_\mu,$$

then

$$[D_\mu, D_\nu]\psi = gF_{\mu\nu}\psi.$$

Thus curvature measures the failure of covariant derivatives to commute.

Theorem

The field strength of a Yang–Mills connection is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + g[A_\mu, A_\nu].$$

The commutator term is responsible for the self-interaction of non-Abelian gauge fields.

This equation should be compared with electromagnetism:

$$F^{U(1)}_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

The difference is not cosmetic. In Yang–Mills theory, even the free-looking field strength is nonlinear in the potential.

10.5 Gauge Transformations

The finite gauge transformation of the connection is

$$A'_\mu = U A_\mu U^{-1} - \frac{1}{g}(\partial_\mu U)U^{-1}.$$

The field strength transforms covariantly:

$$F'_{\mu\nu} = U F_{\mu\nu} U^{-1}.$$

This is the non-Abelian analogue of gauge invariance. Notice the difference: in electromagnetism the field strength itself is gauge invariant, while in Yang–Mills theory it is gauge covariant.

A trace or invariant inner product of two field strengths is gauge invariant. Indeed,

$$\mathrm{tr}(F'_{\mu\nu}F'^{\mu\nu}) = \mathrm{tr}(UF_{\mu\nu}U^{-1}UF^{\mu\nu}U^{-1}) = \mathrm{tr}(F_{\mu\nu}F^{\mu\nu}).$$

This is why the Yang–Mills action is built from a trace or invariant inner product.

Remark

In a non-Abelian theory, the potential A_μ is gauge-dependent, and the field strength $F_{\mu\nu}$ is gauge-covariant rather than gauge-invariant. Gauge-invariant observables must be built by contracting group indices in an invariant way or by using nonlocal objects such as Wilson loops.

An infinitesimal gauge transformation gives

$$\delta_\alpha F_{\mu\nu} = [\alpha, F_{\mu\nu}].$$

Thus $F_{\mu\nu}$ rotates in the adjoint representation of the gauge group.

10.6 Covariant Derivative

The covariant derivative depends on the representation of the field it acts on. For a matter field ψ in a representation R,

$$D_\mu\psi = \partial_\mu\psi + gA^a_\mu R(T_a)\psi.$$

For a Lie-algebra-valued field $X = X^a T_a$, such as $F_{\mu\nu}$, the covariant derivative is taken in the adjoint representation:

$$D_\mu X = \partial_\mu X + g[A_\mu, X].$$

In components,

$$(D_\mu X)^a = \partial_\mu X^a + g f_{bc}{}^a A_\mu^b X^c.$$

The purpose of the covariant derivative is not only to preserve notation. It ensures that equations transform homogeneously under local gauge transformations. For example, the Yang–Mills equation will have the covariant form

$$D_\mu F^{\mu\nu} = J^\nu.$$

Both sides transform in the adjoint representation.

Definition

The adjoint covariant derivative of a Lie-algebra-valued field X is

$$D_\mu X = \partial_\mu X + g[A_\mu, X].$$

It replaces the ordinary derivative whenever the object being differentiated carries non-Abelian gauge charge.

In Abelian theory, the commutator vanishes and the adjoint covariant derivative reduces to the ordinary derivative. This is another way to see why Maxwell theory is structurally simpler than Yang–Mills theory.

10.7 Yang–Mills Action

The Yang–Mills Lagrangian density is

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4}\langle F_{\mu\nu}, F^{\mu\nu}\rangle.$$

In a component normalization where

$$\langle T_a, T_b\rangle = \delta_{ab},$$

this becomes

$$\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu}.$$

The action is

$$S_{\mathrm{YM}}[A] = -\frac{1}{4}\int d^{d+1}x\, F^a_{\mu\nu}F^{a\mu\nu}.$$

The action is gauge invariant because the field strength transforms covariantly and the inner product on the Lie algebra is invariant. It is also Lorentz invariant because all spacetime indices are contracted.

Definition

The *Yang–Mills action* is

$$S_{\mathrm{YM}}[A] = -\frac{1}{4}\int d^{d+1}x\, F^a_{\mu\nu}F^{a\mu\nu}.$$

It is the natural non-Abelian generalization of the Maxwell action.

Expanding the field strength shows that the Yang–Mills action contains terms quadratic, cubic, and quartic in A_μ. Schematically,

$$F^2 \sim (\partial A)^2 + g(\partial A)A^2 + g^2A^4.$$

Thus non-Abelian gauge fields self-interact even without matter.

10.8 Yang–Mills Equations

Varying the Yang–Mills action gives the Yang–Mills field equations. The variation of the curvature is

$$\delta F_{\mu\nu} = D_\mu \delta A_\nu - D_\nu \delta A_\mu.$$

Using antisymmetry of $F^{\mu\nu}$, the variation of the action is

$$\delta S_{\mathrm{YM}} = -\frac{1}{2}\int d^{d+1}x\, \langle F^{\mu\nu}, \delta F_{\mu\nu}\rangle = -\int d^{d+1}x\, \langle F^{\mu\nu}, D_\mu \delta A_\nu\rangle.$$

After integrating by parts covariantly and discarding the boundary term, one obtains

$$\delta S_{\mathrm{YM}} = \int d^{d+1}x\, \langle D_\mu F^{\mu\nu}, \delta A_\nu\rangle.$$

Therefore the vacuum Yang–Mills equation is

$$D_\mu F^{\mu\nu} = 0.$$

In components,

$$(D_\mu F^{\mu\nu})^a = \partial_\mu F^{a\mu\nu} + g f_{bc}{}^a A_\mu^b F^{c\mu\nu} = 0.$$

With matter, the equation becomes

$$D_\mu F^{\mu\nu} = J^\nu,$$

where J^ν is a Lie-algebra-valued matter current. Consistency requires a covariant conservation law,

$$D_\nu J^\nu = 0,$$

rather than the ordinary conservation law $\partial_\nu J^\nu = 0$.

Theorem

The source-free Yang–Mills equation is

$$D_\mu F^{\mu\nu} = 0.$$

It is nonlinear because both D_μ and $F_{\mu\nu}$ depend on the gauge potential A_μ.

This equation is the non-Abelian analogue of the source-free Maxwell equation

$$\partial_\mu F^{\mu\nu} = 0.$$

The ordinary derivative has been replaced by the gauge covariant derivative.

10.9 Bianchi Identity

The Yang–Mills field strength satisfies the covariant Bianchi identity

$$D_{[\lambda} F_{\mu\nu]} = 0.$$

Equivalently,

$$D_\lambda F_{\mu\nu} + D_\mu F_{\nu\lambda} + D_\nu F_{\lambda\mu} = 0.$$

In differential-form notation, this is

$$DF = 0,$$

where D is the exterior covariant derivative.

This identity follows from the definition of curvature and the Jacobi identity for covariant derivatives. Since

$$[D_\mu, D_\nu] = gF_{\mu\nu},$$

the Jacobi identity

$$[D_\lambda, [D_\mu, D_\nu]] + [D_\mu, [D_\nu, D_\lambda]] + [D_\nu, [D_\lambda, D_\mu]] = 0$$

leads directly to the Bianchi identity.

Remark

The Bianchi identity is not an equation of motion. It holds identically for any connection because the field strength is a curvature. The Yang–Mills equation comes from varying the action; the Bianchi identity comes from geometry.

For an Abelian gauge field, D reduces to d, and the Bianchi identity becomes

$$dF = 0,$$

which is the homogeneous Maxwell equation.

10.10 Self-Interaction of Gauge Fields

The most important physical difference between Abelian and non-Abelian gauge fields is self-interaction. In Maxwell theory, the field strength is linear in A_μ. Therefore the free Maxwell action is quadratic in A_μ, and the vacuum equations are linear.

In Yang–Mills theory,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + g[A_\mu, A_\nu].$$

The commutator term makes the field strength nonlinear in the gauge potential. Consequently, the action contains interaction terms even without matter. Schematically,

$$\mathcal{L}_{\text{YM}} \sim (\partial A)^2 + g(\partial A)A^2 + g^2 A^4.$$

The cubic and quartic terms describe gauge-field self-interactions.

Example

Gluons in quantum chromodynamics are quanta of an $SU(3)$ Yang–Mills field. Unlike photons in Abelian electromagnetism, gluons carry non-Abelian color charge. This is the quantum version of the classical fact that Yang–Mills gauge fields interact with themselves.

Classically, self-interaction means that Yang–Mills waves do not simply pass through one another by linear superposition. The field equations are nonlinear, and the gauge field acts as a source for itself.

10.11 Energy-Momentum Tensor of Yang–Mills Fields

The Yang–Mills stress-energy tensor is the natural non-Abelian generalization of the electromagnetic one:

$$T^{\mu\nu} = -\langle F^{\mu\rho}, F^{\nu}{}_{\rho}\rangle + \frac{1}{4}\eta^{\mu\nu}\langle F_{\rho\sigma}, F^{\rho\sigma}\rangle.$$

In component notation with $\langle T_a, T_b \rangle = \delta_{ab}$,

$$T^{\mu\nu} = -F^{a\mu\rho} F^{a\nu}{}_{\rho} + \frac{1}{4}\eta^{\mu\nu} F^{a}_{\rho\sigma} F^{a\rho\sigma}.$$

In vacuum, on shell,

$$\partial_\mu T^{\mu\nu} = 0.$$

This is ordinary spacetime energy-momentum conservation, not a gauge-covariant conservation law, because $T^{\mu\nu}$ is gauge invariant and has no free Lie algebra index.

The energy density is positive for compact gauge groups with positive invariant inner product. In analogy with electromagnetism, one may decompose the field strength into non-Abelian electric and magnetic fields,

$$E^a_i = F^a_{0i},$$

and

$$B^a_i = \frac{1}{2}\epsilon_{ijk} F^a_{jk}$$

up to sign conventions. Then the energy density has the schematic form

$$\mathcal{E} = \frac{1}{2}\left(E^a_i E^a_i + B^a_i B^a_i\right).$$

Definition

The Yang–Mills stress-energy tensor is gauge invariant and measures the energy, momentum, and stresses carried by the non-Abelian gauge field. It is quadratic in $F_{\mu\nu}$, but because $F_{\mu\nu}$ itself is nonlinear in A_μ, it contains nonlinear gauge-field interactions.

This tensor is the source that a Yang–Mills field would contribute to the Einstein equations if coupled to gravity.

10.12 Classical Meaning of Non-Abelian Charge

In electromagnetism, charge is a single conserved number associated with a global $U(1)$ symmetry. In a non-Abelian gauge theory, charge takes values in a Lie algebra or in a representation of the gauge group. It has components,

$$Q = Q^a T_a.$$

These components do not represent independent Abelian charges. They rotate into one another under gauge transformations.

Matter currents in Yang–Mills theory are Lie-algebra-valued:

$$J^\mu = J^{a\mu} T_a.$$

Their conservation law is covariant:

$$D_\mu J^\mu = 0.$$

This means that non-Abelian charge is transported with parallel transport in the internal gauge space. Ordinary derivatives are not sufficient because the internal frame itself can vary from point to point.

Remark

A non-Abelian charge is not just a list of separately conserved numbers. It is a geometric object in internal space whose components depend on the choice of gauge frame. Gauge-covariant conservation replaces ordinary componentwise conservation.

This is the classical seed of a major quantum distinction. In non-Abelian gauge theories, the gauge bosons themselves carry the charge to which they couple. This is why the gauge field participates directly in the dynamics of its own charge distribution.

10.13 Comparison: Abelian vs Non-Abelian Theory

The difference between Maxwell theory and Yang–Mills theory can be summarized as follows.

Feature	Abelian $U(1)$	Non-Abelian Yang–Mills
Gauge group	commutative	noncommutative
Potential	real-valued A_μ	Lie-algebra-valued $A_\mu^a T_a$
Field strength	$\partial_\mu A_\nu - \partial_\nu A_\mu$	$\partial_\mu A_\nu - \partial_\nu A_\mu + g[A_\mu, A_\nu]$
Gauge transformation of F	invariant	covariant
Equation of motion	$\partial_\mu F^{\mu\nu} = J^\nu$	$D_\mu F^{\mu\nu} = J^\nu$
Gauge-field self-interaction	absent	present
Charge conservation	$\partial_\mu J^\mu = 0$	$D_\mu J^\mu = 0$

The Abelian theory is linear in the absence of sources. The non-Abelian theory is nonlinear even in vacuum. This single difference has enormous consequences: self-interactions, nontrivial classical solutions, confinement in quantum chromodynamics, and a much richer gauge structure.

Summary

Yang–Mills theory generalizes electromagnetism by replacing the Abelian $U(1)$ gauge group with a non-Abelian Lie group G. The gauge potential is a Lie-algebra-valued connection, the field strength is $F = dA + gA \wedge A$, and the action is $-\frac{1}{4}\int F^a_{\mu\nu} F^{a\mu\nu}\, d^{d+1}x$. The field equations are $D_\mu F^{\mu\nu} = J^\nu$, and the non-Abelian commutator term makes the gauge field self-interacting.

Chapter 11

Spinor Fields and the Dirac Equation

Remark

Spinor fields are the natural relativistic fields for spin-one-half degrees of freedom. They are not ordinary scalars, vectors, or tensors. Their transformation law is tied to the double cover of the Lorentz group, and their dynamics is governed by a first-order relativistic equation: the Dirac equation.

11.1 Why Spinors Are Needed

Scalar fields describe spin-zero degrees of freedom. Vector fields describe spin-one degrees of freedom, as in electromagnetism. However, many fundamental matter fields are spin-one-half fields. Such fields cannot be represented by ordinary spacetime scalars or vectors. They require spinors.

The need for spinors can be seen from symmetry. Relativistic fields must carry representations of the Lorentz group. Scalars transform trivially, vectors transform with Lorentz matrices $\Lambda^{\mu}{}_{\nu}$, and tensors transform with tensor products of these matrices. Spinors transform under a different kind of representation: a representation of the double cover of the proper Lorentz group.

The proper orthochronous Lorentz group $SO^{+}(1,3)$ has a double cover,

$$SL(2,\mathbb{C}) \longrightarrow SO^{+}(1,3).$$

Spinors transform naturally under $SL(2,\mathbb{C})$. This is why a rotation by 2π can act as multiplication by -1 on a spinor, while a rotation by 4π returns it to itself.

Definition

A *spinor* is a field transforming under a spin representation of the Lorentz group, more precisely under a representation of the double cover of the proper Lorentz group.

In classical field theory, one may first treat spinors formally as fields with components. The deeper quantum statement is that spin-one-half fields must be quantized with anticommutation relations. This chapter focuses on the classical field equations and their symmetries, while indicating where quantum field theory enters.

11.2 Clifford Algebra and Gamma Matrices

The Dirac equation is built from gamma matrices. With the mostly-minus metric convention

$$\eta_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1),$$

the gamma matrices satisfy the Clifford algebra

$$\{\gamma^{\mu}, \gamma^{\nu}\} = \gamma^{\mu}\gamma^{\nu} + \gamma^{\nu}\gamma^{\mu} = 2\eta^{\mu\nu}\mathbb{I}.$$

Thus
$$(\gamma^0)^2 = \mathbb{I}, \qquad (\gamma^i)^2 = -\mathbb{I}$$
for spatial indices $i = 1, 2, 3$.

The gamma matrices are not themselves spacetime vectors. They are constant matrices acting on spinor components, arranged so that contractions such as
$$\gamma^\mu \partial_\mu$$
transform covariantly when the spinor transforms appropriately.

Definition

The *Clifford algebra* associated with Minkowski spacetime is generated by matrices γ^μ satisfying
$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbb{I}.$$

A Dirac spinor in four spacetime dimensions has four complex components:
$$\psi(x) = \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \\ \psi_4(x) \end{pmatrix}.$$
Different explicit representations of the gamma matrices are possible. Physical statements do not depend on the representation. A change of gamma-matrix basis is a similarity transformation.

The key algebraic point is that the Clifford algebra linearizes the Klein–Gordon operator. If ψ satisfies the Dirac equation
$$(i\gamma^\mu \partial_\mu - m)\psi = 0,$$
then applying $(i\gamma^\nu \partial_\nu + m)$ gives
$$(\Box + m^2)\psi = 0.$$
Thus every component of a free Dirac spinor satisfies the Klein–Gordon equation, but the Dirac equation contains additional spinor structure.

11.3 Lorentz Transformations of Spinors

Under a Lorentz transformation
$$x^\mu \longrightarrow x'^\mu = \Lambda^\mu{}_\nu x^\nu,$$
a Dirac spinor transforms as
$$\psi(x) \longrightarrow \psi'(x') = S(\Lambda)\psi(x),$$
where $S(\Lambda)$ is a spinor representation matrix. It is defined so that
$$S(\Lambda)^{-1}\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\nu \gamma^\nu.$$
This identity ensures covariance of the Dirac equation.

Infinitesimally, write
$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu + O(\omega^2),$$
where $\omega_{\mu\nu} = -\omega_{\nu\mu}$. The spinor representation has the form
$$S(\Lambda) = \exp\left(\frac{1}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}\right),$$
where
$$\Sigma^{\mu\nu} = \frac{1}{4}[\gamma^\mu, \gamma^\nu].$$
These matrices are the Lorentz generators in the spinor representation.

Remark

Spinors do not transform like vectors. The Lorentz transformation of a spinor is implemented by matrices $S(\Lambda)$ acting on spinor components, while the spacetime point transforms by $\Lambda^\mu{}_\nu$.

This distinction is central. A spinor index is not a spacetime vector index. It belongs to a different representation, even though it is tied to spacetime Lorentz symmetry.

11.4 Dirac Field and Dirac Adjoint

A Dirac field is a spinor-valued field

$$\psi : M \to \mathbb{C}^4.$$

The ordinary Hermitian conjugate $\psi^\dagger$ does not by itself give a Lorentz scalar when multiplied by ψ. The correct relativistic adjoint is the Dirac adjoint

$$\bar{\psi} = \psi^\dagger \gamma^0.$$

With this definition,

$$\bar{\psi}\psi$$

is a Lorentz scalar, while

$$\bar{\psi}\gamma^\mu\psi$$

is a Lorentz vector.

Definition

The *Dirac adjoint* of a Dirac spinor ψ is

$$\bar{\psi} = \psi^\dagger \gamma^0.$$

It is introduced so that spinor bilinears have simple Lorentz transformation properties.

Important bilinears include

$$\bar{\psi}\psi,$$

which is a scalar, and

$$j^\mu = \bar{\psi}\gamma^\mu\psi,$$

which is a vector current. More bilinears, involving γ^5 and commutators of gamma matrices, appear in advanced treatments of spinor field theory.

11.5 Dirac Lagrangian

The free Dirac Lagrangian density is

$$\mathcal{L}_{\mathrm{D}} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi.$$

Equivalently,

$$\mathcal{L}_{\mathrm{D}} = i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi.$$

This Lagrangian is first order in derivatives. This is one of the main differences between the Dirac field and the scalar field. The scalar Lagrangian contains quadratic derivatives; the Dirac Lagrangian contains one derivative.

For a real action one often uses the symmetrized form

$$\mathcal{L}_{\mathrm{D}} = \frac{i}{2}\left(\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi - \partial_{\mu}\bar{\psi}\gamma^{\mu}\psi\right) - m\bar{\psi}\psi.$$

The two forms differ by a total derivative under suitable assumptions. For classical field equations, the unsymmetrized form is usually sufficient.

Definition

The free Dirac Lagrangian is

$$\mathcal{L}_{\mathrm{D}} = \bar{\psi}(i\gamma^{\mu}\partial_{\mu} - m)\psi.$$

Its Euler–Lagrange equation is the Dirac equation.

The mass term

$$m\bar{\psi}\psi$$

couples left- and right-handed components of the spinor. This becomes especially important when discussing chirality.

11.6 Dirac Equation from the Variational Principle

In the variational principle, ψ and $\bar{\psi}$ are treated as independent fields. Varying the action with respect to $\bar{\psi}$ gives

$$\delta_{\bar{\psi}} S = \int d^{d+1}x\, \delta\bar{\psi}(i\gamma^{\mu}\partial_{\mu} - m)\psi.$$

Therefore the field equation is

$$(i\gamma^{\mu}\partial_{\mu} - m)\psi = 0.$$

This is the Dirac equation.

Varying with respect to ψ gives the adjoint Dirac equation. Starting from

$$\mathcal{L}_{\mathrm{D}} = i\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi - m\bar{\psi}\psi,$$

the variation with respect to ψ gives, after integration by parts,

$$i\partial_{\mu}\bar{\psi}\gamma^{\mu} + m\bar{\psi} = 0.$$

Equivalently,

$$\bar{\psi}(i\overleftarrow{\partial}_{\mu}\gamma^{\mu} + m) = 0,$$

where the arrow indicates that the derivative acts to the left.

Theorem

The free Dirac field satisfies

$$(i\gamma^{\mu}\partial_{\mu} - m)\psi = 0,$$

while the adjoint field satisfies

$$i\partial_{\mu}\bar{\psi}\gamma^{\mu} + m\bar{\psi} = 0.$$

Applying the conjugate first-order operator shows that the Dirac equation implies the Klein–Gordon equation for each component:

$$(\Box + m^2)\psi = 0.$$

Thus the Dirac equation is a relativistic wave equation whose square gives the mass-shell condition.

11.7 Conserved Dirac Current

The free Dirac Lagrangian is invariant under the global phase transformation

$$\psi \longrightarrow e^{i\alpha}\psi, \qquad \bar{\psi} \longrightarrow \bar{\psi}e^{-i\alpha},$$

where α is constant. Noether's theorem gives the conserved current

$$j^\mu = \bar{\psi}\gamma^\mu\psi.$$

Let us verify conservation directly. The Dirac equation is

$$i\gamma^\mu\partial_\mu\psi - m\psi = 0,$$

and the adjoint equation is

$$i\partial_\mu\bar{\psi}\gamma^\mu + m\bar{\psi} = 0.$$

Now compute

$$\partial_\mu j^\mu = \partial_\mu(\bar{\psi}\gamma^\mu\psi) = (\partial_\mu\bar{\psi})\gamma^\mu\psi + \bar{\psi}\gamma^\mu\partial_\mu\psi.$$

Using the field equations, the two terms cancel, giving

$$\partial_\mu j^\mu = 0.$$

Definition

The Dirac current is

$$j^\mu = \bar{\psi}\gamma^\mu\psi.$$

It is conserved on shell:

$$\partial_\mu j^\mu = 0.$$

The time component is

$$j^0 = \bar{\psi}\gamma^0\psi = \psi^\dagger\psi.$$

This is nonnegative. This feature was historically important: the Dirac equation has a positive density associated with its conserved current, unlike the naive probability interpretation of the Klein–Gordon current.

11.8 Coupling to Electromagnetism

A charged Dirac field couples to electromagnetism by replacing the ordinary derivative with a gauge covariant derivative. With the convention

$$D_\mu = \partial_\mu + iqA_\mu,$$

the minimally coupled Dirac Lagrangian is

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi.$$

Under a local $U(1)$ transformation,

$$\psi \longrightarrow e^{i\alpha(x)}\psi,$$

the gauge potential transforms as

$$A_\mu \longrightarrow A_\mu - \frac{1}{q}\partial_\mu\alpha.$$

Then

$$D_\mu\psi \longrightarrow e^{i\alpha(x)} D_\mu\psi.$$

Thus the Lagrangian is locally gauge invariant.

If the electromagnetic field is dynamical, the full Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi.$$

The Dirac equation becomes

$$(i\gamma^\mu D_\mu - m)\psi = 0.$$

Varying the action with respect to A_μ gives Maxwell's equation with Dirac matter current. With the above convention, the electric current is proportional to

$$j^\mu = q\bar{\psi}\gamma^\mu\psi,$$

up to the sign convention used in the interaction term.

Remark

Minimal electromagnetic coupling has the same logic for scalar and spinor matter: local phase symmetry forces the introduction of the gauge potential and the covariant derivative.

The Dirac field is therefore the simplest classical relativistic model of charged spin-one-half matter coupled to electromagnetism.

11.9 Chiral Decomposition

In four spacetime dimensions one defines

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3.$$

It satisfies

$$(\gamma^5)^2 = \mathbb{I}, \qquad \{\gamma^5, \gamma^\mu\} = 0.$$

The chiral projection operators are

$$P_L = \frac{1}{2}(\mathbb{I} - \gamma^5), \qquad P_R = \frac{1}{2}(\mathbb{I} + \gamma^5).$$

They obey

$$P_L^2 = P_L, \qquad P_R^2 = P_R, \qquad P_L P_R = 0, \qquad P_L + P_R = \mathbb{I}.$$

The left- and right-handed parts of a Dirac spinor are

$$\psi_L = P_L\psi, \qquad \psi_R = P_R\psi.$$

For a massless Dirac field, the Dirac equation separates into independent left- and right-handed equations. The mass term couples them:

$$m\bar{\psi}\psi = m(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L).$$

Thus chirality is especially important for massless or approximately massless fermions.

Definition

Chiral projectors split a Dirac spinor into left- and right-handed components:

$$\psi = \psi_L + \psi_R, \qquad \psi_L = P_L\psi, \qquad \psi_R = P_R\psi.$$

Chirality is not just a mathematical refinement. It is essential in the weak interaction, where left- and right-handed fermions transform differently under the gauge group.

11.10 Weyl and Majorana Spinors

A Weyl spinor is a spinor of definite chirality. In four dimensions, it is obtained by imposing

$$\psi = \psi_L$$

or

$$\psi = \psi_R.$$

For massless fields, left- and right-handed Weyl spinors can be treated independently. A mass term generally couples opposite chiralities and therefore cannot be written for a single isolated Weyl spinor in the same way as for a Dirac spinor.

A Majorana spinor satisfies a reality condition relating the spinor to its charge conjugate. Schematically,

$$\psi^c = \psi.$$

The precise form of charge conjugation depends on the gamma-matrix convention. Majorana spinors reduce the number of independent real degrees of freedom.

Remark

Dirac, Weyl, and Majorana spinors are not three unrelated objects. They are different ways of organizing spin-one-half degrees of freedom, depending on chirality, reality conditions, dimension, signature, and gauge representation.

In many physical applications, whether a Majorana condition is possible depends on the gauge charges of the field. A charged field generally cannot be Majorana with respect to the same $U(1)$ charge, because charge conjugation reverses the charge.

11.11 Classical Spinor Fields vs Quantum Fermions

The Dirac equation can be studied as a classical field equation. One may solve it, analyze its currents, couple it to gauge fields, and study its Lorentz covariance. However, the physical interpretation of spinor fields is deeply quantum.

In quantum field theory, spin-one-half fields are fermionic. Their field operators satisfy anticommutation relations, not commutation relations. This is connected with the spin-statistics theorem. A fully consistent quantum treatment of the Dirac field therefore requires fermionic quantization.

In classical variational field theory, one often treats spinor components as Grassmann-valued variables. Grassmann variables anticommute:

$$\theta_1\theta_2 = -\theta_2\theta_1.$$

This formalism anticipates the algebraic structure needed for fermionic quantum fields and path integrals.

Definition

A *Grassmann variable* is an anticommuting variable. Classical fermionic field theory is often formulated using Grassmann-valued spinor fields, especially when preparing for quantization.

For a first course in classical field theory, it is acceptable to first treat the Dirac equation formally as a classical relativistic wave equation. But one should not confuse this formal classical description with the full physics of fermions.

11.12 What Is Classical and What Is Already Anticipating QFT?

Several parts of the Dirac theory are genuinely classical field-theoretic:

1. the Lorentz covariance of the Dirac equation;
2. the construction of gamma matrices and spinor representations;
3. the variational principle for the Dirac action;
4. the conserved current associated with global phase symmetry;
5. minimal coupling to gauge fields.

Other parts point beyond ordinary classical field theory:

1. the interpretation of $\psi^\dagger\psi$ as probability density for a single relativistic particle;
2. the existence of antiparticles;
3. fermionic anticommutation relations;
4. the spin-statistics connection;
5. Grassmann path integrals.

The safest interpretation in this course is the following. The Dirac field is a classical field variable with a Lorentz-covariant first-order action. Its full physical meaning as a fermion field emerges only in quantum field theory.

Summary

Spinor fields carry spin representations of the Lorentz group. The gamma matrices satisfy the Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbb{I}$, and the Dirac equation is $(i\gamma^\mu\partial_\mu - m)\psi = 0$. The Dirac adjoint $\bar{\psi} = \psi^\dagger\gamma^0$ allows Lorentz-covariant bilinears, including the conserved current $j^\mu = \bar{\psi}\gamma^\mu\psi$. Minimal coupling replaces ∂_μ by D_μ, and chiral projectors split a Dirac spinor into left- and right-handed components. The classical formalism is meaningful, but the physical interpretation of fermions belongs to quantum field theory.

Appendix A

Notation and Conventions

Remark

This appendix collects the notational conventions used throughout the notes. It is not a substitute for the main text. Its purpose is to make signs, indices, Fourier conventions, differential-form notation, and gamma-matrix conventions explicit in one place.

A.1 Metric Signatures

Throughout these notes, unless stated otherwise, flat Minkowski spacetime is written with the mostly-minus convention

$$\eta_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1).$$

Thus the invariant interval is

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu = dt^2 - d\mathbf{x}^2.$$

The square of a four-vector $p^\mu = (E, \mathbf{p})$ is

$$p_\mu p^\mu = E^2 - |\mathbf{p}|^2.$$

For a massive relativistic excitation,

$$p_\mu p^\mu = m^2.$$

Partial derivatives are denoted by

$$\partial_\mu = \frac{\partial}{\partial x^\mu}.$$

With the mostly-minus metric,

$$\partial^0 = \partial_0 = \partial_t, \qquad \partial^i = -\partial_i.$$

The d'Alembert operator is

$$\Box = \partial_\mu \partial^\mu = \partial_t^2 - \nabla^2.$$

Definition

The convention used in these notes is

$$\eta_{\mu\nu} = \mathrm{diag}(+ - - -), \qquad \Box = \partial_t^2 - \nabla^2.$$

Changing signature changes several signs in field equations and actions.

Curved spacetime metrics are usually denoted by $g_{\mu\nu}$. In that case, indices are raised and lowered with $g^{\mu\nu}$ and $g_{\mu\nu}$, and the volume density is $\sqrt{|g|}\, d^{d+1}x$, where $g = \det(g_{\mu\nu})$.

A.2 Index Conventions

Greek indices usually denote spacetime indices:

$$\mu, \nu, \rho, \sigma = 0, 1, 2, 3.$$

Latin indices from the middle of the alphabet usually denote spatial indices:

$$i, j, k = 1, 2, 3.$$

Internal indices are often written as

$$a, b, c = 1, \ldots, \dim \mathfrak{g}$$

for a Lie algebra $\mathfrak{g}$. Field-component labels may be written as A, B, C.

The Einstein summation convention is used: repeated upper-lower index pairs are summed over. For example,

$$A_\mu B^\mu = \sum_{\mu=0}^{3} A_\mu B^\mu.$$

Dummy indices may be renamed. Free indices must match on both sides of an equation. For instance,

$$\partial_\mu F^{\mu\nu} = J^\nu$$

is an equation with one free index ν. The index μ is summed.

For Euclidean spatial vectors, bold notation is used:

$$\mathbf{x}, \qquad \mathbf{E}, \qquad \mathbf{B}.$$

The Euclidean dot product is written

$$\mathbf{a} \cdot \mathbf{b} = a_i b_i.$$

A.3 Units and Dimensions

Most field-theory formulas are written in natural units,

$$c = \hbar = 1.$$

In these units, time and length have the same dimension, and mass, energy, frequency, and inverse length have the same dimension. For example,

$$[m] = [E] = [\omega] = [L^{-1}].$$

The action is dimensionless in units $\hbar = 1$:

$$[S] = 1.$$

Therefore the Lagrangian density in $D = d + 1$ spacetime dimensions has mass dimension

$$[\mathcal{L}] = D.$$

For a real scalar field with kinetic term

$$\frac{1}{2}\partial_\mu \phi \partial^\mu \phi,$$

one obtains

$$[\phi] = \frac{D-2}{2}.$$

In four spacetime dimensions, $[\phi] = 1$.

Dimensional analysis is not a decorative check. It is often the fastest way to find missing constants, wrong powers, or inconsistent interaction terms.

A.4 Fourier Transform Conventions

For spatial Fourier transforms, a common convention used in these notes is

$$f(\mathbf{x}) = \int \frac{d^d k}{(2\pi)^d}\, \tilde{f}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{x}},$$

with inverse

$$\tilde{f}(\mathbf{k}) = \int d^d x\, f(\mathbf{x}) e^{-i\mathbf{k}\cdot\mathbf{x}}.$$

For spacetime plane waves, we often write

$$e^{-ik_\mu x^\mu} = e^{-i\omega t + i\mathbf{k}\cdot\mathbf{x}}.$$

With the mostly-minus convention,

$$k_\mu x^\mu = \omega t - \mathbf{k}\cdot\mathbf{x}.$$

For the Klein–Gordon equation,

$$(\Box + m^2)\phi = 0,$$

plane waves obey

$$\omega^2 = |\mathbf{k}|^2 + m^2.$$

Remark

Fourier conventions vary from book to book. The placement of signs in the exponential and factors of 2π must always be checked before comparing propagators or Green's functions across references.

A.5 Levi-Civita Symbols and Tensors

The three-dimensional Levi-Civita symbol is denoted by ϵ_{ijk}, with

$$\epsilon_{123} = +1.$$

It is antisymmetric in all indices. The cross product is

$$(\mathbf{a}\times\mathbf{b})_i = \epsilon_{ijk} a_j b_k.$$

In four dimensions, the Levi-Civita symbol is chosen as

$$\epsilon^{0123} = +1$$

unless stated otherwise. Care is required when lowering indices in Lorentzian signature. With the mostly-minus metric, signs may change when all indices are lowered.

The dual electromagnetic field strength is

$$\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}.$$

Useful Lorentz invariants are

$$F_{\mu\nu}F^{\mu\nu} = 2(|\mathbf{B}|^2 - |\mathbf{E}|^2),$$

and, with the conventions of these notes,

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = -4\mathbf{E}\cdot\mathbf{B}.$$

A.6 Differential Form Conventions

A p-form is written as

$$\alpha = \frac{1}{p!}\alpha_{\mu_1\cdots\mu_p} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_p}.$$

The wedge product is antisymmetric:

$$dx^\mu \wedge dx^\nu = -dx^\nu \wedge dx^\mu.$$

The exterior derivative satisfies

$$d^2 = 0.$$

For electromagnetism,

$$A = A_\mu dx^\mu, \qquad F = dA.$$

For non-Abelian gauge theory,

$$F = dA + gA \wedge A.$$

The Abelian Bianchi identity is

$$dF = 0,$$

while the Yang–Mills Bianchi identity is

$$DF = 0.$$

A.7 Gamma Matrix Conventions

Gamma matrices satisfy

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbb{I}.$$

With the mostly-minus convention,

$$(\gamma^0)^2 = \mathbb{I}, \qquad (\gamma^i)^2 = -\mathbb{I}.$$

The Dirac adjoint is

$$\bar{\psi} = \psi^\dagger\gamma^0.$$

The free Dirac equation is

$$(i\gamma^\mu\partial_\mu - m)\psi = 0.$$

The Dirac current is

$$j^\mu = \bar{\psi}\gamma^\mu\psi.$$

In four spacetime dimensions,

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3,$$

and the chiral projectors are

$$P_L = \frac{1}{2}(\mathbb{I} - \gamma^5), \qquad P_R = \frac{1}{2}(\mathbb{I} + \gamma^5).$$

Summary

The global convention of these notes is mostly-minus signature, natural units, Einstein summation, $\Box = \partial_t^2 - \nabla^2$, and gamma matrices satisfying $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbb{I}$. Whenever comparing with a different reference, check metric signature, Fourier signs, gauge-coupling conventions, and gamma-matrix conventions first.

Appendix B

Variational Calculus for Fields

Remark

The variational calculus of fields is the technical engine behind the action principle, Euler–Lagrange equations, Noether currents, stress-energy tensors, and boundary terms. This appendix collects the core manipulations in one place.

B.1 Variation of a Functional

A functional assigns a number to an entire field configuration. The basic action functional has the form

$$S[\phi] = \int_\Omega d^{d+1}x\, \mathcal{L}(\phi^A, \partial_\mu \phi^A, x).$$

A variation is defined by considering a one-parameter family of fields

$$\phi^A_\varepsilon = \phi^A + \varepsilon \eta^A,$$

where η^A is an arbitrary test variation satisfying whatever boundary conditions are appropriate. The first variation is

$$\delta S[\phi; \eta] = \frac{d}{d\varepsilon} S[\phi + \varepsilon\eta]\Big|_{\varepsilon=0}.$$

For a first-order Lagrangian density,

$$\delta S = \int_\Omega d^{d+1}x \left(\frac{\partial \mathcal{L}}{\partial \phi^A} \delta\phi^A + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \delta(\partial_\mu \phi^A) \right).$$

Since variation and partial differentiation commute,

$$\delta(\partial_\mu \phi^A) = \partial_\mu \delta\phi^A.$$

Thus

$$\delta S = \int_\Omega d^{d+1}x \left(\frac{\partial \mathcal{L}}{\partial \phi^A} \delta\phi^A + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \partial_\mu \delta\phi^A \right).$$

B.2 Integration by Parts

The fundamental step is integration by parts:

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \partial_\mu \delta\phi^A = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \delta\phi^A \right) - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \right) \delta\phi^A.$$

Substituting into δS gives

$$\begin{aligned} \delta S = &\int_\Omega d^{d+1}x \left[\frac{\partial \mathcal{L}}{\partial \phi^A} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \right) \right] \delta\phi^A \\ &+ \int_\Omega d^{d+1}x\, \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \delta\phi^A \right). \end{aligned}$$

Using Gauss's theorem, the second term becomes a boundary integral:

$$\int_{\partial\Omega} d\Sigma_\mu \, \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \delta\phi^A.$$

The bulk coefficient of $\delta\phi^A$ defines the Euler–Lagrange derivative:

$$E_A(\mathcal{L}) = \frac{\partial \mathcal{L}}{\partial \phi^A} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \right).$$

The Euler–Lagrange equations are

$$E_A(\mathcal{L}) = 0.$$

Theorem

For a first-order local action, stationarity under variations with vanishing boundary term gives

$$\frac{\partial \mathcal{L}}{\partial \phi^A} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \right) = 0.$$

B.3 Boundary Terms

Boundary terms are not mistakes. They decide which variational problem is being posed. The general first variation can be written schematically as

$$\delta S = \int_\Omega d^{d+1}x \, E_A(\mathcal{L}) \delta\phi^A + \int_{\partial\Omega} d\Sigma_\mu \, \Theta^\mu(\phi, \delta\phi).$$

For first-order theories,

$$\Theta^\mu(\phi, \delta\phi) = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^A)} \delta\phi^A.$$

Common ways to control boundary terms include:

1. requiring $\delta\phi^A = 0$ on $\partial\Omega$, called Dirichlet boundary conditions;
2. imposing boundary conditions on normal derivatives or momenta;
3. adding a boundary term to the action so that the desired boundary data define a well-posed variational principle;
4. requiring sufficiently fast falloff at infinity.

A total derivative added to the Lagrangian density,

$$\mathcal{L} \longrightarrow \mathcal{L} + \partial_\mu K^\mu,$$

does not change the bulk Euler–Lagrange equations, but it changes the boundary term. Therefore it can change canonical momenta, Noether charges, and the correct boundary conditions.

Remark

A statement such as "total derivatives do not matter" is incomplete. They do not change bulk equations under appropriate assumptions, but they can matter for boundaries, charges, and Hamiltonian generators.

B.4 Functional Derivatives

The functional derivative is defined by

$$\delta F[\phi] = \int d^{d+1}x \, \frac{\delta F}{\delta \phi^A(x)} \delta \phi^A(x).$$

For the action,

$$\frac{\delta S}{\delta \phi^A(x)} = E_A(\mathcal{L})(x)$$

when boundary terms are discarded or controlled.

A basic identity is

$$\frac{\delta \phi^A(y)}{\delta \phi^B(x)} = \delta^A{}_B \delta^{(d+1)}(y-x).$$

For example, if

$$F[\phi] = \int d^d x \, \frac{1}{2} \phi(x)^2,$$

then

$$\delta F = \int d^d x \, \phi(x) \delta \phi(x),$$

so

$$\frac{\delta F}{\delta \phi(x)} = \phi(x).$$

If

$$F[\phi] = \int d^d x \, \frac{1}{2} |\nabla \phi|^2,$$

then

$$\delta F = \int d^d x \, \nabla \phi \cdot \nabla \delta \phi = -\int d^d x \, \nabla^2 \phi \, \delta \phi$$

up to a boundary term. Hence

$$\frac{\delta F}{\delta \phi(x)} = -\nabla^2 \phi(x).$$

B.5 Variation with Constraints

Sometimes variations are constrained. A simple finite-dimensional analogy is the method of Lagrange multipliers. In field theory, suppose one wants to extremize $S[\phi]$ subject to

$$C[\phi] = 0.$$

One introduces a multiplier λ and varies

$$S_\lambda[\phi] = S[\phi] + \lambda C[\phi].$$

For local constraints

$$C(x, \phi, \partial \phi) = 0,$$

one uses a multiplier field $\lambda(x)$:

$$S_\lambda[\phi, \lambda] = S[\phi] + \int d^{d+1}x \, \lambda(x) C(x, \phi, \partial \phi).$$

Variation with respect to λ imposes the constraint, while variation with respect to ϕ gives modified field equations.

Constraints also arise in Hamiltonian field theory when the Legendre transform is degenerate. In that setting, constraints are equations on phase space, such as

$$\chi(\phi, \pi) \approx 0.$$

They must be handled with Dirac's constraint algorithm.

B.6 Variation with Respect to the Metric

The stress-energy tensor can be obtained by varying the matter action with respect to the metric. A common convention is

$$T_{\mu\nu} = -\frac{2}{\sqrt{|g|}}\frac{\delta S_{\text{matter}}}{\delta g^{\mu\nu}}.$$

Equivalently,

$$\delta S_{\text{matter}} = -\frac{1}{2}\int d^{d+1}x\,\sqrt{|g|}\,T_{\mu\nu}\delta g^{\mu\nu}.$$

The variation of the determinant is

$$\delta\sqrt{|g|} = -\frac{1}{2}\sqrt{|g|}\,g_{\mu\nu}\delta g^{\mu\nu}.$$

Also,

$$\delta g_{\mu\nu} = -g_{\mu\rho}g_{\nu\sigma}\delta g^{\rho\sigma}.$$

For a scalar field in curved spacetime,

$$S = -\int d^{d+1}x\sqrt{|g|}\left(\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + V(\phi)\right)$$

with sign conventions adjusted to the chosen metric convention. Varying with respect to $g^{\mu\nu}$ gives the Hilbert stress-energy tensor.

B.7 Common Variational Identities

Useful identities include

$$\delta(AB) = (\delta A)B + A(\delta B),$$
$$\delta(A^{-1}) = -A^{-1}(\delta A)A^{-1},$$

for an invertible matrix A, and

$$\delta\det A = (\det A)\,\mathrm{tr}(A^{-1}\delta A).$$

For gauge theory,

$$\delta F_{\mu\nu} = D_\mu\delta A_\nu - D_\nu\delta A_\mu$$

in the non-Abelian case, while in the Abelian case

$$\delta F_{\mu\nu} = \partial_\mu\delta A_\nu - \partial_\nu\delta A_\mu.$$

For the Dirac field, ψ and $\bar{\psi}$ are varied independently:

$$\delta\bar{\psi} \neq (\delta\psi)^\dagger\gamma^0$$

inside the variational calculation.

Summary

The core variational pattern is always the same: vary the action, integrate by parts, separate bulk terms from boundary terms, impose boundary conditions or add boundary counterterms, and read off the Euler–Lagrange equations from the bulk coefficient of arbitrary variations.

Appendix C

Lie Groups and Lie Algebras

Remark

Gauge theory is written in the language of Lie groups and Lie algebras. This appendix gives only the minimal toolkit needed for the main text: generators, structure constants, representations, adjoint action, the exponential map, and basic examples.

C.1 Groups, Algebras, and Generators

A group is a set with multiplication, an identity element, and inverses. A Lie group is a group that is also a smooth manifold, with smooth multiplication and inversion. Examples include

$$U(1), \qquad SU(2), \qquad SU(3), \qquad SO(3), \qquad SO(1,3).$$

The Lie algebra $\mathfrak{g}$ of a Lie group G is the tangent space at the identity element, equipped with a bracket. For matrix groups, the bracket is the commutator:

$$[X, Y] = XY - YX.$$

A basis of the Lie algebra is denoted

$$T_a, \qquad a = 1, \ldots, \dim \mathfrak{g}.$$

A general Lie-algebra element is

$$X = X^a T_a.$$

In gauge theory, the gauge potential is Lie-algebra-valued:

$$A_\mu = A_\mu^a T_a.$$

This means that for each spacetime index μ, the object A_μ is an internal-space generator-valued field.

Definition

The Lie algebra of a matrix Lie group is the vector space of infinitesimal group generators, equipped with the commutator bracket.

The normalization of generators is conventional. Some texts use Hermitian generators, others use anti-Hermitian generators. In these notes, the Yang–Mills chapter uses anti-Hermitian generators unless stated otherwise.

C.2 Structure Constants

The Lie bracket of basis generators is written

$$[T_a, T_b] = f_{ab}{}^c T_c.$$

The coefficients $f_{ab}{}^{c}$ are the structure constants of the algebra. They are antisymmetric in the lower indices:

$$f_{ab}{}^{c} = -f_{ba}{}^{c}.$$

The Jacobi identity is

$$[T_a, [T_b, T_c]] + [T_b, [T_c, T_a]] + [T_c, [T_a, T_b]] = 0.$$

In terms of structure constants,

$$f_{ab}{}^{d} f_{dc}{}^{e} + f_{bc}{}^{d} f_{da}{}^{e} + f_{ca}{}^{d} f_{db}{}^{e} = 0.$$

For an Abelian Lie algebra, all commutators vanish:

$$[T_a, T_b] = 0.$$

Therefore

$$f_{ab}{}^{c} = 0.$$

The Lie algebra of $U(1)$ is Abelian. The Lie algebras of $SU(2)$ and $SU(3)$ are non-Abelian.

Remark

The nonzero structure constants are the algebraic source of the commutator term in the Yang–Mills field strength:

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f_{bc}{}^{a} A^b_\mu A^c_\nu.$$

C.3 Representations

A representation of a Lie group G is a way of realizing its elements as linear transformations on a vector space. If a matter field ψ takes values in such a vector space, a group element $U \in G$ acts by

$$\psi \longrightarrow U\psi.$$

At the Lie-algebra level, generators act as matrices:

$$T_a \longrightarrow R(T_a).$$

The representation must preserve the bracket:

$$[R(T_a), R(T_b)] = f_{ab}{}^{c} R(T_c).$$

The covariant derivative acting on matter in a representation R is

$$D_\mu \psi = \partial_\mu \psi + g A^a_\mu R(T_a) \psi.$$

Different matter fields may transform in different representations of the same gauge group.

Definition

A representation tells us how abstract group or algebra elements act on actual field components. Gauge fields transform in the adjoint representation, while matter fields may transform in fundamental, adjoint, or other representations.

C.4 Adjoint Representation

The adjoint representation is the representation of a Lie algebra on itself. A Lie-algebra element X acts on another element Y by commutation:

$$\mathrm{ad}_X(Y) = [X, Y].$$

For a basis generator,

$$\mathrm{ad}_{T_a}(T_b) = f_{ab}{}^c T_c.$$

Thus the matrices of the adjoint representation are

$$(\mathrm{ad}_{T_a})_b{}^c = f_{ab}{}^c.$$

Lie-algebra-valued fields such as $F_{\mu\nu}$ transform in the adjoint representation:

$$F_{\mu\nu} \longrightarrow U F_{\mu\nu} U^{-1}.$$

Infinitesimally,

$$\delta_\alpha F_{\mu\nu} = [\alpha, F_{\mu\nu}].$$

The adjoint covariant derivative is

$$D_\mu X = \partial_\mu X + g[A_\mu, X].$$

C.5 Exponential Map

The exponential map connects Lie-algebra elements to Lie-group elements. For a matrix Lie group,

$$U = \exp(\alpha^a T_a).$$

For infinitesimal α^a,

$$U = 1 + \alpha^a T_a + O(\alpha^2).$$

The exponential map allows us to move between finite and infinitesimal transformations.

For $U(1)$, the group element is

$$e^{i\alpha}.$$

For non-Abelian groups, the exponential is matrix-valued, and the order of multiplication matters. In general,

$$e^X e^Y \neq e^{X+Y}$$

when $[X, Y] \neq 0$. The Baker–Campbell–Hausdorff formula starts as

$$e^X e^Y = \exp\left(X + Y + \frac{1}{2}[X, Y] + \cdots\right).$$

C.6 Gauge Transformations

A local gauge transformation is a spacetime-dependent group element

$$U = U(x).$$

Matter fields transform as

$$\psi \longrightarrow \psi' = U\psi.$$

The connection transforms as

$$A'_\mu = UA_\mu U^{-1} - \frac{1}{g}(\partial_\mu U)U^{-1}.$$

The field strength transforms covariantly:

$$F'_{\mu\nu} = UF_{\mu\nu}U^{-1}.$$

Therefore $F_{\mu\nu}$ itself is not gauge invariant in a non-Abelian theory, but traces such as

$$\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$$

are gauge invariant.

For an infinitesimal gauge parameter

$$\alpha = \alpha^a T_a,$$

one obtains

$$\delta_\alpha A_\mu = -\frac{1}{g}\partial_\mu\alpha + [\alpha, A_\mu] = -\frac{1}{g}D_\mu\alpha.$$

The exact sign depends on the convention chosen for the covariant derivative and finite transformation law.

C.7 Examples: $U(1)$, $SU(2)$, $SU(3)$, $SO(3)$, $SO(1,3)$

The group $U(1)$ consists of phases

$$e^{i\alpha}.$$

Its Lie algebra is one-dimensional and Abelian. This is the gauge group of electromagnetism.

The group $SU(2)$ consists of unitary 2×2 matrices with determinant one. Its Lie algebra is three-dimensional. In a Hermitian-generator convention, one often uses Pauli matrices; in an anti-Hermitian convention, one uses multiples of $i\sigma_a$. The structure constants are proportional to ${\epsilon_{ab}}^c$.

The group $SU(3)$ consists of unitary 3×3 matrices with determinant one. Its Lie algebra is eight-dimensional. It is the gauge group of the strong interaction in quantum chromodynamics.

The group $SO(3)$ describes rotations of Euclidean three-dimensional space. Its Lie algebra is closely related to $\mathfrak{su}(2)$, but the global group structure differs: $SU(2)$ is the double cover of $SO(3)$.

The proper orthochronous Lorentz group $SO^+(1,3)$ preserves the Minkowski metric and the orientation of time. Its double cover is $SL(2,\mathbb{C})$, which is why spinors naturally transform under $SL(2,\mathbb{C})$ rather than directly under $SO^+(1,3)$.

Summary

Lie groups describe continuous symmetries; Lie algebras describe their infinitesimal generators. In gauge theory, potentials are Lie-algebra-valued, curvatures contain structure constants, matter fields transform in representations, and gauge-invariant quantities are built by contracting internal indices using invariant structures such as traces.

Appendix D

Differential Forms and Geometry

Remark

Differential forms provide a compact language for integration, Stokes' theorem, Maxwell theory, Yang–Mills theory, and geometric field theories. This appendix collects the minimal definitions and identities used in the notes.

D.1 Forms and Exterior Derivative

A p-form on a manifold is an antisymmetric covariant tensor field of degree p. In coordinates,

$$\alpha = \frac{1}{p!}\alpha_{\mu_1\cdots\mu_p} dx^{\mu_1} \wedge \cdots \wedge dx^{\mu_p}.$$

A zero-form is an ordinary function. A one-form is

$$A = A_\mu dx^\mu.$$

A two-form is

$$F = \frac{1}{2}F_{\mu\nu} dx^\mu \wedge dx^\nu.$$

The exterior derivative maps p-forms to $(p+1)$-forms:

$$d : \Omega^p(M) \to \Omega^{p+1}(M).$$

For a function f,

$$df = \partial_\mu f \, dx^\mu.$$

For a one-form $A = A_\mu dx^\mu$,

$$dA = \frac{1}{2}(\partial_\mu A_\nu - \partial_\nu A_\mu) dx^\mu \wedge dx^\nu.$$

The fundamental identity is

$$d^2 = 0.$$

Definition

The exterior derivative is the coordinate-independent derivative on differential forms. Its defining structural property is $d^2 = 0$.

The identity $d^2 = 0$ is the geometric origin of many homogeneous field equations, including the Bianchi identity $dF = 0$ in electromagnetism when $F = dA$.

D.2 Wedge Product

The wedge product combines a p-form and a q-form into a $(p + q)$-form:

$$\wedge : \Omega^p(M) \times \Omega^q(M) \to \Omega^{p+q}(M).$$

It is graded-antisymmetric:

$$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha$$

for $\alpha \in \Omega^p(M)$ and $\beta \in \Omega^q(M)$. In particular,

$$dx^\mu \wedge dx^\nu = -dx^\nu \wedge dx^\mu.$$

Therefore

$$dx^\mu \wedge dx^\mu = 0$$

with no sum.

The exterior derivative satisfies the graded Leibniz rule:

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta$$

for $\alpha \in \Omega^p(M)$.

For Lie-algebra-valued forms, the wedge product also includes multiplication in the Lie algebra or matrix algebra. Thus for a connection one-form A, the expression

$$A \wedge A$$

contains both form antisymmetry and noncommuting matrix multiplication.

D.3 Hodge Dual

The Hodge dual maps p-forms in n dimensions to $(n-p)$-forms:

$$\star : \Omega^p(M) \to \Omega^{n-p}(M).$$

It depends on the metric and orientation. In components, for a p-form α, the dual has components built using the Levi-Civita tensor and the metric.

In four-dimensional electromagnetism, the field strength is a two-form

$$F = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu,$$

and its Hodge dual is again a two-form:

$$\star F = \frac{1}{2} \widetilde{F}_{\mu\nu} dx^\mu \wedge dx^\nu.$$

The source-free Maxwell equations can be written compactly as

$$dF = 0, \qquad d \star F = 0.$$

With sources, one writes schematically

$$d \star F = \star J.$$

Remark

Unlike the exterior derivative, the Hodge dual is not purely topological. It requires a metric. This is why equations involving $\star$ know about causal or metric structure.

D.4 Stokes' Theorem

Stokes' theorem is the central integration theorem for forms. If α is an $(n-1)$-form on an n-dimensional oriented region Ω, then

$$\int_\Omega d\alpha = \int_{\partial\Omega} \alpha.$$

This is the common geometric origin of the fundamental theorem of calculus, Green's theorem, Gauss's theorem, and the ordinary Stokes theorem of vector calculus.

In variational calculus, integration by parts is a local expression of Stokes' theorem. For example,

$$\int_\Omega d^{d+1}x\, \partial_\mu V^\mu = \int_{\partial\Omega} d\Sigma_\mu V^\mu.$$

Boundary terms in field theory are therefore not artificial. They are required by Stokes' theorem.

Definition

Stokes' theorem converts a bulk integral of an exterior derivative into a boundary integral:

$$\int_\Omega d\alpha = \int_{\partial\Omega} \alpha.$$

This theorem is also the reason why total derivative terms can affect boundary charges while leaving local bulk equations unchanged.

D.5 Connections and Curvature

A connection allows one to differentiate fields that carry internal or geometric indices in a way compatible with local frame changes. In Abelian gauge theory, the connection is the electromagnetic potential

$$A = A_\mu dx^\mu.$$

The curvature is

$$F = dA.$$

In non-Abelian gauge theory, the connection is Lie-algebra-valued:

$$A = A_\mu^a T_a dx^\mu.$$

The curvature is

$$F = dA + gA \wedge A.$$

In components,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + g[A_\mu, A_\nu].$$

The covariant exterior derivative of a Lie-algebra-valued form X is schematically

$$DX = dX + g[A, X],$$

with signs depending on the form degree. In components for an adjoint-valued field,

$$D_\mu X = \partial_\mu X + g[A_\mu, X].$$

Definition

The curvature of a connection is the field strength. It measures the failure of covariant derivatives to commute:

$$[D_\mu, D_\nu] = gF_{\mu\nu}$$

when acting on a field in the relevant representation.

D.6 Bianchi Identities

For electromagnetism,

$$F = dA.$$

Therefore

$$dF = d^2A = 0.$$

This is the Abelian Bianchi identity and gives the homogeneous Maxwell equations.

For a non-Abelian connection,

$$F = dA + gA \wedge A.$$

The Bianchi identity is covariant:

$$DF = 0.$$

In components,

$$D_{[\lambda}F_{\mu\nu]} = 0.$$

This is not an equation of motion. It follows from the definition of curvature.

Bianchi identities also appear in geometry. For the Riemann curvature tensor, the differential Bianchi identity is

$$\nabla_{[\lambda}R_{\mu\nu]\rho}{}^{\sigma} = 0.$$

After contractions it leads to

$$\nabla_\mu G^{\mu\nu} = 0,$$

where $G^{\mu\nu}$ is the Einstein tensor.

D.7 Applications to Maxwell and Yang–Mills Theory

Maxwell theory in form notation is especially compact. The field strength is

$$F = dA.$$

The homogeneous equations are

$$dF = 0.$$

The inhomogeneous equations are

$$d \star F = \star J.$$

The action is

$$S[A] = -\frac{1}{2}\int F \wedge \star F$$

up to conventional normalizations.

Yang–Mills theory replaces $F = dA$ by

$$F = dA + gA \wedge A.$$

The action is

$$S[A] = -\frac{1}{2}\int \langle F \wedge \star F \rangle.$$

The equation of motion is

$$D \star F = \star J,$$

and the Bianchi identity is

$$DF = 0.$$

The form notation makes the analogy between Maxwell and Yang–Mills theory transparent. The difference is that the non-Abelian theory uses a covariant exterior derivative and a curvature with a quadratic connection term.

D.8 Applications to Gravity

Gravity can also be written geometrically using connections and curvature. In ordinary differential geometry, the Levi-Civita connection defines covariant derivatives compatible with the metric and torsion-free condition. Its curvature is the Riemann tensor.

In first-order formulations of gravity, one uses a coframe e^a and a spin connection ω^{ab}. The torsion two-form is

$$T^a = de^a + \omega^a{}_b \wedge e^b,$$

and the curvature two-form is

$$R^{ab} = d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}.$$

The Bianchi identities become

$$DT^a = R^a{}_b \wedge e^b,$$

and

$$DR^{ab} = 0.$$

These formulas show that gauge theory and gravity share a geometric language: connections, curvature, covariant derivatives, Bianchi identities, and boundary terms. The physical interpretation differs, but the mathematical pattern is the same.

Summary

Differential forms package many field-theory structures into compact geometric form. Electromagnetism is $F = dA$, Yang–Mills theory is $F = dA + gA \wedge A$, Bianchi identities are curvature identities, and Stokes' theorem explains why boundary terms are unavoidable in variational principles and charge constructions.

Appendix E

Green's Functions

Remark

Green's functions are the response functions of linear differential equations. They convert differential equations with sources into integral formulas. In field theory they appear as retarded responses, advanced responses, static potentials, and, after quantization, propagators.

E.1 Green's Functions for Ordinary Differential Equations

Consider a linear differential operator L acting on functions of one variable. A Green's function $G(x, y)$ satisfies

$$L_x G(x, y) = \delta(x - y),$$

where L_x means that L acts on the x-dependence. If one wants to solve

$$Lf(x) = s(x),$$

then formally

$$f(x) = \int dy\, G(x, y)s(y)$$

provided the boundary conditions are correctly encoded in G.

The boundary conditions are part of the definition. The same differential operator can have different Green's functions for Dirichlet, Neumann, periodic, retarded, or advanced boundary conditions.

Definition

A Green's function is an inverse kernel for a linear differential operator, subject to specified boundary or causality conditions.

As a simple example, for

$$-\frac{d^2}{dx^2} G(x, y) = \delta(x - y)$$

on an interval, the Green's function depends on what is required at the endpoints. Thus it is misleading to speak of "the" Green's function without specifying the problem.

E.2 Green's Functions for Partial Differential Equations

For a linear partial differential equation

$$L\phi(x) = J(x),$$

a Green's function satisfies

$$L_x G(x, y) = \delta^{(D)}(x - y),$$

where $D = d+1$ is the spacetime dimension or $D = d$ for a purely spatial problem. The solution is formally

$$\phi(x) = \int d^D y \, G(x,y) J(y).$$

If the operator is translation invariant, then

$$G(x,y) = G(x-y).$$

In that case Fourier transform is often the most efficient tool. If

$$L(\partial)G(x) = \delta^{(D)}(x),$$

then in momentum space one has

$$L(ik)\widetilde{G}(k) = 1,$$

so

$$\widetilde{G}(k) = \frac{1}{L(ik)}.$$

Poles of $\widetilde{G}(k)$ encode the characteristic modes of the differential operator.

Remark

The formal inverse $1/L(ik)$ is not yet a full Green's function. One must also specify how poles are treated. That prescription encodes causality or boundary conditions.

E.3 Retarded, Advanced, and Feynman-Type Green's Functions

For hyperbolic equations, different Green's functions encode different causal conditions. The retarded Green's function satisfies

$$G_{\text{ret}}(x-y) = 0$$

unless x lies in or on the future light cone of y. It gives causal response to past sources:

$$\phi(x) = \int d^D y \, G_{\text{ret}}(x-y) J(y).$$

The advanced Green's function satisfies the opposite support condition. It is nonzero only when x lies in or on the past light cone of y. It is useful mathematically but usually not used for causal initial-value evolution.

In quantum field theory, the Feynman propagator appears. It is not simply the classical retarded Green's function. It implements time ordering and a specific pole prescription. Formally, for a scalar field it is associated with

$$\frac{1}{k^2 - m^2 + i\epsilon}$$

up to metric and Fourier conventions.

Definition

Retarded Green's functions encode causal response. Advanced Green's functions encode anti-causal response. Feynman-type Green's functions encode time-ordered quantum propagation.

E.4 Green's Functions for the Wave Equation

The massless wave equation with source is

$$\Box\phi = J,$$

where

$$\Box = \partial_t^2 - \nabla^2.$$

A Green's function satisfies

$$\Box G(x-y) = \delta^{(d+1)}(x-y).$$

The retarded Green's function gives the field produced by sources in the past. In three spatial dimensions, the retarded solution has support on the light cone. This reflects sharp propagation at the speed of light.

For a source J, the retarded solution is schematically

$$\phi_{\text{ret}}(x) = \int d^{d+1}y\, G_{\text{ret}}(x-y)J(y).$$

If initial data are also present, the full solution is the sum of the homogeneous solution determined by initial data and a particular solution determined by the source.

Remark

A Green's function solves the inhomogeneous problem with a point source. General sources are handled by superposition, but only because the differential equation is linear.

E.5 Green's Functions for the Klein–Gordon Equation

The sourced Klein–Gordon equation is

$$(\Box + m^2)\phi = J.$$

A Green's function satisfies

$$(\Box_x + m^2)G(x-y) = \delta^{(d+1)}(x-y).$$

In Fourier representation, one formally writes

$$G(x) = \int \frac{d^{d+1}k}{(2\pi)^{d+1}} \frac{e^{-ik_\mu x^\mu}}{-k_\mu k^\mu + m^2}$$

with a pole prescription.

The massive case differs from the massless wave equation in two important ways. First, the dispersion relation is

$$\omega^2 = |\mathbf{k}|^2 + m^2.$$

Second, the static Green's function in three spatial dimensions decays exponentially:

$$G_{\text{static}}(r) \propto \frac{e^{-mr}}{r}.$$

This is the Yukawa behavior. The length scale is

$$\ell_m = \frac{1}{m}.$$

In the massless limit,

$$\frac{e^{-mr}}{r} \longrightarrow \frac{1}{r}.$$

Thus a massive field has short-range static response, while a massless field has long-range static response.

E.6 Physical Interpretation as Response to Sources

The most useful physical interpretation of a Green's function is response to a point source. If

$$L_x G(x, y) = \delta(x - y),$$

then $G(x, y)$ is the field at x produced by a unit source placed at y, with the specified boundary or causal prescription.

For a general source $J(y)$, linearity gives

$$\phi(x) = \int dy\, G(x, y) J(y).$$

This says that the field is a superposition of responses to infinitesimal point sources.

In electromagnetism, retarded Green's functions give retarded potentials. In scalar field theory, they give causal response to scalar sources. In boundary value problems, Green's functions encode how boundaries influence fields. In quantum field theory, propagators encode how disturbances and correlations are transmitted.

Summary

A Green's function is an inverse kernel for a linear differential operator plus a choice of boundary or causal prescription. Retarded Green's functions describe causal response, static Green's functions describe potentials, and the Klein–Gordon Green's function displays the difference between massive short-range response and massless long-range response.

Appendix F

Suggested Problems

Remark

These problems are designed to accompany the notes. They are not meant to be a separate problem book. Their purpose is to force active use of the definitions, variational principles, conservation laws, and examples developed in the main chapters.

F.1 Variational Principle

1. Derive the Euler–Lagrange equation for a real scalar field with Lagrangian density
$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi).$$
Carefully identify the boundary term.

2. Consider the Lagrangian density
$$\mathcal{L} = \frac{1}{2}(\partial_t\phi)^2 - \frac{c^2}{2}(\partial_x\phi)^2.$$
Derive the wave equation and state what boundary conditions make the variational principle well posed on a finite interval.

3. Show that adding a total derivative
$$\mathcal{L} \longrightarrow \mathcal{L} + \partial_\mu K^\mu$$
does not change the Euler–Lagrange equations, but changes the boundary term in the first variation.

4. Let
$$S[q] = \int_{t_1}^{t_2} dt\left(\frac{1}{2}m\dot{q}^2 - V(q)\right).$$
Derive Newton's equation from the variational principle and compare the calculation with the field-theory derivation.

5. For a theory with two real fields ϕ and χ, derive the Euler–Lagrange equations for
$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi + \frac{1}{2}\partial_\mu\chi\partial^\mu\chi - V(\phi,\chi).$$

F.2 Noether Currents

1. For a massless real scalar field with
$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi,$$
show that the constant shift symmetry
$$\phi \longrightarrow \phi + a$$
gives the conserved current $J^\mu = \partial^\mu\phi$.

2. For a complex scalar field with

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi,$$

derive the Noether current associated with global $U(1)$ phase rotations.

3. Derive the canonical stress-energy tensor for a real scalar field and verify that it is conserved on shell when the Lagrangian density has no explicit spacetime dependence.

4. Show explicitly that the conservation equation

$$\partial_\mu J^\mu = 0$$

implies conservation of

$$Q(t) = \int d^d x \, J^0(t, \mathbf{x})$$

when the flux through spatial infinity vanishes.

5. Explain why a gauge symmetry should not be interpreted in exactly the same way as a global symmetry. Use electromagnetism as the example.

F.3 Scalar Fields

1. Starting from

$$(\Box + m^2)\phi = 0,$$

insert a plane wave ansatz and derive the dispersion relation

$$\omega^2 = |\mathbf{k}|^2 + m^2.$$

2. Compute the phase velocity and group velocity of a massive Klein–Gordon mode. Discuss the limits $|\mathbf{k}| \gg m$ and $|\mathbf{k}| \ll m$.

3. For

$$V(\phi) = \frac{1}{2} m^2 \phi^2 + \frac{\lambda}{4} \phi^4,$$

determine for which signs of m^2 and λ the potential is bounded below.

4. For the double-well potential

$$V(\phi) = \frac{\lambda}{4} (\phi^2 - v^2)^2,$$

find the vacua and compute the mass squared of small fluctuations around each vacuum.

5. Consider a static source coupled to a massive scalar field. Explain why the static response is short-ranged and identify the length scale.

F.4 Electromagnetism

1. Starting from

$$\mathbf{E} = -\nabla \Phi - \partial_t \mathbf{A}, \qquad \mathbf{B} = \nabla \times \mathbf{A},$$

show that the two homogeneous Maxwell equations are automatically satisfied.

2. Prove that the gauge transformation

$$\mathbf{A} \longrightarrow \mathbf{A} + \nabla \chi, \qquad \Phi \longrightarrow \Phi - \partial_t \chi$$

leaves $\mathbf{E}$ and $\mathbf{B}$ unchanged.

3. Vary the Maxwell action

$$S[A] = \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - J_\mu A^\mu \right)$$

and derive

$$\partial_\mu F^{\mu\nu} = J^\nu.$$

4. Show that gauge invariance of the source coupling requires

$$\partial_\mu J^\mu = 0.$$

5. Derive the wave equation for **E** and **B** in vacuum and show that electromagnetic waves are transverse.

F.5 Yang–Mills Theory

1. For a non-Abelian gauge field

$$A_\mu = A_\mu^a T_a,$$

derive the component expression

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f_{bc}{}^a A_\mu^b A_\nu^c.$$

2. Show that if the Lie algebra is Abelian, the Yang–Mills field strength reduces to the Maxwell field strength.

3. Starting from

$$S_{\mathrm{YM}} = -\frac{1}{4} \int d^{d+1}x \, F_{\mu\nu}^a F^{a\mu\nu},$$

derive the source-free Yang–Mills equation

$$D_\mu F^{\mu\nu} = 0.$$

4. Verify the Bianchi identity

$$D_{[\lambda} F_{\mu\nu]} = 0$$

using the Jacobi identity for covariant derivatives.

5. Explain in words why non-Abelian gauge fields self-interact while the electromagnetic field does not self-interact at the classical Maxwell level.

F.6 Spinor Fields

1. Using the Clifford algebra

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \mathbb{I},$$

show that the Dirac equation implies the Klein–Gordon equation for each component of ψ.

2. Derive the Dirac equation by varying

$$\mathcal{L}_{\mathrm{D}} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi$$

with respect to $\bar{\psi}$.

3. Derive the adjoint Dirac equation by varying with respect to ψ.
4. Verify directly that
$$j^\mu = \bar{\psi}\gamma^\mu\psi$$
is conserved on shell.
5. Show that minimal electromagnetic coupling gives
$$(i\gamma^\mu D_\mu - m)\psi = 0.$$
Explain how local $U(1)$ symmetry determines the form of D_μ.

F.7 Spontaneous Symmetry Breaking

1. Analyze the minima of
$$V(\phi) = \frac{\lambda}{4}(\phi^2 - v^2)^2.$$
Explain why the potential is symmetric but an individual vacuum need not be.
2. Expand $\phi = v + \eta$ around one vacuum of the double-well potential and compute the quadratic part of the Lagrangian for η.
3. For a complex scalar field with potential
$$V(\phi) = \lambda(\phi^*\phi - v^2/2)^2,$$
describe the set of vacua geometrically.
4. Explain the difference between explicit symmetry breaking and spontaneous symmetry breaking.

F.8 Solitons

1. For a static real scalar field in one spatial dimension, show that the energy functional is
$$E[\phi] = \int dx \left(\frac{1}{2}(\partial_x\phi)^2 + V(\phi)\right).$$
2. For the double-well potential, explain why finite-energy configurations must approach vacua as $x \to \pm\infty$.
3. Define a topological sector for the one-dimensional double-well model using the limiting values of ϕ at spatial infinity.
4. Explain why a configuration connecting $-v$ at one end to $+v$ at the other end cannot be continuously deformed into the trivial vacuum while keeping finite-energy boundary behavior.

F.9 Gravity as Field Theory

1. Compare the role of the metric $g_{\mu\nu}$ in gravity with the role of a scalar field ϕ in scalar field theory. What makes gravity special?

2. Starting from the idea that matter couples to the metric through the stress-energy tensor, explain the meaning of
$$T_{\mu\nu} = -\frac{2}{\sqrt{|g|}}\frac{\delta S_{\text{matter}}}{\delta g^{\mu\nu}}.$$

3. In first-order language, define the torsion two-form
$$T^a = de^a + \omega^a{}_b \wedge e^b$$
and curvature two-form
$$R^{ab} = d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}.$$
Compare these expressions with Yang–Mills curvature.

4. Explain why diffeomorphism invariance is more subtle than an ordinary internal gauge symmetry.

F.10 Boundaries and Charges

1. Starting from a conserved current
$$\partial_\mu J^\mu = 0,$$
derive the relation between the time derivative of the charge in a spatial region and the flux through the boundary.

2. Give an example where a current is locally conserved but the charge in a finite region changes with time.

3. Explain why boundary terms in the variation of the action matter for defining Hamiltonian generators.

4. For electromagnetism, explain why gauge transformations that do not vanish at the boundary may be associated with charges.

5. Discuss the statement: "The physical charge is often more robust than the local representative of the Noether current." Use improvement terms as part of the explanation.

Summary

The problems in this appendix are intended to reinforce the central skills of classical field theory: varying actions, controlling boundary terms, deriving currents, identifying symmetries, reading physical content from field equations, and distinguishing local equations from global charges.

Appendix G

Worked Examples and Common Pitfalls

Remark

This appendix is designed as a bridge between reading the notes and doing field theory independently. It collects worked calculations and common mistakes that recur throughout classical field theory.

G.1 Worked Example: Scalar Field Equation from the Action

Consider the real scalar action

$$S[\phi] = \int d^{d+1}x \left(\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi) \right).$$

The variation is

$$\delta S = \int d^{d+1}x \left(\partial_\mu\phi\partial^\mu\delta\phi - V'(\phi)\delta\phi \right).$$

Integrating the first term by parts gives

$$\int d^{d+1}x\, \partial_\mu\phi\partial^\mu\delta\phi = -\int d^{d+1}x\, \partial_\mu\partial^\mu\phi\, \delta\phi + \text{boundary term.}$$

If the boundary term vanishes, then

$$\delta S = -\int d^{d+1}x\, \left(\Box\phi + V'(\phi) \right) \delta\phi.$$

Since $\delta\phi$ is arbitrary in the bulk, the Euler–Lagrange equation is

$$\Box\phi + V'(\phi) = 0.$$

For

$$V(\phi) = \frac{1}{2}m^2\phi^2,$$

this becomes

$$(\Box + m^2)\phi = 0.$$

Remark

Common mistake: forgetting the boundary term. The bulk equation follows only after the boundary term has been made harmless by boundary conditions, falloff conditions, or an additional boundary term in the action.

G.2 Worked Example: Noether Current of a Complex Scalar Field

The free complex scalar field has Lagrangian density

$$\mathcal{L} = \partial_\mu\phi^*\partial^\mu\phi - m^2\phi^*\phi.$$

It is invariant under the global phase transformation

$$\phi \longrightarrow e^{i\alpha}\phi, \qquad \phi^* \longrightarrow e^{-i\alpha}\phi^*.$$

For infinitesimal α, remove the common parameter and write

$$\Delta\phi = i\phi, \qquad \Delta\phi^* = -i\phi^*.$$

The derivatives of the Lagrangian are

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi^*, \qquad \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} = \partial^\mu \phi.$$

Therefore the Noether current is

$$\begin{aligned} J^\mu &= \partial^\mu \phi^*(i\phi) + \partial^\mu \phi(-i\phi^*) \\ &= i(\phi\partial^\mu \phi^* - \phi^*\partial^\mu \phi). \end{aligned}$$

On shell,

$$\partial_\mu J^\mu = 0.$$

The corresponding charge is

$$Q = \int d^d x \, J^0.$$

Remark

Common mistake: confusing a global phase symmetry with a gauge symmetry. A global symmetry has a constant parameter and usually gives a conserved Noether charge. A gauge symmetry has an arbitrary function as parameter and represents redundancy of description; its charges require more careful boundary analysis.

G.3 Worked Example: Maxwell Equation from the Action

Start with

$$S[A] = \int d^4x \left(-\frac{1}{4} F_{\mu\nu}F^{\mu\nu} - J_\mu A^\mu \right),$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

The variation of the field strength is

$$\delta F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu.$$

Using antisymmetry of $F^{\mu\nu}$,

$$\delta \left(-\frac{1}{4} F_{\mu\nu}F^{\mu\nu} \right) = -F^{\mu\nu}\partial_\mu \delta A_\nu.$$

Integrating by parts gives

$$\delta S = \int d^4x \, (\partial_\mu F^{\mu\nu} - J^\nu) \, \delta A_\nu + \text{boundary term}.$$

Thus the field equation is

$$\partial_\mu F^{\mu\nu} = J^\nu.$$

Gauge invariance of the source term requires

$$\partial_\mu J^\mu = 0.$$

Indeed, under $A_\mu \to A_\mu + \partial_\mu \chi$,

$$\delta S_{\text{int}} = -\int d^4x \, J^\mu \partial_\mu \chi = \int d^4x \, \chi \partial_\mu J^\mu + \text{boundary term}.$$

Remark

Common mistake: treating the potential A_μ as directly observable. The field strength $F_{\mu\nu}$ is gauge invariant in Abelian theory; the potential contains redundancy.

G.4 Worked Example: Why the Dirac Equation Squares to Klein–Gordon

Let the free Dirac equation be

$$(i\gamma^\mu\partial_\mu - m)\psi = 0.$$

Apply the operator $(i\gamma^\nu\partial_\nu + m)$ from the left:

$$(i\gamma^\nu\partial_\nu + m)(i\gamma^\mu\partial_\mu - m)\psi = 0.$$

Expanding gives

$$-\gamma^\nu\gamma^\mu\partial_\nu\partial_\mu\psi - m^2\psi = 0,$$

because the cross terms cancel. Since $\partial_\nu\partial_\mu$ is symmetric in μ, ν, only the anticommutator part of $\gamma^\nu\gamma^\mu$ matters:

$$\gamma^\nu\gamma^\mu\partial_\nu\partial_\mu = \frac{1}{2}\{\gamma^\nu, \gamma^\mu\}\partial_\nu\partial_\mu = \eta^{\nu\mu}\partial_\nu\partial_\mu = \Box.$$

Therefore

$$(\Box + m^2)\psi = 0.$$

Each component of a free Dirac spinor satisfies the Klein–Gordon equation, but the Dirac equation contains additional first-order spinor information.

Remark

Common mistake: saying that the Dirac equation is simply the square root of the Klein–Gordon equation and stopping there. It is more than that: it also carries a spinor representation of the Lorentz group and a conserved positive Dirac current.

G.5 Common Pitfall: Equation Without Model

An equation becomes a physical model only after its variables and assumptions are specified. The expression

$$\partial_t^2\phi - c^2\nabla^2\phi = 0$$

may describe a string, sound, a scalar perturbation, or an idealized relativistic wave. The formal equation alone does not tell us what ϕ means, what c represents, or what boundary conditions are allowed.

A safe reading checklist is:

1. identify the field;
2. identify the domain;
3. identify the parameters and dimensions;
4. identify initial and boundary data;
5. identify the symmetries;
6. identify the conserved quantities;
7. identify what is gauge and what is physical.

G.6 Common Pitfall: Sign Conventions

Metric signature changes signs. Fourier conventions change signs. Gauge-coupling conventions change signs. Gamma-matrix conventions change signs. The same theory can look different in two books because the conventions differ.

For these notes the default convention is

$$\eta_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1), \qquad \Box = \partial_t^2 - \nabla^2.$$

With this convention the Klein–Gordon equation is

$$(\Box + m^2)\phi = 0.$$

Before comparing formulas with another source, check the signature and Fourier convention first.

G.7 Common Pitfall: Local Conservation vs Conserved Charge

A local conservation law has the form

$$\partial_\mu J^\mu = 0.$$

In space-plus-time notation this is

$$\partial_t \rho + \nabla \cdot \mathbf{J} = 0.$$

The charge in a region R is

$$Q_R(t) = \int_R d^d x\, \rho(t, \mathbf{x}).$$

Its time derivative is

$$\frac{dQ_R}{dt} = -\int_{\partial R} \mathbf{J} \cdot d\mathbf{S}.$$

Thus a locally conserved current does not imply that the charge inside every finite region is constant. It implies that charge changes only by flux through the boundary. Total charge is conserved only when the boundary flux vanishes or when the region is all space with suitable falloff.

G.8 Common Pitfall: Gauge Choice vs Physics

A gauge condition is a choice of description, not an additional physical law. For example,

$$\partial_\mu A^\mu = 0$$

is the Lorenz gauge, while

$$\nabla \cdot \mathbf{A} = 0$$

is the Coulomb gauge. These conditions restrict the representative A_μ, not the physical electromagnetic field itself.

Gauge-invariant or gauge-covariant quantities carry the physical content. In Abelian electromagnetism, $F_{\mu\nu}$ is gauge invariant. In non-Abelian gauge theory, $F_{\mu\nu}$ transforms covariantly, and gauge-invariant quantities are constructed by traces, invariant inner products, or suitable nonlocal objects.

G.9 Common Pitfall: Classical Spinor Field vs Fermion

The Dirac equation can be treated as a classical relativistic field equation, but fermions are quantum objects. The physical spin-statistics connection and anticommutation relations belong to quantum field theory. In a classical action formalism, spinor components are often treated as Grassmann-valued variables to prepare for quantization.

The safe statement is this: the classical Dirac action, Lorentz covariance, current conservation, and gauge coupling can be studied classically. The full interpretation of the Dirac field as a fermion field is quantum.

Summary

Most technical errors in classical field theory come from a small number of sources: ignoring boundary terms, confusing global and gauge symmetries, changing sign conventions without noticing, mistaking gauge choices for physics, and forgetting that an equation is not a complete model until its variables, assumptions, and boundary data have been specified.

List of Figures